

Nonparametric regression with Cauchy priors

by

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Abstract

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In this dissertation, we study Bayesian inference methods for the Gaussian sequence model, where the problem is to estimate an unknown sequence from noisy observations. This model is fundamental to nonparametric function estimation and time series analysis, and finds application across diverse scientific disciplines.

Our goal is to develop flexible methods that do not impose restrictive assumptions on the structure of the unknown sequence, and that are able to adapt to different degrees of smoothness. Moreover, we aim to construct efficient algorithms that do not require any hyperparameter tuning and that provide automatic uncertainty quantification. To achieve these goals, we leverage properties of the Cauchy distribution, and of the Bayesian state-space model framework.

In Chapter 1, we focus on “weak smoothness” assumptions formulated in terms of the k^{th} -order differences of the unknown sequence for $k \in \mathbb{N}$. We motivate the use of Cauchy priors through the principle of Maximum Entropy, and propose two algorithms for posterior inference: an efficient Gibbs sampler and an approximate filtering-smoothing method. Through simulations, we compare the performance of our method to other approaches from the statistical literature. The Gibbs sampler implementation of our model demonstrates favorable empirical convergence rates across diverse test functions while maintaining linear computational complexity. Applications to real-world datasets, including financial returns, temperature anomalies, and economic indicators, underscore the method’s versatility and effectiveness compared to other Bayesian and frequentist alternatives.

Chapter 2 introduces several extensions of the method discussed in Chapter 1. We first present the *Cauchy local linear trend* model, and generalize it to the *Cauchy decomposition* model. Both these approaches rely on decomposing the unknown sequence into multiple components, each characterized by Cauchy priors on differences of distinct order. Next, we extend the method from Chapter 1 to allow for irregularly spaced observations by casting

the problem in an infinite-dimensional space and leveraging Cauchy process priors. We also introduce the *latent Cauchy AR(k)* model, which further relaxes the weak smoothness assumptions of 1, leading to a method that is particularly suitable for sequences described by linear differential equations with shocks. We conclude by extending the first-order model from Chapter 1 to handle bivariate observations, broadening its applicability to multivariate settings. For each extension, we develop Gibbs samplers for efficient posterior inference. Moreover, we evaluate the performance of our methods through simulation studies.

Chapter 3 moves into a theoretical exploration of the frequentist properties of Cauchy priors in the context of the Normal means model characterized by nearly black signals. In particular, we derive bounds on the mean squared error of the posterior mean, and demonstrate that it attains minimax risk when the sparsity level is known. This analysis lays the groundwork for future research on theoretical guarantees for the more complex models developed in Chapters 1 and 2.

This dissertation contributes to Bayesian methodology by offering practical, flexible, and computationally efficient tools for sequence estimation. The methods presented herein are broadly applicable in contexts where signals must be estimated from noisy data without imposing strong smoothness constraints.

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Chapter 1

Basic Models

1.1 Introduction

We study the problem of estimating an unknown sequence $\theta_1, \dots, \theta_n$ from data $y_1, \dots, y_n$ under the model:

$$y_i = \theta_i + \epsilon_i \quad \text{with } \epsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2). \quad (1.1)$$

Model (1.1) is often referred to as the Gaussian sequence model and is fundamental to nonparametric function estimation (see e.g., the book length treatment [Johnstone \(2019\)](#)) and to time series analysis (see e.g., [Durbin and Koopman \(2012\)](#); [Kitagawa and Gersch \(1996\)](#)). In nonparametric function estimation, $\theta_1, \dots, \theta_n$ are the values of an unknown function $f^* : [0, 1] \rightarrow \mathbb{R}$ at fixed design points $i/n, 1 \leq i \leq n$ i.e., $\theta_i = f^*(i/n)$. In time series, $\theta_1, \dots, \theta_n$ denote the underlying trend in an observed regularly sampled time series $y_1, \dots, y_n$.

For effective estimation in (1.1), one needs to make assumptions on the structure of the unknown vector $\theta = (\theta_1, \dots, \theta_n)$. In this Chapter, we shall focus on “smoothness” assumptions which are specified in terms of differences of θ . Before describing our main assumption, let us first introduce our notation for differences. We use $D^{(1)}\theta := (\theta_2 - \theta_1, \dots, \theta_n - \theta_{n-1})$ for the vector of first differences of θ . Higher order differences are defined recursively via $D^{(k)}\theta = D^{(1)}(D^{(k-1)}\theta)$ for $k \geq 2$. Note that the length of the vector $D^{(k)}\theta$ equals $n - k$ where n is the length of θ .

We work in the Bayesian paradigm and assume a Cauchy prior for the components of $D^{(k)}\theta$ for a fixed k . Specifically, we assume the prior:

$$(D^{(k)}\theta)_1, \dots, (D^{(k)}\theta)_{n-k} \stackrel{\text{i.i.d.}}{\sim} \text{Cauchy}(0, \tau) \quad (1.2)$$

for an unknown hyperparameter τ . Recall that the Cauchy distribution with location u and scale τ is given by:

$$f_{\text{Cauchy}}(x; u, \tau) = \frac{\tau}{\pi((x - u)^2 + \tau^2)}$$

The use of the Cauchy prior for the differences $D^{(k)}\theta$ has previously appeared, for example, in Kitagawa and Gersch (1996, Section 8.4), Lang et al. (2002), Markkanen et al. (2019) and Suuronen et al. (2022); these papers focused on obtaining posterior samples for $\theta_1, \dots, \theta_n$. There are also papers that used Cauchy priors and focused on obtaining posterior maximizers (MAP) (see, for example, Karakuş et al. (2020) and Karakuş and Achim (2020)). Other works have approached the estimation of θ in similar settings using alternative prior choices, such as Faulkner and Minin (2018), Liu et al. (2017), and Roualdes (2015).

Here is a description of our contributions to this problem. In Section 1.2, we provide a rationale for considering the Cauchy distribution for the differences $(D^{(k)}\theta)_i$. We start with popular frequentist methodology for incorporating smoothness constraints on θ in terms of constraints on the ℓ_p norm of $D^{(k)}\theta$ for $p \in [1, \infty)$. We then introduce a constraint that is weaker (i.e., less stringent) than all these ℓ_p norm constraints. We then argue that a Bayesian version of this constraint together with the principle of Maximum Entropy leads to the Cauchy prior (1.2). This suggests that the Cauchy prior is suitable for dealing with sequences that satisfy smoothness in a very weak sense. In Section 1.3, we cast our model in state-space form, and take advantage of its special structure to develop an algorithm providing inference through approximate filtering and smoothing. We also review other alternative inference methods, in particular a fast Gibbs sampler that leverages the state-space structure of the problem to achieve tremendous efficiency gains. Finally, in Sections 1.4 and 1.5, we use synthetic and real data to investigate the performance of our methodology in comparison to other frequentist and Bayesian methods. In particular, we inspect convergence rates of different methods under different assumptions on the data generating process, and we highlight that the time complexity of Cauchy estimation via Gibbs sampling compares favorably with respect to trend filtering. The latter fact is particularly interesting, considering that usually Bayesian methods are believed to be more computationally expensive than their frequentist counterparts.

1.2 Cauchy Prior Rationale

In this section, we provide some reasoning leading to the Cauchy prior (1.2). We start with a review of the usual frequentist approach to incorporating smoothness constraints on θ .

Smoothness Constraints and Frequentist Methods

In frequentist methods, smoothness is usually dealt with by imposing a hard constraint on an appropriate ℓ_p norm of $D^{(k)}\theta$ for fixed choice of k and p . Specifically, the following constraint is quite popular:

$$\|D^{(k)}\theta\|_p \leq V(n-k)^{\frac{1}{p}-k} \quad \text{for fixed } p \in [1, \infty] \text{ and constant } V > 0 \quad (1.3)$$

where $\|x\|_p := (\sum_i |x_i|^p)^{1/p}$ if $p \in [1, \infty)$ and $\|x\|_\infty := \max_i |x_i|$.

Given the constraint (1.3) for some p , a natural frequentist approach for estimating the unknown vector $(\theta_1, \dots, \theta_n)$ from the data $(y_1, \dots, y_n)$ is given by the following constrained least squares estimator:

$$\hat{\theta}(k, p, V) = \underset{\theta: \|D^{(k)}\theta\|_p \leq V(n-k)^{\frac{1}{p}-k}}{\operatorname{argmin}} \sum_{i=1}^n (y_i - \theta_i)^2. \quad (1.4)$$

$\hat{\theta}(k, p, V)$ is also the maximum likelihood estimator under the assumed Gaussian distribution for the errors in (1.1) and the constraint (1.3). Typically k and p are selected to be some standard values (e.g., $k = 2$ and $p = 1$) by the user and V is selected by some form of cross validation. The optimization problem in (1.4) is convex and can be computed efficiently using software for convex optimization.

Let us now provide some more intuition for the smoothness constraint (1.3). When $p = \infty$, it becomes $\max_{1 \leq i \leq n-k} |(D^{(k)}\theta)_i| \leq V(n-k)^{-k}$ which stipulates that the entries of $D^{(k)}\theta$ never exceed $V(n-k)^{-k}$ in absolute value. For finite $p \in [1, \infty)$, (1.3) becomes

$$\frac{1}{n-k} \sum_{i=1}^{n-k} |(D^{(k)}\theta)_i|^p \leq (V(n-k)^{-k})^p, \quad (1.5)$$

which stipulates that the absolute values of the entries of $D^{(k)}\theta$ raised to the power p are bounded by $(V(n-k)^{-k})^p$ on average. Of these, let us highlight the constraint for $p = 1$:

$$\frac{1}{n-k} \sum_{i=1}^{n-k} |(D^{(k)}\theta)_i| \leq V(n-k)^{-k} \quad (1.6)$$

which stipulates that the absolute values of the entries of $D^{(k)}\theta$ do not exceed $V(n-k)^{-k}$ on average.

Because ℓ_p norms with respect to probability measures monotonically increase with p , the constraint (1.3) becomes progressively stronger as p increases. The strongest of this family of constraints (for fixed $V(n-k)^{-k}$) corresponds to $p = \infty$, and the weakest constraint of this family is (1.6) corresponding to $p = 1$. There exist many sequences θ that satisfy the $p = 1$ constraint (1.6) but do not satisfy (1.3) for every $p > 1$. As a concrete example, suppose θ is given by the piecewise constant vector:

$$\theta = (0, \dots, 0, V, \dots, V). \quad (1.7)$$

Then, for $k = 1$, the differenced vector $D^{(1)}\theta = (0, \dots, 0, V, 0, \dots, 0)$ clearly satisfies the ℓ_1 constraint (1.6) but does not satisfy (1.3) for any $p > 1$.

Because of the mild nature of the constraint (1.6) (compared to (1.3) for $p > 1$) and the resultant flexibility to work for a wider set of sequences $\theta_1, \dots, \theta_n$, the estimator (1.4) for $p = 1$ (i.e., $\hat{\theta}(k, 1, V)$) along with its penalized variant:

$$\hat{\theta}_{\text{pen}}(k, 1, \lambda) = \operatorname{argmin} \left\{ \theta : \sum_{i=1}^n (y_i - \theta_i)^2 + \lambda \|D^{(k)}\theta\|_1 \right\} \quad (1.8)$$

have been quite popular in time series and nonparametric regression. They are known by the name total variation regularization for $k = 1$ (see e.g., [Rudin et al. \(1992\)](#)) and by the name trend filtering for $k \geq 1$ (see e.g., [Kim et al. \(2009\)](#); [Mammen and van de Geer \(1997\)](#); [Tibshirani \(2014\)](#)).

If one is willing to tolerate non-convexity, then it is possible to obtain even milder constraints than (1.6). Our main example for this will be the following constraint:

$$\frac{1}{n-k} \sum_{i=1}^{n-k} \log \left(1 + \frac{(n-k)^{2k}}{V^2} |(D^{(k)}\theta)_i|^2 \right) \leq \log 4. \quad (1.9)$$

This constraint also stipulates that the components of $D^{(k)}\theta$ are not too large in some average sense. But this sense is actually much weaker than (1.6). Indeed, every θ that satisfies (1.6) also satisfies (1.9), because of the following elementary inequality (applied to $m = n - k$ and $u_i = \frac{(n-k)^k}{V} (D^{(k)}\theta)_i$):

$$\frac{1}{m} \sum_{i=1}^m \log(1 + u_i^2) \leq \frac{2}{m} \sum_{i=1}^m \log(1 + |u_i|) \leq 2 \log \left(1 + \frac{1}{m} \sum_{i=1}^m |u_i| \right). \quad (1.10)$$

The first inequality above is a consequence of $1 + u_i^2 \leq (1 + |u_i|)^2$ and the second inequality is a consequence of the concavity of the logarithm function.

On the other hand, there are many sequences $\theta_1, \dots, \theta_n$ that satisfy (1.9) but fail to satisfy (1.6). As an explicit example, consider a vector θ that is constant at c , except for one entry that equals $V + c$. Taking $k = 1$, we obtain $D^{(1)}\theta = (0, \dots, 0, V, -V, 0, \dots, 0)$. Then (1.6) is violated because

$$\frac{1}{n-1} \sum_{i=1}^{n-1} |(D^{(1)}\theta)_i| = \frac{2V}{n-1} \quad (1.11)$$

which is strictly larger than $V/(n-1)$ for all $n \geq 2$. On the other hand, θ will satisfy (1.9) for all large n . Indeed, the left-hand side of (1.9) is

$$\frac{1}{n-k} \sum_{i=1}^{n-k} \log \left(1 + \frac{(n-k)^{2k}}{V^2} |(D^{(k)}\theta)_i|^2 \right) = \frac{2}{n-1} \log(1 + (n-1)^2)$$

which can be checked to be strictly smaller than $\log 4$ for all $n \geq 6$. In fact, its asymptotic order is $(\log n)/n = o(1)$ that is much smaller than the right-hand side of (1.9), which is a constant. Therefore θ satisfies (1.9) quite comfortably as long as n is not too small.

We shall refer to (1.9) as the “Cauchy” constraint because it can be checked to be equivalent to

$$\frac{-1}{n-k} \sum_{i=1}^{n-k} \log f_{\text{Cauchy}} \left((D^{(k)}\theta)_i; 0, \frac{V}{(n-k)^k} \right) \leq \log \left(4\pi \frac{V}{(n-k)^k} \right). \quad (1.12)$$

Analogous to (1.4), one can define the constrained least squares estimator with the constraint (1.9):

$$\hat{\theta}_{\text{cauchy}}(k, V) = \operatorname{argmin} \left\{ \sum_{i=1}^n (y_i - \theta_i)^2 : \frac{1}{n-k} \sum_{i=1}^{n-k} \log \left(1 + \frac{(n-k)^{2k}}{V^2} |(D^{(k)}\theta)_i|^2 \right) \leq \log 4. \right\} \quad (1.13)$$

and also define the analogous penalized version. Computationally, these estimators are harder compared to (1.4) and (1.8) because of the nonconvexity of the set of θ satisfying (1.9). However efficient algorithms exist along with experimental demonstrations of the superiority of (1.13) compared to (1.4) and (1.8), and applications to imaging problems (see, for example, Karakuş et al. (2020) and Karakuş and Achim (2020)). For a general overview of nonconvex penalties and constraints in sequence and function estimation problems, see Mohammad-Djafari (2012) and Nikolova (2011).

Bayesian Methods with Maximum Entropy Priors

Let us now come to Bayesian methods. Unlike frequentist methods that work with hard constraints such as (1.3) and (1.9), Bayesian methods need specification of priors for the unknown sequence $\theta_1, \dots, \theta_n$. The Maximum Entropy principle (see e.g., Cover and Thomas (2006, Chapter 11), Jaynes (2003, Chapter 11), Sivia and Skilling (2006, Chapter 5), Von der Linden et al. (2014, Chapter 11)) offers a logical approach to convert the hard constraints (1.3) and (1.9) into prior distributions for θ .

Let us first consider (1.3) for a fixed finite $1 \leq p < \infty$. As we already mentioned, this constraint stipulates that the absolute values of the entries of $D^{(k)}\theta$ raised to the power p are bounded by $(V(n-k)^{-k})^p$ on average. Here, “on average” refers to averaging with respect to the index i corresponding to the entries $(D^{(k)}\theta)_i$ of $D^{(k)}\theta$. From the Bayesian perspective, it is natural to change the meaning of “average” to be with respect to the prior uncertainty as opposed to the indices corresponding to entries of $D^{(k)}\theta$. This leads to the following “Bayesian” version of (1.5):

$$\mathbb{E} \left| (D^{(k)}\theta)_i \right|^p \leq (V(n-k)^{-k})^p \quad \text{for each } i = 1, \dots, n-k \quad (1.14)$$

where the expectation is with respect to the prior over θ . Both (1.5) and (1.14) stipulate that the absolute values of the entries of $D^{(k)}\theta$ raised to the power p are bounded by $(V(n-k)^{-k})^p$ on average. But the average is with respect to the entries of the vector $D^{(k)}\theta$ in the former while it is with respect to the prior in the latter. The constraint in (1.14) applies to the prior distribution (unlike (1.3) that applies to the true sequence $\theta_1, \dots, \theta_n$). Just as (1.3), the Bayesian constraints (1.14) also get progressively more stringent as p grows from 1 to ∞ with the weakest constraint corresponding to $p = 1$.

Inequality (1.14) presents a constraint on the set of allowable priors. We now invoke the principle of maximum entropy to find the prior that maximizes entropy among all priors

satisfying (1.14). The maximizing entropy prior distribution on $(D^{(k)}\theta)_1, \dots, (D^{(k)}\theta)_{n-k}$ subject to (1.14) is given by (using e.g., Cover and Thomas (2006, Theorem 12.1.1)):

$$(D^{(k)}\theta)_1, \dots, (D^{(k)}\theta)_{n-k} \stackrel{\text{i.i.d.}}{\sim} f_{\text{MaxEnt}}^{(p)}\left(\cdot; \frac{V}{(n-k)^k}\right) \quad (1.15)$$

with $f_{\text{MaxEnt}}^{(p)}(x; \tau) = \frac{p^{1-(1/p)}}{2\tau\Gamma(1/p)} \exp\left(-\frac{|x|^p}{p\tau^p}\right).$

Note that prior independence between $(D^{(k)}\theta)_1, \dots, (D^{(k)}\theta)_{n-k}$ is also a consequence of the maximum entropy principle (the constraint (1.14) itself is silent about independence across i).

When $p = 1$, the density $f_{\text{MaxEnt}}^{(p)}(\cdot; V(n-k)^{-k})$ is the Laplace distribution with location 0 and scale $V(n-k)^{-k}$. When $p = 2$, the density $f_{\text{MaxEnt}}^{(p)}(\cdot; V(n-k)^{-k})$ is the Normal distribution with mean 0 and standard deviation $V(n-k)^{-k}$.

For $p = \infty$, the constraint (1.3) is $\max_{1 \leq i \leq n-k} |(D^{(k)}\theta)_i| \leq V(n-k)^{-k}$ which means that $(D^{(k)}\theta)_i$ never exceeds $V(n-k)^{-k}$. From the Bayesian perspective, this can be reinterpreted to mean that the prior distribution for $(D^{(k)}\theta)_i$ is supported on $[-V(n-k)^{-k}, V(n-k)^{-k}]$ for each $i = 1, \dots, n-k$. The prior that maximizes entropy subject to this support restriction is:

$$(D^{(k)}\theta)_1, \dots, (D^{(k)}\theta)_{n-k} \stackrel{\text{i.i.d.}}{\sim} f_{\text{MaxEnt}}^{(\infty)}(\cdot; V(n-k)^{-k}) = \text{Uniform}[-V(n-k)^{-k}, V(n-k)^{-k}].$$

Now let us consider the constraint (1.9) that, as argued earlier, is weaker than (1.3) for every p . To derive the Bayesian prior corresponding to (1.9), we first reinterpret the averaging over i in (1.9) as an average over the prior distribution to obtain the Bayesian constraint:

$$\mathbb{E} \log \left(1 + \frac{(n-k)^{2k}}{V^2} |(D^{(k)}\theta)_i|^2 \right) \leq \log 4 \quad \text{for every } i = 1, \dots, n-k. \quad (1.16)$$

As for (1.9) and (1.3), the Bayesian constraint (1.16) is also weaker than the Bayesian ℓ_p constraint (1.14) for every $p \in [1, \infty)$ (the same proof given in (1.10) applies with $\frac{1}{m} \sum_{i=1}^m$ replaced by prior expectation).

The principle of maximum entropy then leads to the centered Cauchy prior with scale parameter $V(n-k)^{-k}$:

$$(D^{(k)}\theta)_1, \dots, (D^{(k)}\theta)_{n-k} \stackrel{\text{i.i.d.}}{\sim} \text{Cauchy}(0, V(n-k)^{-k}). \quad (1.17)$$

A proof of this fact can be found in Kapur (1989). Equation (1.17) corresponds to (1.2) for $\tau = V(n-k)^{-k}$. The following table summarizes the Bayesian constraints with respect to which entropy is maximized by each of the four priors: uniform, normal, Laplace and Cauchy.

Our focus is on the Cauchy prior because its corresponding constraint is much milder, and hence more widely applicable, compared to the other constraints in Table 1.1. We will henceforth use the shorthand notation $C(0, \tau)$ for $\text{Cauchy}(0, \tau)$.

Bayesian Constraint	Entropy Maximizing Prior
$\mathbb{E} (D^{(k)}\theta)_i \leq \tau$ for all i	$(D^{(k)}\theta)_i \stackrel{\text{i.i.d.}}{\sim} \text{Laplace}(0, \tau)$
$\mathbb{E}((D^{(k)}\theta)_i ^2) \leq \tau^2$ for all i	$(D^{(k)}\theta)_i \stackrel{\text{i.i.d.}}{\sim} N(0, \tau^2)$
$\text{supp}(D^{(k)}\theta)_i \in [-\tau, \tau]$ for all i	$(D^{(k)}\theta)_i \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}[-\tau, \tau]$
$\mathbb{E} \log \left(1 + \frac{ (D^{(k)}\theta)_i ^2}{\tau^2} \right) \leq \log 4$ for all i	$(D^{(k)}\theta)_i \stackrel{\text{i.i.d.}}{\sim} \text{Cauchy}(0, \tau)$

Table 1.1: Constraints and their corresponding entropy maximizing priors.

1.3 Posterior Inference

Sections 1.1 and 1.2 lead to the following simple model

$$\begin{aligned} y_i &= \theta_i + \epsilon_i, & \epsilon_i &\stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2) \\ (D^{(k)}\theta)_j &= \eta_j, & \eta_j &\stackrel{\text{i.i.d.}}{\sim} C(0, \tau) \end{aligned} \quad (1.18)$$

with $i = 1, \dots, n$, and $j = 1, \dots, n - k$. We complete the prior specification by letting $\theta_1, \dots, \theta_k \stackrel{\text{i.i.d.}}{\sim} N(0, L)$, with L an arbitrarily large positive constant, and by placing Jeffreys' priors on the scale parameters (i.e. $p(\sigma) \propto 1/\sigma$ and $p(\tau) \propto 1/\tau$). We will refer to this model as *latent Cauchy k^{th} -order differences* model. Given this model, we are interested in inference based on the posterior distribution $p(\theta | y)$. This posterior distribution cannot be derived in closed form, so we need to resort to simulation or approximation techniques. For computational purposes, it is relevant to notice that (1.18) is a linear state-space model (see as a reference Kitagawa (2020), Chapters 9 and 14). A linear state-space model is a model for an l -variate time series y_t given by the combination of the following observation model and system model

$$\begin{aligned} y_t &= F_t x_t + v_t & (\text{observation model}) \\ x_t &= G_t x_{t-1} + H_t w_t & (\text{system model}) \end{aligned} \quad (1.19)$$

where x_t is an m -dimensional unobservable vector, referred to as the *state*. v_t and w_t are white noise vectors with covariance matrices Q_t and R_t , respectively. F_t, G_t , and H_t are deterministic matrices, possibly depending on t . In order to express (1.18) as a state-space model, we first re-write its second equation as

$$\theta_{j+k} = \sum_{l=1}^k \phi_l \theta_{j+k-l} + \eta_j \quad (1.20)$$

where $-\phi_l$ is equal to the $(k+1-l)^{th}$ entry of the first row of $D^{(k)}$, for $l = 1, \dots, k$. Next, letting $x_{j+k} := (\theta_{j+k}, \dots, \theta_{j+1})^\top$, we can express (1.18) as

$$\begin{aligned} y_i &= (1 \quad \mathbf{0}^\top) x_i + \epsilon_i & \epsilon_i &\stackrel{iid}{\sim} N(0, \sigma^2) \\ x_{j+k} &= \begin{pmatrix} \phi_{1:(k-1)} & \phi_k \\ I_{k-1} & \mathbf{0} \end{pmatrix} x_{j+k-1} + \begin{pmatrix} \eta_j \\ \mathbf{0} \end{pmatrix} & \eta_j &\stackrel{iid}{\sim} C(0, \tau) \end{aligned} \quad (1.21)$$

with $i = 1, \dots, n$, and $j = 1, \dots, n-k$, and with I_{k-1} denoting the identity matrix of dimension $k-1$, $\mathbf{0}$ the $(k-1) \times 1$ vector of zeros, and $\phi_{1:(k-1)}$ the vector $(\phi_1, \dots, \phi_{k-1})$.

Notice that $\eta_j \stackrel{i.i.d}{\sim} C(0, \tau)$ can be equivalently expressed as a scale mixture of Normals:

$$\begin{aligned} \eta_j | \lambda_j^2 &\stackrel{ind}{\sim} N(0, \tau^2 \lambda_j^2) \\ \lambda_j^2 &\stackrel{i.i.d}{\sim} IG(0.5, 0.5) \end{aligned} \quad (1.22)$$

for every $j = 1, \dots, n-k$ (see e.g. Equation (28.13) in [Murphy \(2023\)](#)). We use $IG(\alpha, \beta)$ to denote the Inverse-Gamma distribution with shape α and scale β , defined as

$$IG(x; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} \exp\left(-\frac{\beta}{x}\right) I\{x > 0\},$$

where $\Gamma(\cdot)$ denotes the gamma function.

We observe that (1.21) is a state-space model with $l = 1$ and $m = k$, where the matrices G_t and Q_t do not depend on t , and are such that $F_t = F = (1 \quad \mathbf{0}^\top)$, and $Q_t = Q = \sigma^2$. Moreover, $G_t = \begin{pmatrix} \phi_{1:(k-1)} & \phi_k \\ I_{k-1} & \mathbf{0} \end{pmatrix}$ for $t > k$, while $G_t = \begin{pmatrix} \mathbf{0}^\top & 0 \\ I_{k-1} & \mathbf{0} \end{pmatrix}$ for $t \leq k$. The system error w_t has dimension equal to 1 and variance $R_t = \tau^2 \lambda_{t-k}^2$ for $t > k$, and $R_t = L$ for $t \leq k$ with L an arbitrarily large positive constant. Finally, H_t is constant and equal to $(1 \quad \mathbf{0}^\top)$.

This special structure of the problem allows us to use sequential algorithms for posterior inference. We will next introduce two different estimation approaches: a Gibbs sampler, similar to the one proposed by [Lang et al. \(2002\)](#), and a novel sequential approximate method. We will also summarize other alternative approaches to posterior estimation.

Gibbs Sampler

In order to obtain the complete conditional distributions needed to set up a Gibbs sampler, we take advantage of (1.22), i.e. of the characterization of the Cauchy distribution as a scale mixture of Gaussian distributions. The introduction of auxiliary variables $\lambda^2 = (\lambda_1^2, \dots, \lambda_{n-k}^2)$ allows us to obtain closed form expressions for the complete conditional distributions of each variable in the model $(\lambda_1^2, \dots, \lambda_{n-k}^2, \tau^2, \sigma^2, \theta_1, \dots, \theta_n)$.

We report these distributions next, while deferring their mathematical derivations to Appendix A.1. In what follows, we will refer to the complete conditional distribution of a parameter, say α , using the notation $p(\alpha|\cdot)$.

Each auxiliary parameter λ_j^2 , $j = 1, \dots, n - k$, is conditionally independent with

$$p(\lambda_j^2 | \cdot) = IG \left(1, \frac{(D^{(k)}\theta)_j^2 + \tau^2}{2\tau^2} \right).$$

The complete conditional distributions of τ^2 and σ^2 are respectively

$$p(\tau^2 | \cdot) = IG \left(\frac{n - k}{2}, \sum_{j=1}^{n-k} \frac{(D^{(k)}\theta)_j^2}{2\lambda_j^2} \right),$$

and

$$p(\sigma^2 | \cdot) = IG \left(\frac{n}{2}, \frac{\|y - \theta\|_2^2}{2} \right).$$

Let Λ be the diagonal matrix having j^{th} diagonal entry equal to λ_j^2 , for $j = 1, \dots, n - k$. The complete conditional distribution of the vector θ is then given by

$$p(\theta | \cdot) = N(\mu, \Sigma)$$

with mean and covariance

$$\mu = \frac{1}{\sigma^2} \Sigma y \quad \text{and} \quad \Sigma = \left(\frac{1}{\sigma^2} I_n + \frac{1}{\tau^2} D^{(k)\top} \Lambda^{-1} D^{(k)} \right)^{-1}.$$

The Gibbs sampler alternates between sampling the vectors θ and λ^2 , and the scalars τ^2 and σ^2 from their complete conditional distributions. Notice that θ is n -dimensional, and λ^2 is $(n - k)$ -dimensional, so they might seem computationally expensive to sample for large dataset sizes. The vector λ^2 is actually fast to sample, as its entries are conditionally i.i.d., while that is not the case for θ , which is conditionally distributed as a multivariate normal with covariance matrix Σ . In the latter case, the computational bottleneck lies in the inversion of the $n \times n$ precision matrix

$$P = \frac{1}{\sigma^2} I_n + \frac{1}{\tau^2} D^{(k)\top} \Lambda^{-1} D^{(k)}$$

which is needed to obtain the covariance Σ . This operation, if performed naively, would have time complexity of $O(n^3)$, which would lead to a very slow Gibbs sampler, unable to scale to large dataset sizes. To improve efficiency, we take advantage of the approach in [Rue \(2001\)](#) that has only $O(n)$ complexity for banded precision matrices. Notice that in our case the matrix $D^{(k)}$ has banded structure, with bandwidth equal to $k + 1$, leading to a banded precision matrix P with bandwidth $b_k = 2k + 1$. Sampling θ from its conditional distribution then involves the following steps:

1. Performing Cholesky factorization of the precision matrix

$$P = LL^\top,$$

which is efficiently computed in nb_k^2 floating point operations.

2. Solving the linear equations

- a) $Lv = y/\sigma^2$

- b) $L^\top a = v$

- c) $L^\top b = z$, where $z_i \stackrel{\text{i.i.d.}}{\sim} N(0, 1)$, for $i = 1, \dots, n$.

3. Letting $\theta = a + b$.

The algorithm above allows us to sample θ without directly inverting the precision matrix P . It does so by taking advantage of the Cholesky factors to separately obtain, and then combine, the mean of $p(\theta|\cdot)$, i.e. $a = \Sigma y/\sigma^2$, and a vector $b \sim N(0, \Sigma)$. Solving the three equations in step (2) takes in total $4nb_k$ floating point operations.

The resulting Gibbs sampler is described as Algorithm 1.

Approximate Method

We will now outline a novel approach that does not involve MCMC sampling, and that introduces approximations in the context of filtering and smoothing algorithms for state-space models. We will start off by briefly summarizing the general structure of these algorithms, and then describe the approximation scheme proposed for model (1.21). Since these algorithms are usually performed in the context of time series problems, for the rest of this Section we will adopt the standard time series indexing by t .

Let's consider a linear state-space model as in (1.19). Filtering and smoothing algorithms provide a framework (see Kitagawa (2020) Chapter 14) that enables inference on the latent, scalar or vector-valued, state x_t by recursive computation of the three integrals reported in Table 1.2. When both $p(x_t|x_{t-1})$ and $p(y_t|x_t)$ are linear and Gaussian, these integrals can be computed in closed form, and each of the conditional distributions in Table 1.2 is Gaussian too. The recursive algorithm obtained leads to the well-known Kalman filter and smoother equations. If we drop the normality assumption, the recursive closed form structure is lost, and we need an approximate procedure to evaluate the integrals, or a method to simulate from the conditional distributions that circumvents the need to compute the integrals.

In the specific state-space model (1.21) under consideration, the observation y_t is univariate, while the state $x_t = (\theta_t, \dots, \theta_{t-k+1})$ is a vector of dimension $k \times 1$ for every t . Throughout this Section, we primarily work with the vector x_t , although we occasionally focus on θ_t , particularly in the discussion on smoothing. Additionally, we adopt the shorthand notation $y_{a:b}$ to refer to the vector $(y_a, \dots, y_b)$ for some integers a and b with $a < b$.

Our approximation method is based on a two step procedure. First, moving forward in time, we iterate the following for every t :

Algorithm 1: Gibbs Sampler for the latent Cauchy k^{th} -order differences model

Input: data $y = (y_1, \dots, y_n)$, and order k .

Initialize $\sigma^{2(0)}$, $\tau^{2(0)}$, $\lambda_1^{2(0)}, \dots, \lambda_{n-k}^{2(0)}$ to some positive constants (we let $\sigma^{2(0)} = 10$, $\tau^{2(0)} = 0.01$, and $\lambda_j^{2(0)} = 1$ for $j = 1, \dots, n - k$).

For $m = 1, \dots, M$ (with M a large positive integer):

1. Let

$$\Lambda^{(m-1)} = \text{diag}(\lambda^{2(m-1)})$$

and

$$P^{(m-1)} = \frac{1}{\sigma^{2(m-1)}} I_n + \frac{1}{\tau^{2(m-1)}} D^{(k)\top} \Lambda^{-1(m-1)} D^{(k)},$$

and perform the Cholesky factorization

$$P^{(m-1)} = LL^\top$$

2. Solve

a) $Lv = y/\sigma^{2(m-1)}$

b) $L^\top a = v$

c) $L^\top b = z$, where $z \sim N(\mathbf{0}, I_n)$

3. Let

$$\theta^{(m)} = a + b$$

and

$$\beta^{(m)} = D^{(k)} \theta^{(m)}$$

4. Sample

$$\lambda_j^{2(m)} \stackrel{\text{ind}}{\sim} IG \left(1, \frac{\beta_j^{2(m)} + \tau^{2(m-1)}}{2\tau^{2(m-1)}} \right), \quad j = 1, \dots, n - k$$

$$\tau^{2(m)} \sim IG \left(\frac{n - k}{2}, \sum_{i=1}^{n-k} \frac{\beta_i^{2(m)}}{2\lambda_i^{2(m)}} \right)$$

$$\sigma^{2(m)} \sim IG \left(\frac{n}{2}, \frac{\|y - \theta^{(m)}\|_2^2}{2} \right)$$

Table 1.2: Filtering and smoothing recursions

$$\text{One-step ahead prediction : } p(x_t|y_{1:t-1}) = \int p(x_t|x_{t-1})p(x_{t-1}|y_{1:t-1})dx_{t-1}$$

$$\text{Filtering : } p(x_t|y_{1:t}) = \frac{p(y_t|x_t)p(x_t|y_{1:t-1})}{\int p(y_t|x)p(x_t|y_{1:t-1})dx_t}$$

$$\text{Smoothing : } p(x_t|y_{1:n}) = p(x_t|y_{1:t}) \int \frac{p(x_{t+1}|y_{1:n})p(x_{t+1}|x_t)}{p(x_{t+1}|y_{1:t})}dx_{t+1}$$

- Assume the filtering distribution at time t can be approximated by a mixture of Gaussian distributions;
- Given the filtering approximation, perform a one-step ahead prediction for $t + 1$;
- Compute the mean and the variance of the filtering distribution for $t + 1$, and approximate it with a mixture of Gaussian distributions having the same mean and variance.

Secondly, we obtain samples from the smoothing distribution using a backward sampling procedure.

Filtering

Assuming a diffuse prior on $x_k = (\theta_k, \dots, \theta_1)$ will lead to the following conditional distribution

$$p(x_k|y_{1:k}) = N(x_k; (y_k, \dots, y_1)^\top, \sigma^2 I_k) \quad (1.23)$$

In our filtering algorithm, where the goal is to approximate $p(x_t|y_{1:t})$, we shall start the iterations at $t = k$, and use (1.23) as the initialization.

Let's assume that the filtering distribution at time t is approximated by a mixture of Normal distributions, i.e.

$$p(x_t|y_{1:t}) = \sum_{i=1}^{n_t} w_t^{(i)} N(x_t; m_{t|t}^{(i)}, V_{t|t}^{(i)}). \quad (1.24)$$

At $t = k$, we let

$$n_k = 1, \quad m_{k|k}^{(1)} = (y_k, \dots, y_1)^\top, \quad V_{k|k}^{(1)} = \sigma^2 I_k. \quad (1.25)$$

We shall obtain a recursive scheme for updating the parameters n_{t+1} , $\{w_{t+1}^{(i)}, 1 \leq i \leq n_{t+1}\}$, $\{m_{t+1|t+1}^{(i)}, V_{t+1|t+1}^{(i)} \text{ for } 1 \leq i \leq n_{t+1}\}$ given their values at time t .

Notice that in the following mathematical derivations, the notation assumes $k > 1$. It is straightforward to adapt all equations to the case $k = 1$, as it corresponds to only considering the first entry in each vector and matrix introduced below. We will henceforth use the notation $z[-k]$ to refer to a vector z without the k^{th} entry, and $z[k]$ to refer to k^{th} entry of that vector. Similarly, we will use the notation $Z[, -k]$ to refer to the matrix Z without its k^{th} column, $Z[-k,]$ to refer to matrix Z without its k^{th} row, and $Z[-k, -k]$ to refer to matrix Z without its k^{th} row and column.

Let $G := \begin{pmatrix} \phi_{1:(k-1)} & \phi_k \\ I_{k-1} & \mathbf{0} \end{pmatrix}$, and $\phi := (\phi_1, \dots, \phi_k)^\top$. Given $p(x_t|y_{1:t})$, the one-step ahead prediction for $t + 1$ is

$$\begin{aligned} p(x_{t+1}|y_{1:t}) &= \int p(x_{t+1}|y_{1:t}, \eta_{t+1})p(\eta_{t+1}|y_{1:t})d\eta_{t+1} \\ &= \int p(x_{t+1}|y_{1:t}, \eta_{t+1})p(\eta_{t+1})d\eta_{t+1} \\ &= \sum_{i=1}^{n_t} w_t^{(i)} \int N(x_{t+1}; \tilde{m}_t^{(i)}, \tilde{V}_t^{(i)})C(\eta_{t+1}; 0, \tau)d\eta_{t+1} \end{aligned} \quad (1.26)$$

where

$$\tilde{m}_t^{(i)} = Gm_{t|t}^{(i)} + \begin{pmatrix} \eta_{t+1} \\ \mathbf{0} \end{pmatrix} = \begin{pmatrix} \phi^\top m_{t|t}^{(i)} + \eta_{t+1} \\ m_{t|t}^{(i)}[-k] \end{pmatrix}$$

and

$$\tilde{V}_t^{(i)} = GV_{t|t}^{(i)}G^\top = \begin{pmatrix} \phi^\top V_{t|t}^{(i)}\phi & \phi^\top V_{t|t}^{(i)}[-k] \\ V_{t|t}^{(i)}[-k,]\phi & V_{t|t}^{(i)}[-k, -k] \end{pmatrix}.$$

Notice that $\tilde{m}_t^{(i)}$ depends on η_{t+1} while the covariance matrix $\tilde{V}_t^{(i)}$ does not. The filtering distribution at $t + 1$ is then given by

$$\begin{aligned} p(x_{t+1}|y_{1:t+1}) &\propto p(y_{t+1}|x_{t+1})p(x_{t+1}|y_{1:t}) \\ &= \sum_{i=1}^{n_t} w_t^{(i)} \int N(y_{t+1}; x_{t+1}[1], \sigma^2)N(x_{t+1}; \tilde{m}_t^{(i)}, \tilde{V}_t^{(i)})C(\eta_{t+1}; 0, \tau)d\eta_{t+1}. \end{aligned}$$

Using Normal-Normal conjugacy, we can write

$$\begin{aligned} N(y_{t+1}; x_{t+1}[1], \sigma^2)N(x_{t+1}; \tilde{m}_t^{(i)}, \tilde{V}_t^{(i)}) &= \\ N(y_{t+1}; \phi^\top m_{t|t}^{(i)} + \eta_{t+1}, \phi^\top V_{t|t}^{(i)}\phi + \sigma^2)N(x_{t+1}; \check{m}_{t+1}^{(i)}, \check{V}_{t+1}^{(i)}) \end{aligned}$$

where $\check{m}_{t+1}^{(i)}$ and $\check{V}_{t+1}^{(i)}$ are given by:

$$\check{m}_{t+1}^{(i)} = \begin{pmatrix} \phi^\top m_{t|t}^{(i)} + \eta_{t+1} \\ m_{t|t}^{(i)}[-k] \end{pmatrix} + \frac{y_{t+1} - \phi^\top m_{t|t}^{(i)} - \eta_{t+1}}{\sigma^2 + \phi^\top V_{t|t}^{(i)}\phi} \begin{pmatrix} \phi^\top V_{t|t}^{(i)}\phi \\ V_{t|t}^{(i)}[-k,]\phi \end{pmatrix}$$

and

$$\check{V}_{t+1}^{(i)} = \begin{pmatrix} \phi^\top V_{t|t}^{(i)} \phi & \phi^\top V_{t|t}^{(i)}[, -k] \\ V_{t|t}^{(i)}[-k,]\phi & V_{t|t}^{(i)}[-k, -k] \end{pmatrix} - \frac{1}{\sigma^2 + \phi^\top V_{t|t}^{(i)} \phi} \begin{pmatrix} b\phi^\top V_{t|t}^{(i)} \phi \\ V_{t|t}^{(i)}[-k,]\phi \end{pmatrix} \begin{pmatrix} \phi^\top V_{t|t}^{(i)} \phi & \phi^\top V_{t|t}^{(i)}[, -k] \end{pmatrix}.$$

Notice that $\check{m}_{t+1}^{(i)}$ depends linearly on η_{t+1} while $\check{V}_{t+1}^{(i)}$ does not. The filtering distribution can therefore be expressed as

$$p(x_{t+1}|y_{1:t+1}) \propto \sum_{i=1}^{n_t} w_t^{(i)} \int N(x_{t+1}; \check{m}_{t+1}^{(i)}, \check{V}_{t+1}^{(i)}) C(\eta_{t+1}; 0, \tau) N(y_{t+1}; \phi^\top m_{t|t}^{(i)} + \eta_{t+1}, \phi^\top V_{t|t}^{(i)} \phi + \sigma^2) d\eta_{t+1}.$$

The constant of proportionality for the filtering distribution above is obtained by integrating the right-hand side with respect to x_{t+1} . This integration boils down to the convolution of a Normal and a Cauchy distribution, which corresponds to the Voigt profile (more details on this distribution are deferred to Appendix A.2). We will henceforth use the notation $V(x; \sigma, \tau)$ to refer to the Voigt profile obtained from convolving a zero-centered Normal distribution with scale σ , and a zero-centered Cauchy with scale τ .

The filtering distribution can thus be expressed as

$$p(x_{t+1}|y_{1:t+1}) = \frac{\sum_{i=1}^{n_t} w_t^{(i)} \int N(x_{t+1}; \check{m}_{t+1}^{(i)}, \check{V}_{t+1}^{(i)}) C(\eta_{t+1}; 0, \tau) N(y_{t+1}; \phi^\top m_{t|t}^{(i)} + \eta_{t+1}, \phi^\top V_{t|t}^{(i)} \phi + \sigma^2) d\eta_{t+1}}{\sum_{i=1}^{n_t} w_t^{(i)} V\left(y_{t+1} - \phi^\top m_{t|t}^{(i)}; \sqrt{\phi^\top V_{t|t}^{(i)} \phi + \sigma^2}, \tau\right)}.$$

The density

$$g(\eta; y, \tau, s) := \frac{C(\eta; 0, \tau) N(y; \eta, s^2)}{V(y; s, \tau)} \quad (1.27)$$

can be used to re-write the filtering distribution as

$$p(x_{t+1}|y_{1:t+1}) = \sum_{i=1}^{n_t} \left[\frac{w_t^{(i)} V\left(y_{t+1} - \phi^\top m_{t|t}^{(i)}; s^{(i)}, \tau\right)}{\sum_{i=1}^{n_t} w_t^{(i)} V\left(y_{t+1} - \phi^\top m_{t|t}^{(i)}; s^{(i)}, \tau\right)} \right] \int N\left(x_{t+1}; \check{m}_{t+1}^{(i)}, \check{V}_{t+1}^{(i)}\right) g\left(\eta_{t+1}; y_{t+1} - \phi^\top m_{t|t}^{(i)}, s^{(i)}, \tau\right) d\eta_{t+1}. \quad (1.28)$$

An important aspect of the density g is that its moments can be written in terms of the Voigt functions for which efficient algorithms exist. In Appendix A.2, we show how to derive

the mean $\mu_g(y, s, \tau)$ and the variance $\sigma_g^2(y, s, \tau)$ of a random variable distribute according to $g(\eta; y, s, \tau)$. Using this, the above recursion equation can be further approximated to yield practical filtering algorithms. In this Section, we will only focus on the Normal approximation algorithm, which approximates g with a Normal distribution whose mean and variance are identical to those of g . More sophisticated algorithms can be derived, e.g. based on approximating g with a Normal mixture. In practice, we observed that approximating g with a two-component Normal mixture leads to a substantial increase in computational complexity in comparison to the one-component approximation, and that this drawback is not accompanied by a significant increase in precision. A description of the two-component mixture approximation is reported in Appendix A.3.

Replacing $g(\eta; y, s, \tau)$ by $N(\eta; \mu_g(y, s, \tau), \sigma_g^2(y, s, \tau))$ in (1.28), we obtain

$$p(x_{t+1}|y_{1:t+1}) \approx \sum_{i=1}^{n_t} \left[\frac{w_t^{(i)} V(y_{t+1} - \phi^\top m_{t|t}^{(i)}; s^{(i)}, \tau)}{\sum_{i=1}^{n_t} w_t^{(i)} V(y_{t+1} - \phi^\top m_{t|t}^{(i)}; s^{(i)}, \tau)} \right] N(x_{t+1}; m_{t+1|t+1}^{(i)}, V_{t+1|t+1}^{(i)}) \quad (1.29)$$

where

$$m_{t+1|t+1}^{(i)} = \check{m}_{t+1}^{(i)}(\mu_g(y_{t+1} - \phi^\top m_{t|t}^{(i)}, s^{(i)}, \tau))$$

and

$$V_{t+1|t+1}^{(i)} = \check{V}_{t+1}^{(i)} + \frac{\sigma_g^2(y_{t+1} - \phi^\top m_{t|t}^{(i)}, s^{(i)}, \tau)}{(\sigma^2 + \phi^\top V_{t|t}^{(i)} \phi)^2} \begin{pmatrix} \sigma^4 & -\sigma^2 \phi^\top V_{t|t}^{(i)} [-k] \\ -\sigma^2 V_{t|t}^{(i)} [-k,] \phi & V_{t|t}^{(i)} [-k,] \phi \phi^\top V_{t|t}^{(i)} [-k] \end{pmatrix}.$$

The approximation (1.29) shows that if we start with an n_t -component Normal mixture approximation at time t , we again get an n_t -component Normal mixture approximation at time $t + 1$. Because we initiate the recursions by a Normal distribution at time $t = k$ (i.e., a Normal mixture with just one component), the approximation (1.29) would just produce Normal distributions at all subsequent times. This algorithm is summarized as Algorithm 2.

Connection with Assumed Density Filtering

The (one-component) Normal approximation we proposed is a specific instance of Assumed Density Filtering (ADF), a general approach to approximate inference in the context of state space models (see e.g. [Murphy \(2023\)](#), Section 8.6). In ADF, one chooses a family $\mathcal{Q}$ of tractable distributions (e.g. Gaussian), and assumes that, for any t , the filtering distribution $p(x_t|y_{1:t})$ can be well approximated by a distribution $q_t(x_t)$ where $q_t \in \mathcal{Q}$. The approximate filtering distribution at $t + 1$ is then obtained as

$$p_{t+1}(x_{t+1}) \propto p(y_{t+1}|x_{t+1}) \int p(x_{t+1}|x_t) q_t(x_t) dx_t$$

Algorithm 2: Normal Approximation of Filtering Distribution

Description: This algorithm approximates the filtering distribution $p(x_t|y_{1:t})$ by a normal distribution $N(m_{t|t}, V_{t|t})$ for each $t = k, k+1, \dots, n$.

Input: data $y_1, \dots, y_n$, $\tau > 0$ and $\sigma > 0$.

Define: $\mu_g(y, \tau, s)$ and $\sigma_g^2(y, \tau, s)$ to be the mean and variance of a random variable with density (1.27).

Initialization: Initialize at $t = k$ with $m_{k|k} = (y_k, \dots, y_1)^\top$ and $V_{k|k} = \sigma^2 I_k$.

Recursion (if $k > 1$): Repeat the following for $t = k, k+1, \dots, n$

1. Define the following quantities:

$$s = \sqrt{\sigma^2 + \phi^\top V_{t|t} \phi},$$

$$\check{m}_{t+1}(\eta) = \begin{pmatrix} \phi^\top m_{t|t} + \eta \\ m_{t|t}[-k] \end{pmatrix} + \frac{y_{t+1} - \phi^\top m_{t|t} - \eta}{s^2} \begin{pmatrix} \phi^\top V_{t|t} \phi \\ V_{t|t}[-k,]\phi \end{pmatrix},$$

$$\check{V}_{t+1} = \begin{pmatrix} \phi^\top V_{t|t} \phi & \phi^\top V_{t|t}[, -k] \\ V_{t|t}[-k,]\phi & V_{t|t}[-k, -k] \end{pmatrix} - \frac{1}{s^2} \begin{pmatrix} \phi^\top V_{t|t} \phi \\ V_{t|t}[-k,]\phi \end{pmatrix} \begin{pmatrix} \phi^\top V_{t|t} \phi & \phi^\top V_{t|t}[, -k] \end{pmatrix}$$

2. Update:

$$m_{t+1|t+1} = \check{m}_{t+1}(\mu_g(y_{t+1} - \phi^\top m_{t|t}, s, \tau))$$

$$V_{t+1|t+1} = \check{V}_{t+1} + \frac{\sigma_g^2(y_{t+1} - \phi^\top m_{t|t}, s, \tau)}{s^4} \begin{pmatrix} \sigma^4 & -\sigma^2 \phi^\top V_{t|t}[, -k] \\ -\sigma^2 V_{t|t}[-k,]\phi & V_{t|t}[-k,]\phi \phi^\top V_{t|t}[, -k] \end{pmatrix}$$

Recursion (if $k = 1$): Repeat the following for $t = k, k+1, \dots, n$

Update:

$$m_{t+1|t+1} = \frac{\sigma^2}{\sigma^2 + V_{t|t}} \left(m_{t|t} + \mu_g(y_{t+1} - m_{t|t}, \sqrt{\sigma^2 + V_{t|t}}, \tau) \right) + \frac{V_{t|t}}{\sigma^2 + V_{t|t}} y_{t+1}$$

and

$$V_{t+1|t+1} = \frac{\sigma^2}{\sigma^2 + V_{t|t}} V_{t|t} + \left(\frac{\sigma^2}{\sigma^2 + V_{t|t}} \right)^2 \sigma_g^2(y_{t+1} - m_{t|t}, \sqrt{\sigma^2 + V_{t|t}}, \tau)$$

where $\int p(x_{t+1}|x_t)q_t(x_t)dx_t$ is the exact one-step-ahead predictive distribution under the assumption that $q_t(x_t)$ is the filtering distribution at t . Since $p_{t+1}(x_{t+1})$ is not guaranteed to belong to $\mathcal{Q}$, ADF seeks the best tractable approximation by solving

$$q_{t+1}(x_{t+1}) = \arg \min_{q \in \mathcal{Q}} \mathbb{KL}(p_{t+1}(x_{t+1}) || q(x_{t+1})).$$

If the distributions in $\mathcal{Q}$ belong to the exponential family, solving the above optimization problem is equivalent to finding the distribution $q_{t+1} \in \mathcal{Q}$ having the same moments as p_{t+1} (for a proof of this fact, see [Murphy \(2023\)](#), Chapter 5.1.4). This equivalence implies that our Approximate Filter is an ADF method, since at the filtering step we perform moment matching using a Normal distribution, which belongs to the exponential family.

Parameter Estimation

In Algorithm 2, the parameters τ and σ^2 are assumed to be given. In practice, we provide point estimates for these parameters by finding the combination of τ and σ^2 that maximizes the approximate marginal likelihood. Ideally, we would like to solve the following optimization problem

$$\arg \max_{\tau, \sigma^2} p(y_{1:n} | \tau^2, \sigma^2) = \arg \max_{\tau, \sigma^2} \left[p(y_{1:k} | \tau, \sigma^2) \prod_{t=k}^{n-1} p(y_{t+1} | y_{1:t}, \tau, \sigma^2) \right]. \quad (1.30)$$

Since we assumed a diffuse Normal prior on $\theta_1, \dots, \theta_k$, the factor $p(y_{1:k} | \tau, \sigma^2)$ is approximately constant in τ and σ^2 , so we can drop it from the optimization problem. We get a tractable approximation of $p(y_{t+1} | y_{1:t}, \tau, \sigma^2)$ by observing that

$$\begin{aligned} y_{t+1} &= \theta_{t+1} + \epsilon_{t+1} \\ &= \phi^\top x_t + \eta_{t+1} + \epsilon_{t+1}. \end{aligned}$$

Because $p(x_t | y_{1:t}) \approx N(m_{t|t}, V_{t|t})$ and $(\eta_{t+1}, \epsilon_{t+1})$ are independent of $y_1, \dots, y_t$, we deduce

$$p(y_{t+1} | y_{1:t}, \tau, \sigma^2) \approx V(y_{t+1} - \phi^\top m_{t|t}; \sqrt{\sigma^2 + \phi^\top V_{t|t} \phi}, \tau).$$

So we can approximate (1.30) as

$$\arg \max_{\tau, \sigma^2} \left[\prod_{t=k}^{n-1} V(y_{t+1} - \phi^\top m_{t|t}; \sqrt{\sigma^2 + \phi^\top V_{t|t} \phi}, \tau) \right]. \quad (1.31)$$

Smoothing

We now study the problem of approximating the smoothing distribution $p(x_t | y_{1:n})$. For $t = n$, this coincides with the filtering distribution, which can be obtained applying Algorithm 2. Following standard practice, we employ a simulation-based backward sampling procedure

to approximate smoothing distributions. For a textbook treatment of this topic, see [Särkkä and Svensson \(2023\)](#), Section 11.2, and [Chopin et al. \(2020\)](#), Section 12.3.2.

The recursion scheme is based on the following observation

$$\begin{aligned} p(x_t \mid y_{1:n}) &\propto \int p(x_t \mid x_{t+1}, y_{1:n}) p(x_{t+1} \mid y_{1:n}) dx_{t+1} \\ &\propto \int p(x_t \mid x_{t+1}, y_{1:t}) p(x_{t+1} \mid y_{1:n}) dx_{t+1}. \end{aligned}$$

Assuming we have access to M samples from $p(x_{t+1} \mid y_{1:n})$, denoted as $x_{t+1|n}^{(1)}, \dots, x_{t+1|n}^{(M)}$, we can approximate the smoothing distribution above as

$$p(x_t \mid y_{1:n}) \propto \sum_{i=1}^M p(x_t \mid x_{t+1|n}^{(i)}, y_{1:t}). \quad (1.32)$$

Recall that $x_t = (\theta_t, \dots, \theta_{t-k+1})$, which implies that $x_{t+1} = (\theta_{t+1}, \dots, \theta_{t-k+2})$, and $x_{t+1}[-1] = (\theta_t, \dots, \theta_{t-k+2}) = x_t[-k]$. Therefore, (1.32) can be factored as follows

$$p(x_t \mid y_{1:n}) \propto \sum_{i=1}^M p(\theta_{t-k+1} \mid x_{t+1|n}^{(i)}, y_{1:t}) \delta_{\{x_t[-k] - x_{t+1|n}^{(i)}[-1]\}},$$

where δ denotes the Dirac delta distribution. Therefore, the key to obtain a recursive scheme to sample from $p(x_t \mid y_{1:n})$ is to determine the conditional distribution $p(\theta_{t-k+1} \mid x_{t+1}, y_{1:t})$ for $t = n-1, n-2, \dots, k$, which is done in the following result.

Lemma 1.1. Fix $t \geq k$. Assume that the filtering distribution $p(x_t \mid y_{1:t})$ is given by $N(m_{t|t}, V_{t|t})$. First assume that $k \geq 2$, and let

$$m_{t|t} = \begin{pmatrix} m_{t|t}[-k] \\ m_{t|t}[k] \end{pmatrix} \quad \text{and} \quad V_{t|t} = \begin{pmatrix} V_{t|t}[-k, -k] & V_{t|t}[-k, k] \\ V_{t|t}[k, -k] & V_{t|t}[k, k] \end{pmatrix}.$$

Also, let

$$f_{t|t}(u) := m_{t|t}[k] + V_{t|t}[k, -k] (V_{t|t}[-k, -k])^{-1} (u - m_{t|t}[-k]) \quad \text{for } (k-1) \times 1 \text{ vectors } u,$$

$$r_{t|t} := \sqrt{V_{t|t}[k, k] - V_{t|t}[k, -k] (V_{t|t}[-k, -k])^{-1} V_{t|t}[-k, k]},$$

and

$$b(x_{t+1}) = b(\theta_{t+1}, \dots, \theta_{t-k+2}) := \frac{\theta_{t+1}}{\phi_k} - \frac{\phi_{k-1}\theta_{t-k+2} + \dots + \phi_1\theta_t}{\phi_k}.$$

Then

$$p(\theta_{t-k+1} \mid x_{t+1}, y_{1:t}) \propto C\left(\theta_{t-k+1}; b(x_{t+1}), \frac{\tau}{|\phi_k|}\right) N\left(\theta_{t-k+1}; f_{t|t}(x_{t+1}[-1]), r_{t|t}^2\right). \quad (1.33)$$

For $k = 1$, the formula is given by

$$p(\theta_t \mid \theta_{t+1}, y_{1:t}) \propto C\left(\theta_t; \frac{\theta_{t+1}}{\phi_1}, \frac{\tau}{|\phi_1|}\right) N\left(\theta_t; m_{t|t}, V_{t|t}\right). \quad (1.34)$$

In Appendix A.4, this result is stated and proved for the more general case in which the filtering distribution $p(x_t \mid y_{1:t})$ is given by the mixture $\sum_{i=1}^{n_t} w_t^{(i)} N(m_{t|t}^{(i)}, V_{t|t}^{(i)})$. We leverage Lemma 1.1 to develop a backward sampling scheme that allows us to obtain samples from $p(x_t \mid y_{1:n})$ for every $t = n, n-1, \dots, k$. The idea is that, given a sample $x_{t+1|n}^{(1)}, \dots, x_{t+1|n}^{(M)}$ from $p(x_{t+1} \mid y_{1:n})$, we use importance sampling to get samples from $p(\theta_{t-k+1} \mid x_{t+1|n}^{(i)}, y_{1:t})$ for every $i = 1, \dots, M$. Indeed, since we cannot directly sample from (1.33) or (1.34), we instead

1. Generate $\tilde{\theta}_{t-k+1|n}^{(1)}, \dots, \tilde{\theta}_{t-k+1|n}^{(J)}$ from a proposal density

$$q(\theta_{t-k+1} \mid x_{t+1|n}^{(i)}, y_{1:t}).$$

2. For $j = 1, \dots, J$, compute weights w_j equal to

$$\frac{p(\tilde{\theta}_{t-k+1}^{(j)} \mid x_{t+1|n}^{(i)}, y_{1:t})}{q(\tilde{\theta}_{t-k+1}^{(j)} \mid x_{t+1|n}^{(i)}, y_{1:t})}.$$

3. Normalize the J weights to sum to 1.

4. Sample $\theta_{t-k+1|n}^{(i)}$ from a discrete distribution on $\tilde{\theta}_{t-k+1|n}^{(1)}, \dots, \tilde{\theta}_{t-k+1|n}^{(J)}$ with weights obtained in the previous step, and let $x_{t|n}^{(i)} = (x_{t+1|n}^{(i)}[-1] \quad \theta_{t-k+1|n}^{(i)})^\top$.

Given the results from Lemma 1.1, it is computationally convenient to let

$$q(\theta_{t-k+1} \mid x_{t+1|n}^{(i)}, y_{1:t}) = \begin{cases} N\left(\theta_{t-k+1}; f_{t|t}(x_{t+1|n}^{(i)}[-1]), r_{t|t}^2\right) & \text{for } k > 1 \\ N\left(\theta_{t-k+1}; m_{t|t}, V_{t|t}\right) & \text{for } k = 1, \end{cases}$$

which is easy to sample from, and leads to the following simple weights

$$w_j = \begin{cases} C\left(\tilde{\theta}_{t-k+1|n}^{(i)}; b(x_{t+1|n}^{(i)}), \frac{\tau}{|\phi_k|}\right) & \text{for } k > 1 \\ C\left(\tilde{\theta}_{t-k+1|n}^{(i)}; \frac{x_{t+1|n}^{(i)}}{\phi_1}, \frac{\tau}{|\phi_1|}\right) & \text{for } k = 1. \end{cases}$$

The resulting algorithm is summarized as Algorithm 3.

Algorithm 3: Backward Sampling

Description: This algorithm provides M samples from $p(x_t|y_{1:n})$ for

$t = n, n-1, \dots, k$.

Input: $\tau > 0$, filtering means and variances $\{m_{t|t}, V_{t|t}\}$ for $t = 1, \dots, n$, and large positive constants M and J .

Recursion (if $k > 1$): Iterate the following for $i = 1, \dots, M$:

1. Draw a sample $x_{n|n}^{(i)}$ from $N(m_{n|n}, V_{n|n})$
2. For $t = n-1, n-2, \dots, k$:
 - a) sample $\tilde{\theta}_{t-k+1|n}^{(j)} \stackrel{\text{i.i.d.}}{\sim} N\left(f_{t|t}\left(x_{t+1|n}^{(i)}[-1]\right), r_{t|t}^2\right)$ for $j = 1, \dots, J$
 - b) compute weights $w_j = C\left(\tilde{\theta}_{t-k+1|n}^{(j)}; b\left(x_{t+1|n}^{(i)}\right), \frac{\tau}{|\phi_k|}\right)$ for $j = 1, \dots, J$
 - c) normalize the weights: $\tilde{w}_j = \frac{w_j}{\sum_{l=1}^J w_l}$ for $j = 1, \dots, J$
 - d) draw a sample $\theta_{t-k+1|n}^{(i)}$ from the discrete distribution on $\tilde{\theta}_{t-k+1|n}^{(1)}, \dots, \tilde{\theta}_{t-k+1|n}^{(J)}$ with probabilities $\tilde{w}_1, \dots, \tilde{w}_J$
 - e) Let $x_{t|n}^{(i)} = (x_{t+1|n}^{(i)}[-1] \quad \theta_{t-k+1|n}^{(i)})^\top$
3. Return $x_{k|n}^{(i)}, x_{k+1|n}^{(i)}, \dots, x_{n|n}^{(i)}$

Recursion (if $k = 1$): same as recursion for $k > 1$ with the following modifications in step 2:

- let $m_{t|t}$ and $V_{t|t}$ be the mean and variance of the Normal in a)
 - change the center of the Cauchy to $\frac{x_{t+1|n}^{(i)}}{\phi_k}$ in b)
 - let $x_{t|n}^{(i)} = \theta_{t-k+1|n}^{(i)}$ in e)
-

Other Methods

Other methods could be used to fit model (1.18). One could, for example, consider MCMC methods that do not directly take advantage of the state-space structure of the problem. This has been done, in the context of a model similar to (1.18), by [Faulkner and Minin \(2018\)](#). The authors provide posterior inference by Hamiltonian Monte Carlo (HMC, see [Betancourt \(2017\)](#) for a review on the topic). As [Faulkner and Minin \(2018\)](#) notice, HMC has the benefit of being a particularly practical choice, given its ability to handle a wide variety of models and its automatic implementation through available software, such as `stan` ([Carpenter et al. \(2017\)](#)). The main drawback of HMC is its computational intensity, which is significantly higher compared to the other estimation methods discussed. This is because HMC requires calculating the gradient of the log-posterior density multiple times within each iteration to generate a proposed value. While this process can be computationally expensive, it generally achieves good mixing efficiency on a per-iteration basis. We will discuss more about this issue when comparing estimation methods on simulated and real datasets in Sections 1.4 and 1.5.

1.4 Simulations

In this Section, we will introduce different simulated datasets, for which we will compare the performance of the latent Cauchy k^{th} -order differences model with that of other popular approaches.

We will focus on testing the Cauchy method for $k = 1, 2$, and 3 . The Cauchy prior assumption on the k^{th} -order differences of θ is particularly appropriate when we expect these differences to be around zero most of the times, while allowing for sporadic large deviations from it (in absolute value). Therefore, $k = 1$ is particularly suitable for data generated from piecewise constant functions, while $k = 2$ and $k = 3$ will work well, respectively, for data generated from piecewise linear and piecewise quadratic functions. With these considerations in mind, we choose six different data generating functions, two functions for each order k . We will test the Cauchy method with $k = 1$ on the *Blocks* function, a piecewise constant function appearing in [Donoho and Johnstone \(1994\)](#), and on the *Bump* function, a function that is defined to be always 0 except for one point, which equals 1. The Cauchy method with $k = 2$ is tested on data generated by piecewise linear functions: one without jumps, which we will refer to as *Sawtooth* function, and the other with jumps, which we will term *Jumpy Sawtooth*. Finally, for $k = 3$, we will consider a *Hill* function, which previously appeared in [Sadhanala and Tibshirani \(2019\)](#), and a *Jumpy Hill* function, which introduces a discontinuity (jump) in the Hill function. Noisy data will be generated by, first, evaluating the functions at n regularly spaced design points, and then adding i.i.d. Gaussian noise.

The six functions considered are reported in Figure 1.1.

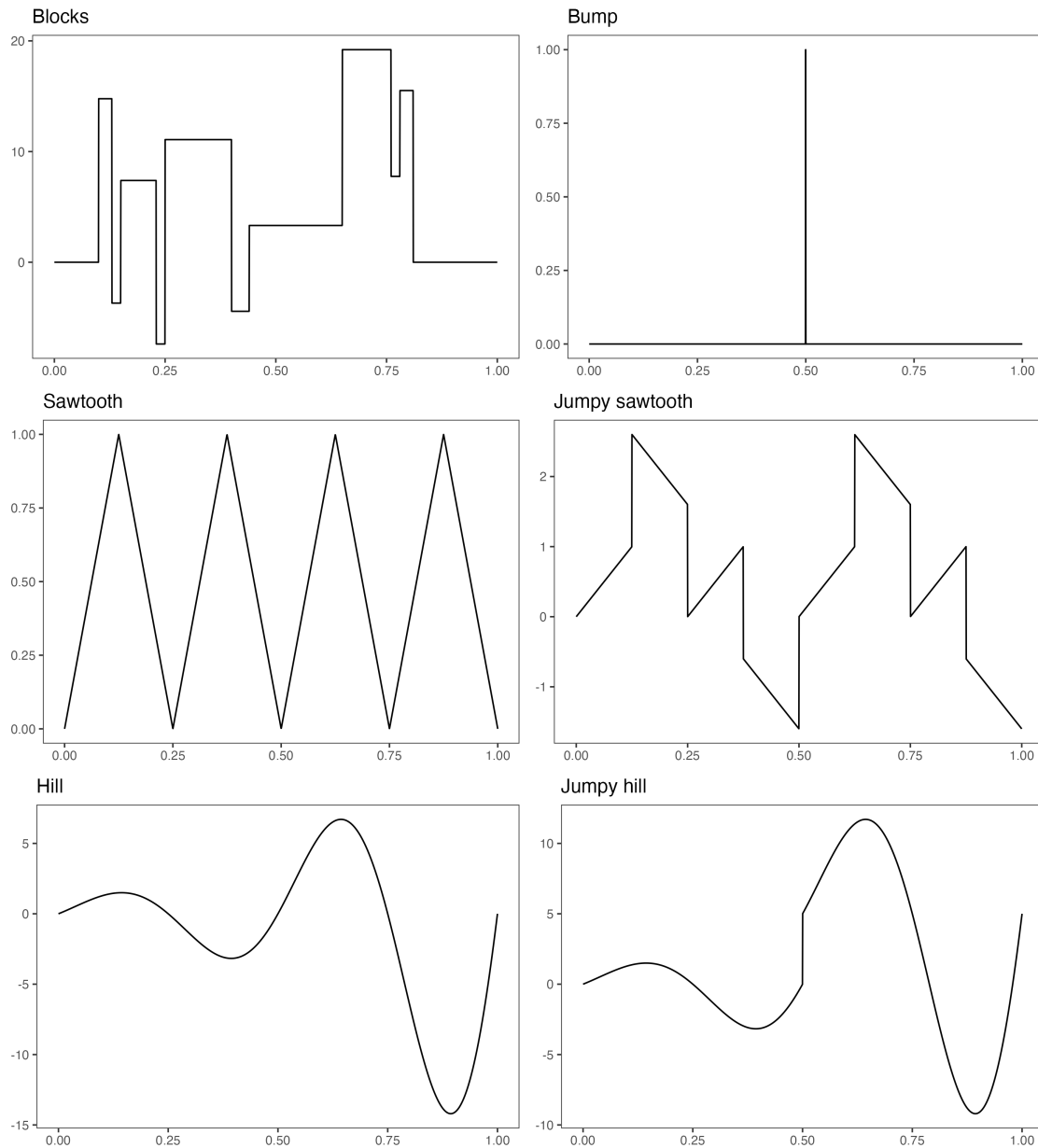


Figure 1.1: Functions used for the simulations

Methods Compared

We compare the latent Cauchy k^{th} -order differences method, implemented via Gibbs sampling and by approximate smoothing, to a frequentist approach, trend filtering, and to two other Bayesian methods, a Gaussian hidden Markov model fitted by Kalman smoothing, and shrinkage prior Markov random fields (SPMRFs). These different approaches are described

in detail next. For all the Bayesian methods, we estimate θ using posterior means.

For trend filtering, we consider two distinct estimation approaches, which rely on different degrees of information on the data generating function. In one case, we use the library **CVXR** (Fu et al. (2017)) to solve the following optimization problem

$$\min_{\theta} \|y - \theta\|_2^2 \quad \text{subject to} \quad \|D^{(k)}\theta\|_1 \leq V^*, \quad (1.35)$$

where V^* is assumed to be known, and set equal to the ℓ_1 norm of the k^{th} -order differences of the true function evaluated at the design points. We also consider the less optimistic scenario in which V^* is not known, and is selected by cross-validation. For this purpose, we take advantage of functions provided by the R library **genlasso** (Arnold et al. (2022)).

In Section 1.2, we showed that the maximum entropy prior on the entries of $D^{(k)}\theta$, subject to a Bayesian version of the ℓ_2 constraint, is the Normal distribution. The ℓ_2 constraint is stronger than the Cauchy constraint (1.16), and therefore leads to a prior that is suitable for a smaller class of functions, which are smooth in a strong sense. Henceforth, the model

$$\begin{aligned} y_i &= \theta_i + \epsilon_i, & \epsilon_i &\stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2) \\ (D^{(k)}\theta)_j &= \eta_j, & \eta_j &\stackrel{\text{i.i.d.}}{\sim} N(0, \tau^2) \end{aligned} \quad (1.36)$$

is less widely applicable than the Cauchy one, but, at the same time, it has the benefit of having a closed form posterior that can be efficiently derived using Kalman filtering and smoothing algorithms. We will fit this method to our simulated data, and compare it to the Cauchy approach in terms of efficiency and estimation accuracy.

The last method we consider was introduced by Faulkner and Minin (2018). This paper proposes a set of models similar to ours:

$$\begin{aligned} y_i &= \theta_i + \epsilon_i, & \epsilon_i &\stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2) \\ (D^{(k)}\theta)_j &= \eta_j, & \eta_j &\stackrel{\text{i.i.d.}}{\sim} p_{\text{shrink}} \end{aligned} \quad (1.37)$$

with p_{shrink} being a Laplace or a Horseshoe distribution. The authors refer to these models as shrinkage prior Markov random fields (SPMRFs), and provide an R library, **spmrf**, to fit them. The fitting strategy proposed by Faulkner and Minin (2018) is substantially different from ours in that it does not take advantage of the Markovian structure of the problem. Indeed, **spmrf** performs posterior inference by leveraging the general-purpose Hamiltonian Monte Carlo methods offered by **stan** (Carpenter et al. (2017)). Notice that, as mentioned in Section 1.1, model (1.37) with a Laplace prior is the Bayesian counterpart of trend filtering, and both methods are expected to perform well on data generated by functions that are smooth in the ℓ_1 sense (but not weaker). On the other hand, the Horseshoe prior is characterized by a pole at the origin and by Cauchy-like tails. These characteristics suggest that its use in the context of (1.37) would lead to posterior estimates that are mostly similar to the ones obtained by the Cauchy model, but with a stronger shrinkage towards zero.

Convergence Study

To investigate convergence rates, we consider 10 different values for the number n of observed data points. We choose values for n that range from 200 to 2000, and that are equispaced on the log-scale. For each value of n and each test function, we simulate 20 datasets by first evaluating the function at n equispaced design points on $[0, 1]$, and then adding Gaussian noise. The variance of the Gaussian noise is set in such a way that the signal-to-noise ratio is always equal to 7. Cauchy estimation (via Gibbs sampling and via the approximate method), Kalman smoothing, and trend filtering (with known V^* and with V^* selected by cross-validation) are fit to each dataset. Notice that we do not include SPMRFs in this convergence study given the long computational times needed to fit these models on relatively large datasets (see Section 1.5 for some examples).

All the methods considered involve a differencing order k , which is decided based on the data generating function, as we described at the beginning of this Section. For each fit, we compute the corresponding mean squared error (MSE). Finally, for each combination of function and estimation method, we average the MSEs for fixed n , and we plot the $\log$ of the obtained average MSEs against $\log(n)$. From the resulting plots, reported in Figure 1.2, we can observe that

- Cauchy estimation via Gibbs sampling has lowest mean MSE for most values of $\log(n)$, with the exception of the *Sawtooth* and *Hill* case, where it is slightly outperformed by trend filtering with known V^* and by Kalman smoothing, respectively.
- The approximate method for Cauchy estimation leads to slightly higher average MSE than Gibbs sampling for the functions estimated with $k = 1$, and for the *Hill* function. In these cases, the convergence rates are also similar. On the other hand, in the *Sawtooth* and *Jumpy Hill* cases, the convergence is considerably worse and similar to the one observed for Kalman smoothing.
- Kalman smoothing seem to perform well only for the *Hill* function.
- Trend filtering, with known or estimated V^* performs comparably to Cauchy estimation via Gibbs sampling only in the *Sawtooth* and *Hill* cases.

Notice that the *Hill* function is “smoother” than the other ones, in the sense that it does not have jumps or sharp edges, and is therefore the only case in which the Kalman smoother performs well. Moreover, all the functions on the right-hand side of Figure 1.1 are not smooth in an ℓ_2 nor in an ℓ_1 sense, which explains why both trend filtering and Kalman smoothing show convergence rates that are very poor in comparison to Cauchy estimation (via Gibbs sampling). The slopes, estimated via linear regression, of each line appearing in Figure 1.2 are reported in Table 1.3.

Figures 1.3 and 1.4 present the point estimates obtained by applying the different methods to noisy Blocks and Sawtooth data with $n = 1000$.

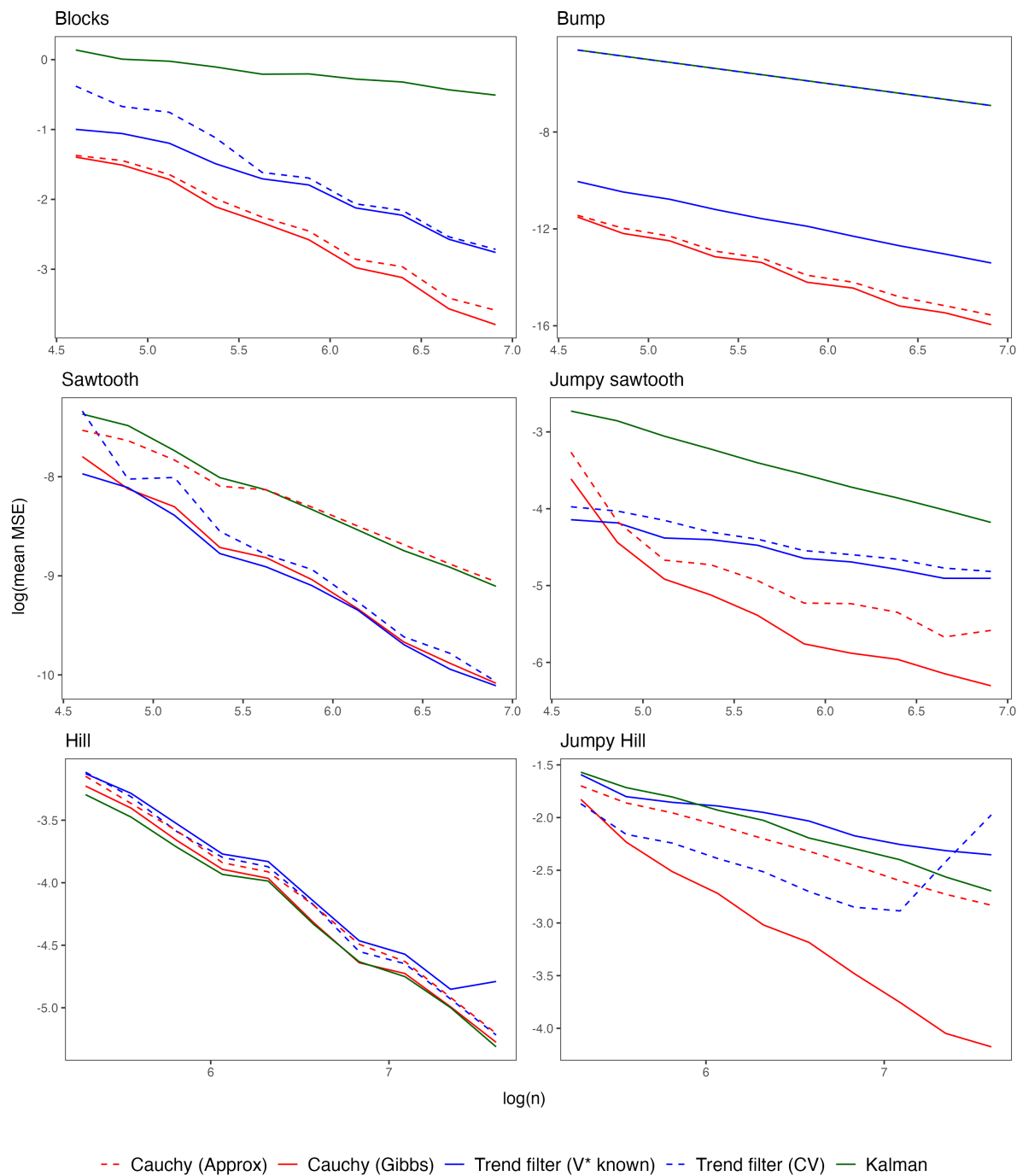


Figure 1.2: Convergence rates for different k^{th} -order methods. First row: $k = 1$, second row: $k = 2$, third row: $k = 3$.

Table 1.3: Slopes of $\log(\text{mean MSE})$ vs $\log(n)$ for different k^{th} -order methods

k	Function	Cauchy (Gibbs)	Cauchy (Approx)	Trend filter (known V^*)	Trend filter (CV)	Kalman
1	Blocks	-1.11	-1.01	-0.84	-0.96	-0.28
	Bump	-2.00	-1.88	-1.48	-1.00	-1.00
2	Sawtooth	-0.97	-0.72	-0.96	-1.01	-0.76
	Jumpy Sawtooth	-0.58	-0.44	-0.29	-0.21	-0.58
3	Hill	-0.89	-0.87	-0.85	-0.90	-0.86
	Jumpy Hill	-1.00	-0.49	-0.32	-0.18	-0.48

For the Blocks data (1.3), the estimates obtained by the two Cauchy algorithms are nearly identical and closely capture the piecewise constant structure of the true function. The trend filtering approaches produce similar results, but tend to detect more segments than actually exist in the underlying function. Kalman smoothing, on the other hand, overfits the data, leading to excessively wiggly estimates.

In the Sawtooth case (1.4), the two Cauchy algorithms yield less overlapping fits: the approximate method over-smooths the sharp edges, resembling the behavior of the Kalman smoother, whereas the Gibbs (and trend filtering) estimates successfully recover the sharp transitions. The over-smoothing obtained by the approximate method highlight its limitations in comparison to the Gibbs approach.

Computation Times

For each experiment carried out to study convergence rates, we recorded computation times. Figure 1.5 shows the time each method needed to produce estimates for the *Hills* data, averaged over each dataset of size n . Kalman smoothing and trend filtering with known V^* are the fastest approaches, and their computation times scale well with n . Gibbs sampling for Cauchy estimation and trend filtering with cross-validation scale similarly, and, for large n , the first is always faster than the latter. The approximate method has computation times that increase too fast with n : despite being comparable to the other methods for small n , it is the slowest method for $n = 2000$. Notice that the Gibbs sampler was run with a single chain and 5000 iterations, so its speed can be further improved using multiple shorter chains run in parallel. From our studies of convergence and computation time, we conclude that:

- Kalman smoothing is very fast and efficient, but is only suitable for data that are reasonably modeled by smooth functions without jumps and sharp edges.
- Trend filtering with known V^* is also computationally efficient, and is appropriate for data that are reasonably modeled by ℓ_1 -smooth functions. Its main drawback is that it requires to specify the ℓ_1 norm of the k^{th} -order differences of θ as an input. This

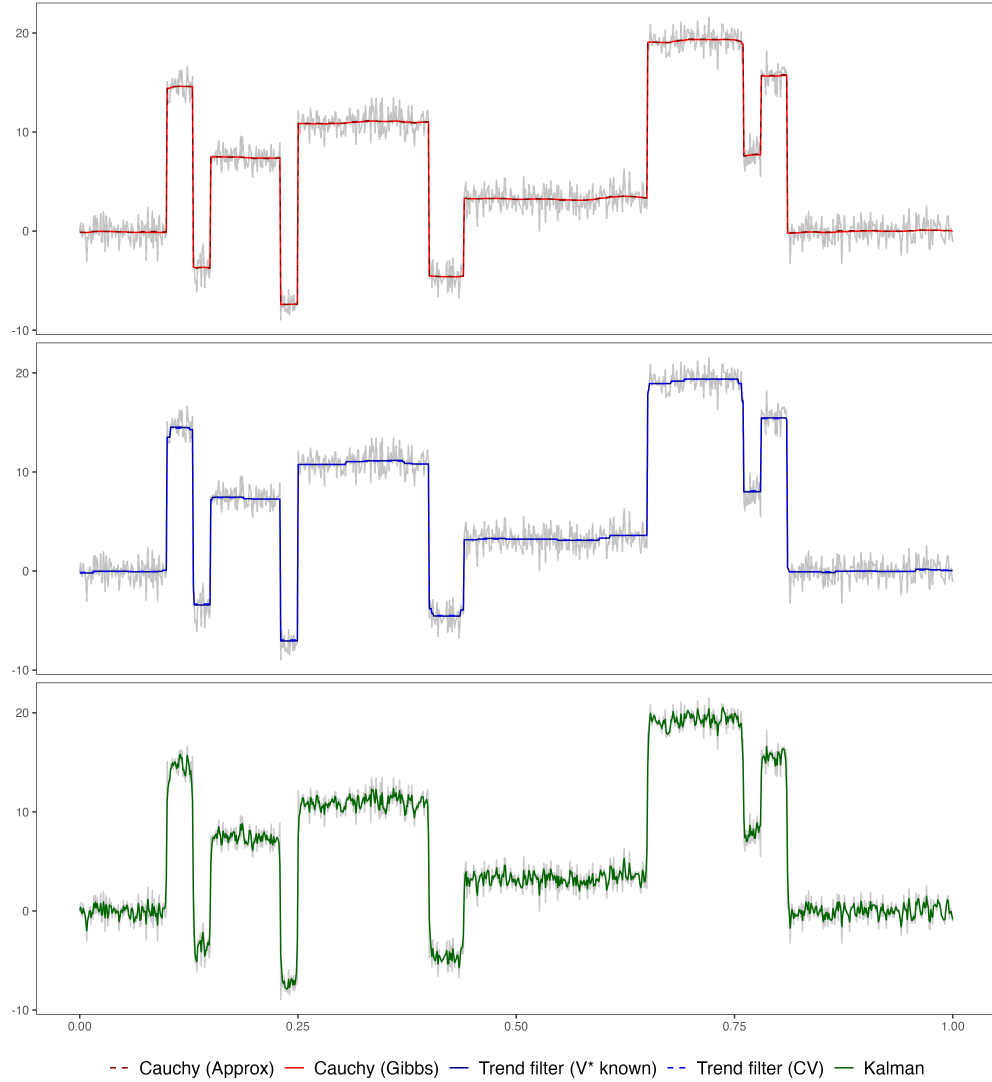


Figure 1.3: Point estimates for noisy Blocks data. From top to bottom: Cauchy, trend filtering, and Kalman estimates ($k = 1$).

quantity is hardly known in practice, which is why trend filtering with cross-validation (CV) is commonly used instead. The latter approach is similarly suitable for ℓ_1 -smooth functions, but has larger computation times, comparable to Gibbs sampling for the Cauchy method. It is worth noting that the computation times for trend filtering with CV can be improved. The library we use, **genlasso**, employs a full path algorithm to compute trend filtering estimates, i.e. it solves the optimization problem as V^* varies on $(0, \infty)$. This approach is particularly expensive, especially if combined with CV. A more efficient alternative is to compute solutions over a discrete grid of V^* values, as

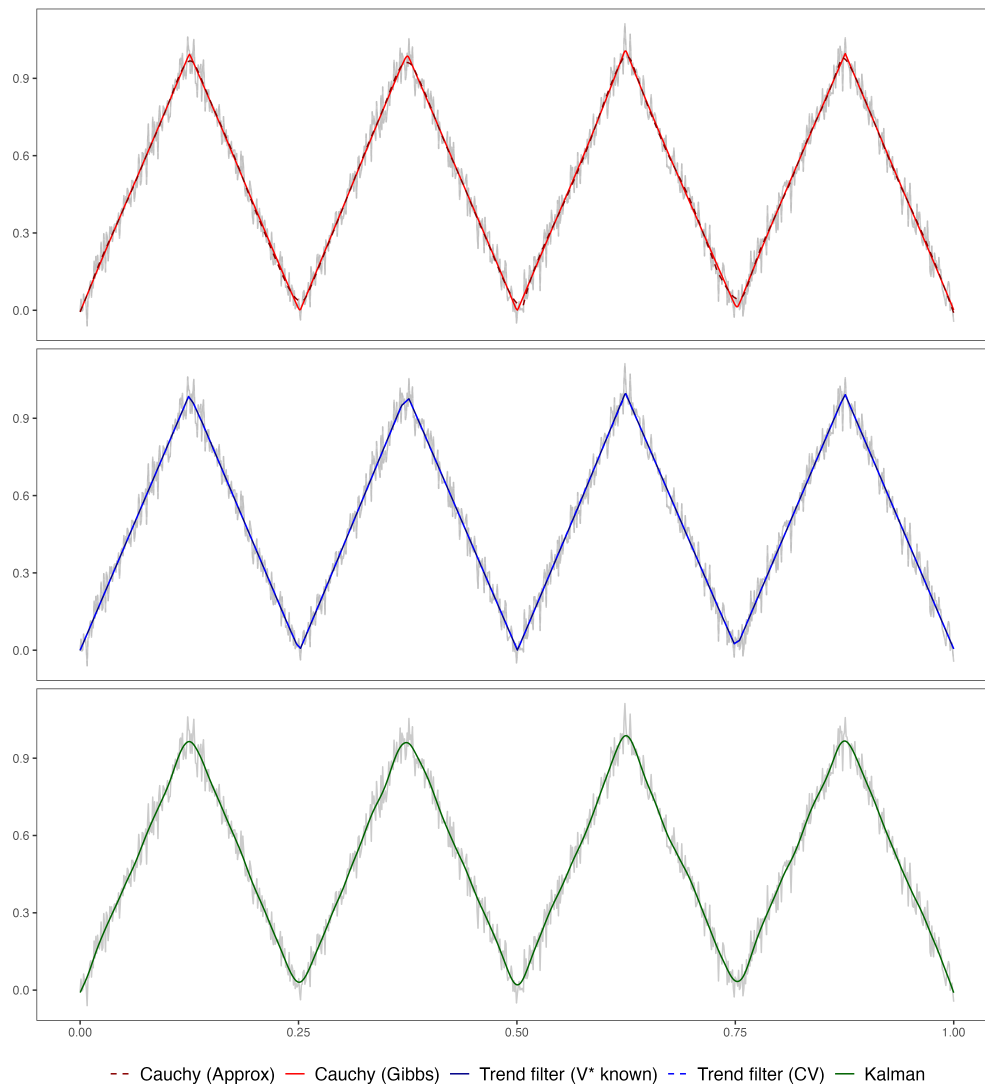


Figure 1.4: Point estimates for noisy Sawtooth data. From top to bottom: Cauchy, trend filtering, and Kalman estimates ($k = 2$).

implemented in the R library `glmgen`. Additionally, when selecting V^* we currently choose the value that minimizes the CV error. More regularized estimates could be obtained by selecting the smallest value of V^* such that the CV error is within one standard error of the minimum CV error.

- In terms of the Cauchy model, the approximate method is inferior to Gibbs sampling both in terms of convergence rates and computation times, thus Gibbs sampling should be preferred. The latter approach has optimal convergence for data generated by a variety of functions, from strongly smooth to non-smooth in the ℓ_1 sense.

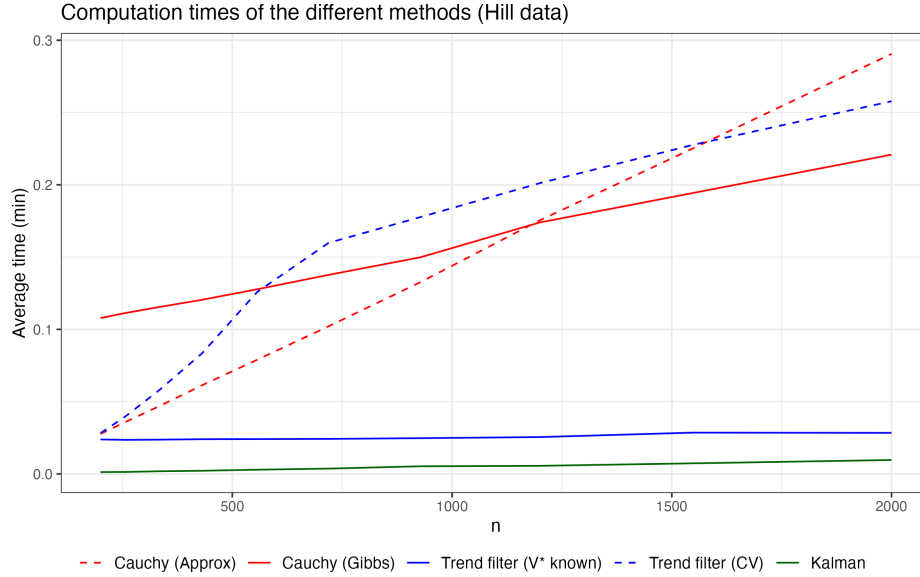


Figure 1.5: Average computation times for different methods on *Hill* data ($k = 3$).

1.5 Empirical Applications

In this Section, we will consider real-world data coming from different domains: finance, climatology, and economics. All the examples considered are time-series data, and have different time resolution: some are reported daily, other monthly, other yearly. Our goal will be to denoise the data, and infer the trend. For each example we will implement the latent Cauchy k^{th} -order differences model (1.18) via Gibbs sampling, trend filtering (1.35) with cross-validated regularization parameter, Kalman smoothing under model (1.36), and SPMRFs (1.37) with Laplace and Horseshoe prior. The differencing order k will be selected based on the data application: data that seem to arise from a piecewise constant function will be fitted with $k = 1$; if a piecewise linear behavior is expected, k will be set to 2; whereas, if the data seem to be noisy versions of an approximately piecewise quadratic function, k will be set to 3. We will use the same number of iterations (5000, with a burn-in period of 1000 iterations) for the Gibbs sampler used in Cauchy estimation and for the HMC used for SPMRF. Notice that, when fitting SPMRF, we often obtain warnings from `stan` about the presence of several divergent transitions and exceeded maximum treedepth, which hinder the reliability of SPMRF posterior estimates.

For each data example, we provide three plots showing the trend estimates: in all plots, the original data are in gray, and, in the first case, we overlay the posterior means and 95% credible intervals obtained by Cauchy estimation; in the second, we overlay the point estimates obtained via trend filtering and Kalman smoothing; in the third we overlay the posterior means and 95% credible intervals obtained by Laplace and Horseshoe SPMRF. Note that in the latter case we distinguish the credible intervals by using dashed lines for

Laplace, and by using colored bands for the Horseshoe.

New York Stock Exchange

We will start by considering an example of financial time series data provided by the R library `astsa`. The data consists of daily returns of the New York Stock Exchange (NYSE) from February 2, 1984 to December 31, 1991. The returns over time are plotted in Figure 1.6.

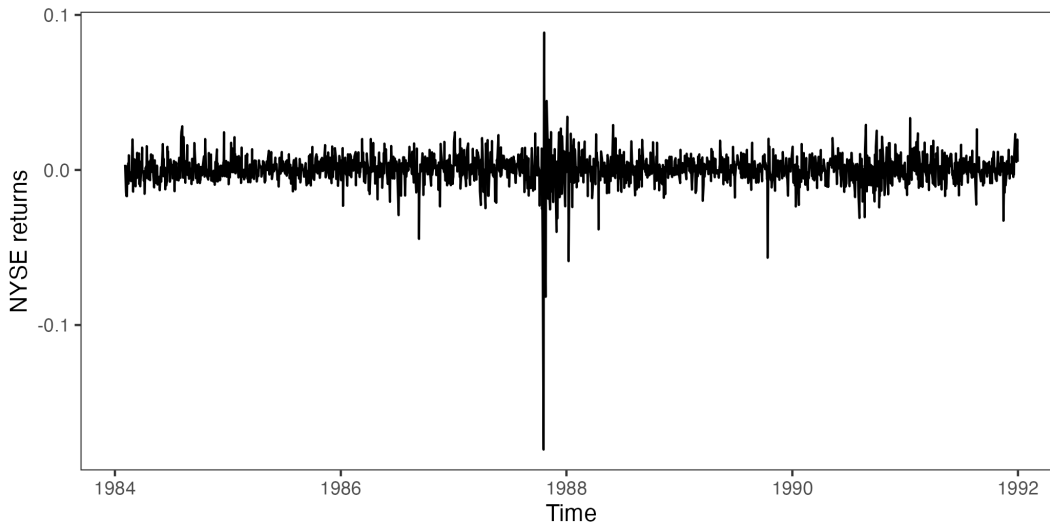
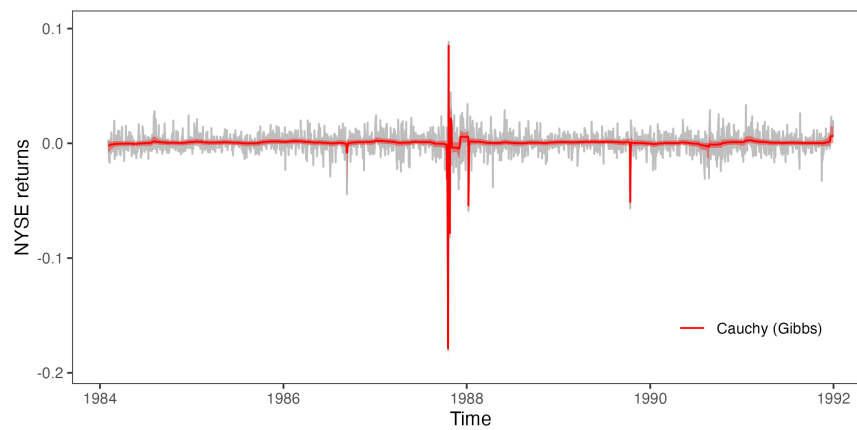
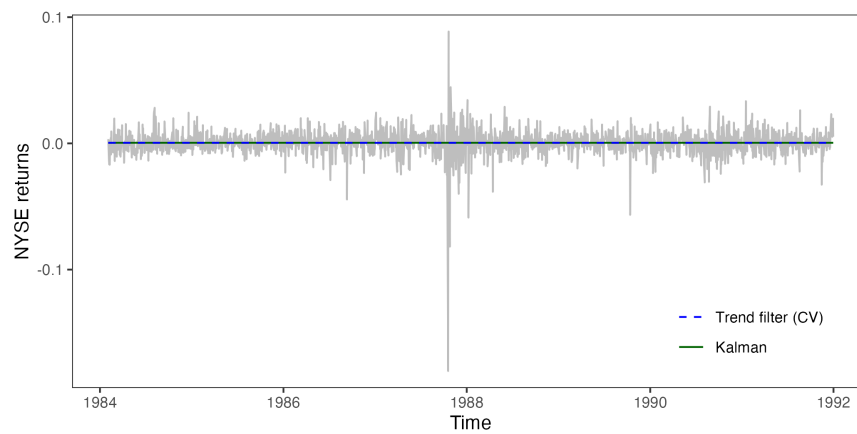


Figure 1.6: Returns of the NYSE. The data are daily value weighted market returns from February 2, 1984 to December 31, 1991.

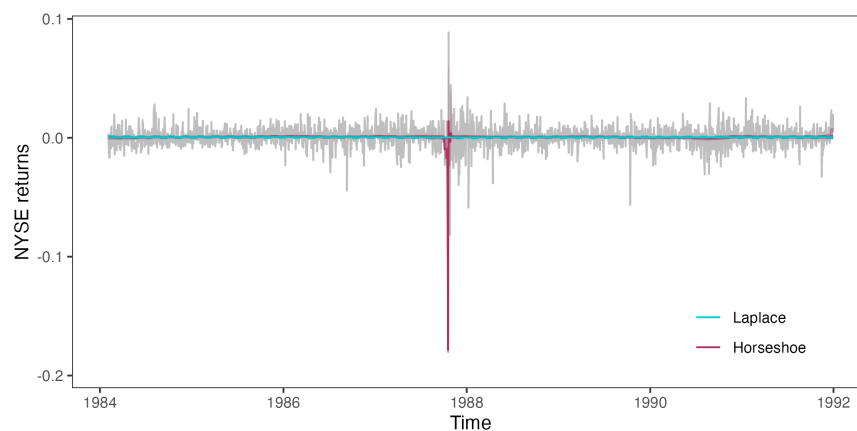
The returns hover around zero and have similar volatility for most of the time considered, but there are some large deviations from zero right before 1988. These deviations correspond to the financial crash of October 19, 1987. In Figure 1.7 we compare different methods to carry out this task. All methods model the first order differences of the function, i.e. $k = 1$ in (1.18), (1.35), (1.36), and (1.37). Comparing the three plots, we can observe that Cauchy estimation is able to adapt to the local changes in the data: it shrinks towards zero most of the returns, while at the same time being able to detect significant changes, like the October 19, 1987 crash. Trend filtering, Kalman smoothing, and Laplace SPMRF are not as locally-adaptive, and they oversmooth the time series, estimating the trend to be constant. On the other hand, SPMRF with Horseshoe prior gives posterior estimates that capture the crash, but are otherwise mostly shrunk towards zero. As expected, the Horseshoe approach performs more shrinkage than the Cauchy one. In terms of computation times, Kalman estimates are obtained in 0.01 minutes, trend filtering in 0.18, Cauchy in 0.19, Laplace SPMRF in 3.43, and Horseshoe SPMRF in 104.84. The number of data points equals 2000.



(a) Cauchy estimation: posterior means and 95% credible intervals



(b) Trend filtering and Kalman smoothing



(c) SPMRF estimation: posterior means and 95% credible intervals

Figure 1.7: Trend estimation of NYSE returns

Temperature Anomalies

We will next consider data on temperature anomalies, available on the website of the [NOAA National Centers for Environmental information](#). In particular, we will focus on global land and ocean average temperature anomalies for the month of August from 1850 to 2024. The anomalies are reported in Celsius, and are computed with respect to the 1901-2000 average. Figure 1.8 shows the original data. Estimation methods are all set to model the second

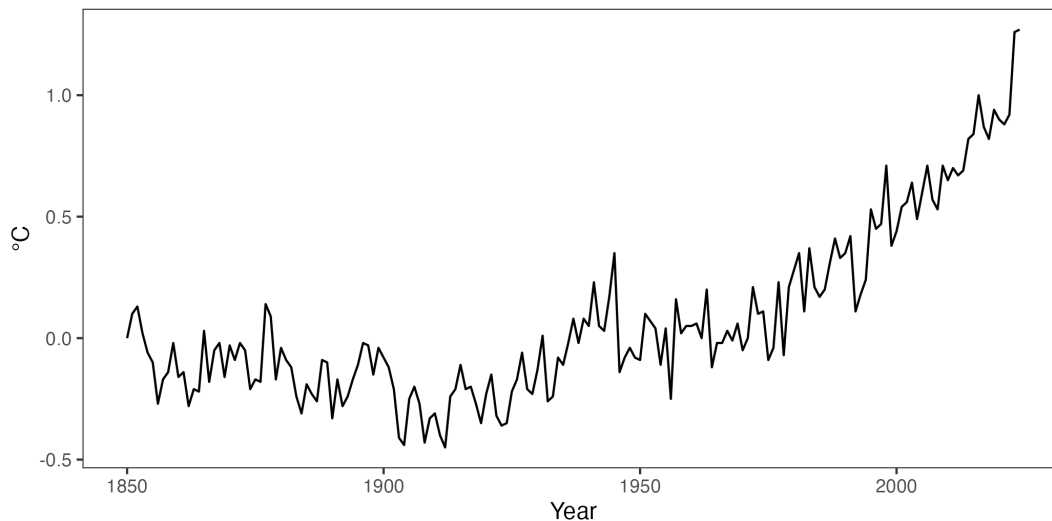
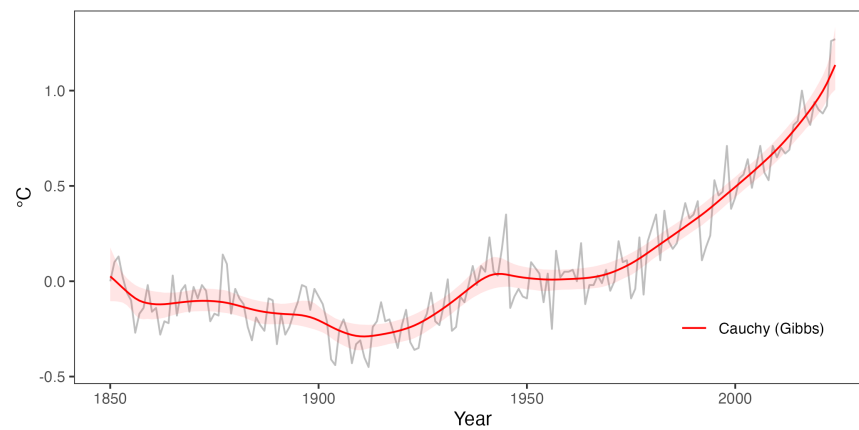
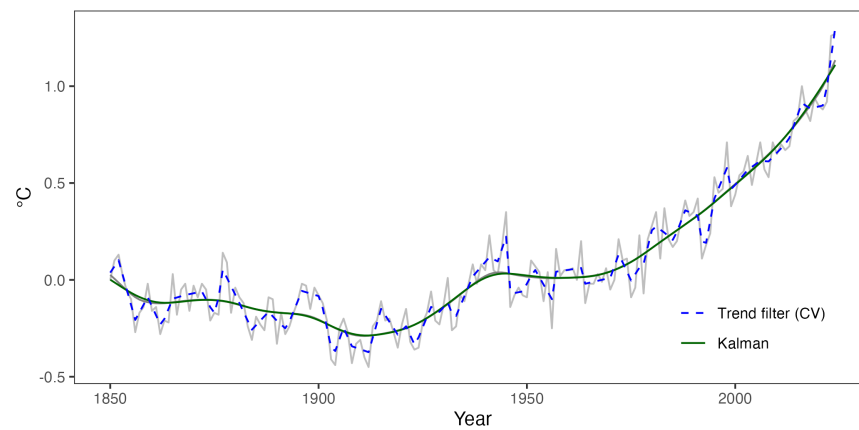


Figure 1.8: Global land and ocean average temperature anomalies for the month of August from 1850 to 2024. The anomalies are computed with respect to the 1901-2000 average.

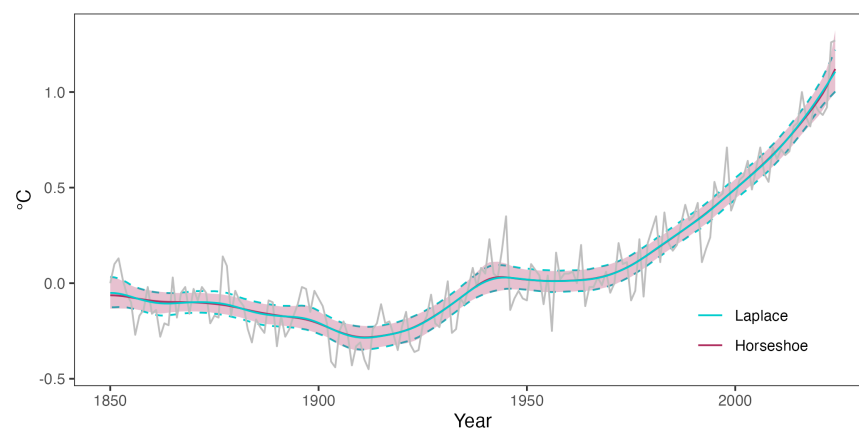
order differences (i.e. $k = 2$). This specific choice of k is made for illustrative purposes rather than scientific considerations, as a piecewise quadratic fit would be more justifiable. When comparing the resulting fits in 1.9, we observe that the posterior estimates from the Bayesian methods are all very similar to each other. Trend filtering, on the other hand, provides a different fit, following the data more closely and leading to an estimate that is not as smooth. The similarity between the Bayesian estimates in this setting can be motivated by the fact that the data does not seem to call for a trend with sharp edges or jumps. Indeed, the data seem to be well-modeled by an ℓ_2 -smooth function, for which the Normal prior used in Kalman smoothing is an appropriate choice. The Cauchy prior, being appropriate for a larger class of functions, can adapt and fit an ℓ_2 -smooth function, when a weaker notion of smoothness is not warranted by the data. The SPMRF fits show a similar ability to recover strongly smooth trends. In terms of computation times, Kalman estimates are obtained in 0.001 minutes, trend filtering in 0.02, Cauchy in 0.11, Laplace SPMRF in 7.48, and Horseshoe SPMRF in 14.13. The number of data points equals 175.



(a) Cauchy estimation: posterior means



(b) Trend filtering and Kalman smoothing



(c) SPMRF estimation: posterior means

Figure 1.9: Trend estimation of global temperature anomalies

Real Gross Domestic Product

In relation to the fit comparisons for the temperature anomalies example, we will now consider a situation in which the same differencing order ($k = 2$) leads to a Cauchy fit that is much closer to trend filtering than to Kalman smoothing.

We will consider U.S. real Gross Domestic Product (GDP) data available on the website of the [Federal Reserve Economic Data \(FRED\)](#) under the code GDPC1. For real GDP, we refer to GDP adjusted for inflation to reflect the actual purchasing power and real growth over time. It is expressed in chained 2017 dollars. The data, visualized in Figure 1.10, span 1994 to 2024, and are reported quarterly. The data do not seem very noisy, and seem to follow a piecewise linear trend. A noticeable downward spike in GDP occurs after the onset of the COVID-19 pandemic. Looking at Figure 1.11, we can observe that the Cauchy, SPMRF,

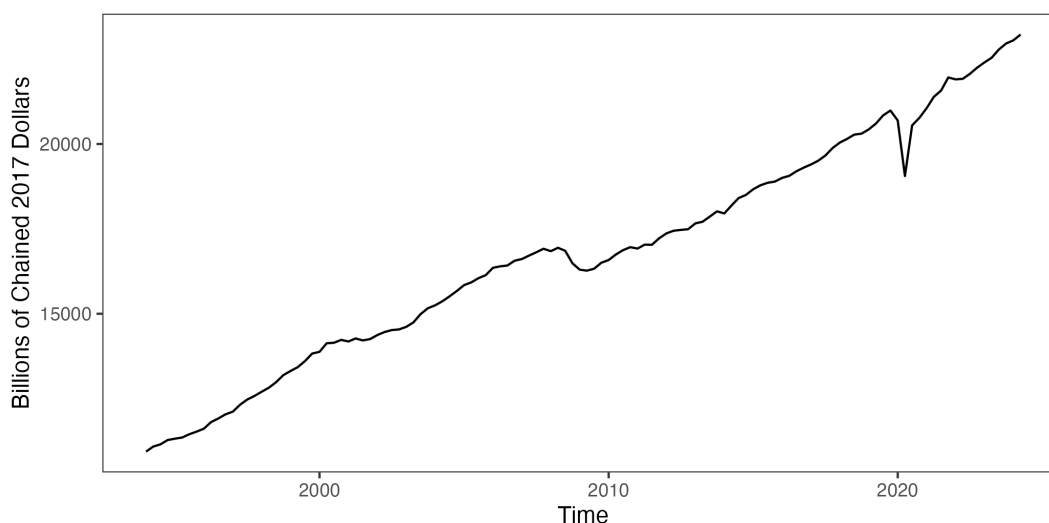
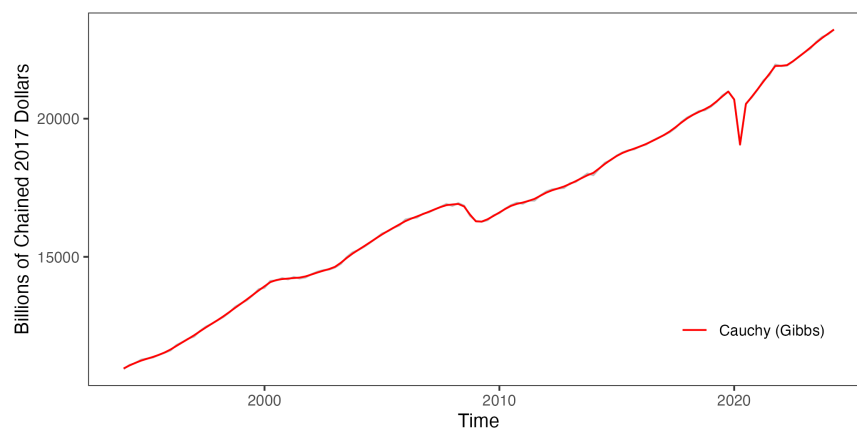
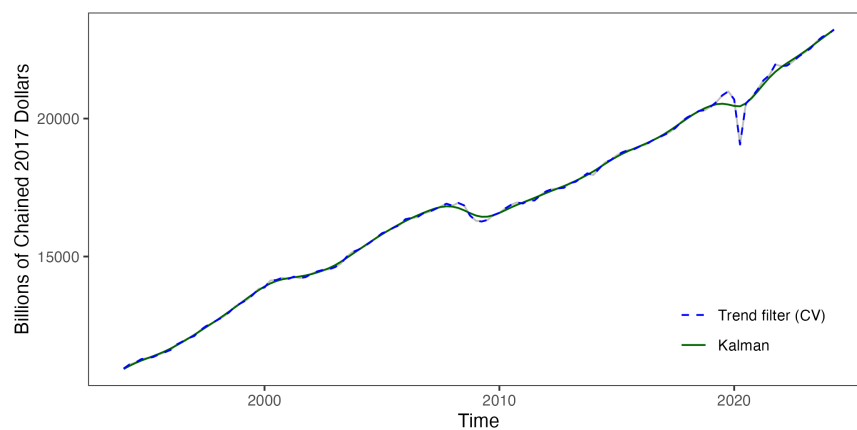


Figure 1.10: Real GDP of the United States. Quarterly data from 1994 to 2024.

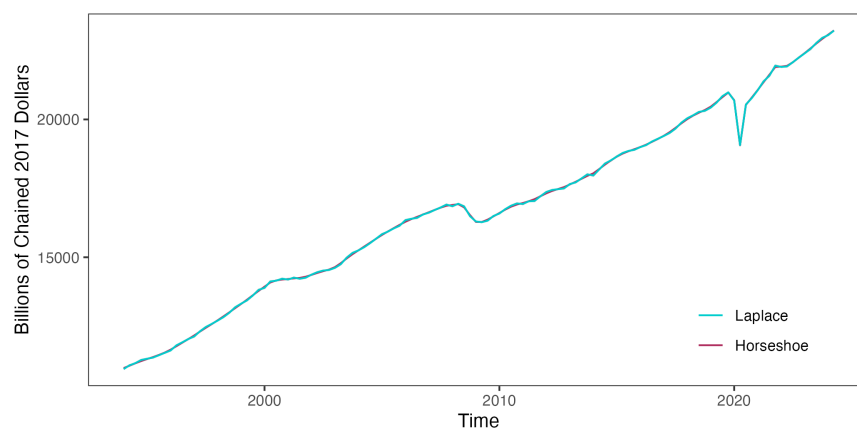
and trend filter fits are very similar: they all slightly smooth the data, while leaving the downward COVID-19 peak unshrunk. The Kalman fit, on the other hand, provides much more smoothing, and the peak is almost completely ignored. In this example, the data seem to be better modeled as generated by a function that is smooth in a sense that is weaker than ℓ_2 , and therefore the Cauchy fit deviates from the Kalman one. Notice that, in this case, we are only plotting posterior means for Cauchy and SPMRF, given the narrowness of the credible intervals. In terms of computation times, Kalman estimates are obtained in 0.002 minutes, trend filtering in 0.01, Cauchy in 0.11, Laplace SPMRF in 28.59, and Horseshoe SPMRF in 28.35. The number of data points equals 122.



(a) Cauchy estimation: posterior means



(b) Trend filtering and Kalman smoothing



(c) SPMRF estimation: posterior means

Figure 1.11: Trend estimation of U.S. real GDP

Unemployment Rate

To conclude, we will consider unemployment rate data also available on the website of the [Federal Reserve Economic Data \(FRED\)](#) with the name UNRATE. The data captures the monthly percentage of the total U.S. labor force that is unemployed but actively seeking employment and willing to work. We use all the data available, from 1948 to the present (2024). Data are shown in Figure 1.12, and are characterized by multiple peaks, with a particularly remarkable one corresponding to the onset of the COVID-19 pandemic. All the

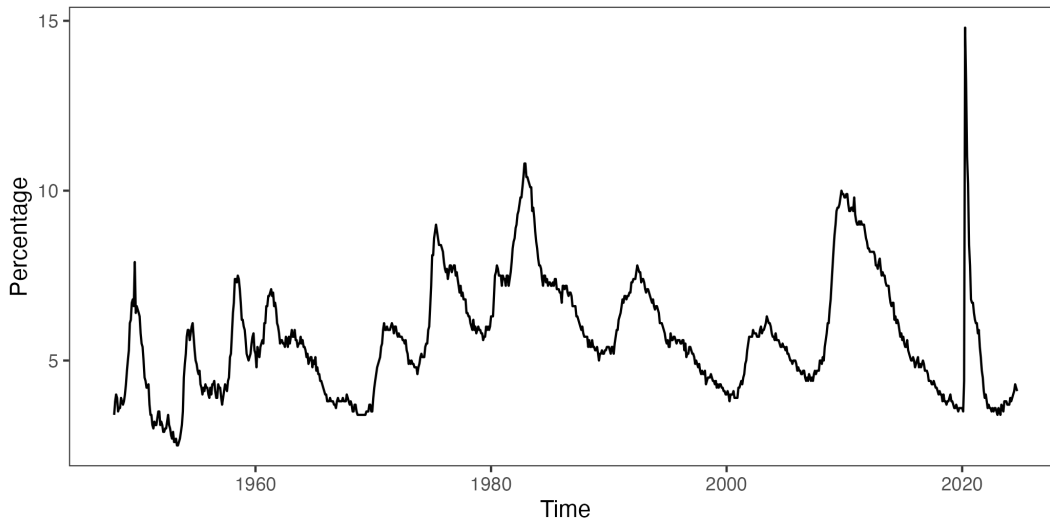
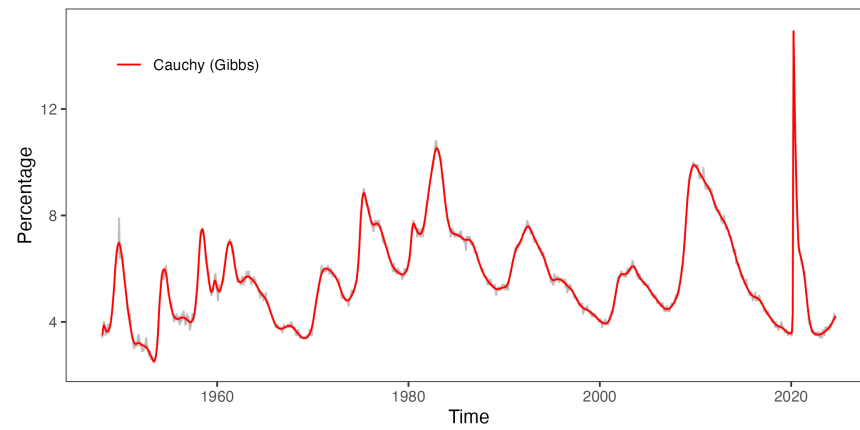
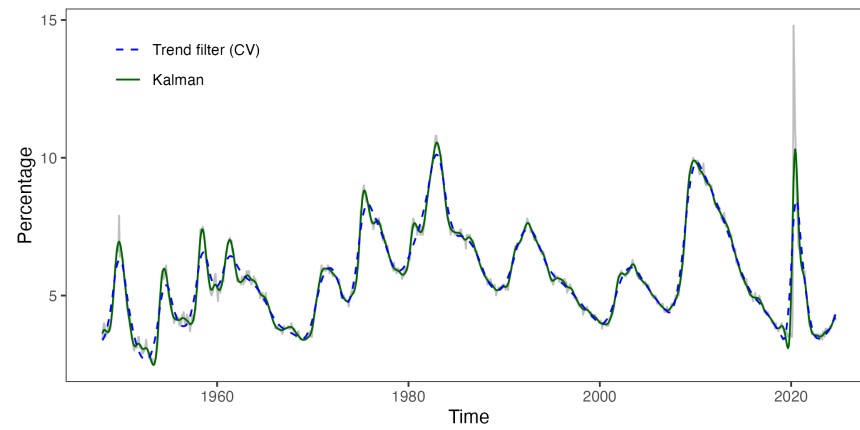


Figure 1.12: Unemployment rate in the United States. Monthly percentages from 1948 to 2024.

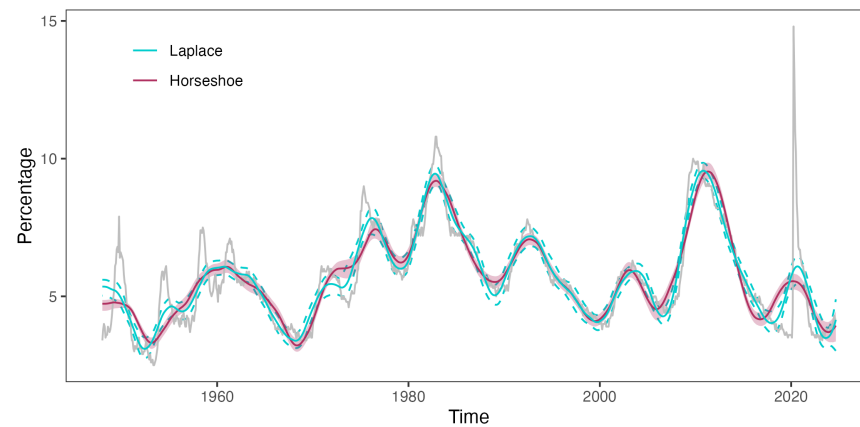
fitted methods are set to model third order differences ($k = 3$). The Cauchy fit smooths the data while leaving most of the peaks unshrunk, in particular the stark COVID-19 peak (see Figure 1.13a). On the other hand, all the other methods perform more smoothing, and considerably shrink the COVID-19 peak (see Figure 1.13b). We conjecture that the inability of the Horseshoe SPMRF to recover the COVID-19 peak could be attributed to HMC issues rather than the model itself. Indeed, given the characteristics of the Horseshoe distribution, we would have expected the corresponding SPMRF posterior estimates to be closer the the ones obtained via Cauchy estimation. In terms of computation times, Kalman estimates are obtained in 0.005 minutes, trend filtering in 0.17, Cauchy in 0.15, Laplace SPMRF in 57.76, and Horseshoe SPMRF in 56.63. The number of data points equals 921.



(a) Cauchy estimation: posterior means and 95% credible intervals



(b) Trend filtering and Kalman smoothing



(c) SPMRF estimation: posterior means and 95% credible intervals

Figure 1.13: Trend estimation of U.S. unemployment rate

Chapter 2

Extensions

2.1 Introduction

In this Chapter, we present several extensions of model (1.18) from Chapter 1.

We begin by introducing models that do not require selection of a specific differencing order k in (1.18). These models decompose the unknown θ into the sum of K vectors, with the k^{th} vector characterized by a Cauchy prior on its k^{th} -order differences, for $k = 1, \dots, K$. First, we introduce the *Cauchy local linear trend* model, which assumes $K = 2$, and then generalize to the *Cauchy decomposition* model, which accommodates arbitrary K .

Next, we extend (1.18) to accomodate cases where $\theta_1, \dots, \theta_n$ are observed at irregularly spaced design points. To address this, we assume that, for every $i = 1, \dots, n$, $\theta_i = \theta(t_i)$, where $\theta(t)$ is a real-valued function defined for $t \geq 0$, and the design points satisfy $0 \leq t_1 < \dots < t_n$. Inference is performed by discretizing a model for the function $\theta(t)$. We leverage the Cauchy process to specify priors on the infinite-dimensional space $\{\theta(t), t \geq 0\}$.

As a further generalization, we introduce the *latent Cauchy autoregressive (AR)* model. For given k , this model relaxes the assumptions on the unknown θ , modeling its evolution with an $AR(k)$ model with Cauchy-distributed errors. This approach is particularly well-suited for scenarios where the latent process is expected to evolve according to some linear differential equation most of the time, but occasional shocks, disrupting the expected dynamics, are allowed to happen.

Finally, we describe an extension of (1.18) allowing for bivariate observations.

2.2 Local Linear Trend

In Chapter 1, we used a Cauchy prior on the k^{th} -order differences of θ . Notice that the order k was not subject to inference, and its choice was left to the user. In Sections 1.4 and 1.5, we set $k = 1$ when we wanted an approximately piecewise constant fit, $k = 2$ for an approximately piecewise linear fit, and $k = 3$ for an approximately piecewise quadratic fit. A more principled approach to the choice of k would be to compute the marginal likelihood

$p(y)$ for different orders k , and to select the order leading to the largest marginal likelihood. This approach is a traditional Bayesian model selection method (see Section 3.8 of [Murphy \(2023\)](#) for more details on the topic). In practice, deriving the marginal likelihood accurately and efficiently is non-trivial for MCMC methods (see Section 13.10 of [Gelman et al. \(2013\)](#) for possible strategies and relative challenges). Moreover, it is possible that a specific value of k is more suited for a subset of the dataset, while another value is better suited for a different subset. For example, consider noisy data generated by a function that is partly piecewise constant and partly piecewise linear. In that case, it would be better not to choose between $k = 1$ and $k = 2$, either arbitrarily or by model selection, but to use a model that learns from the data when the piecewise constant behavior is better suited than the piecewise linear, and vice versa. This goal can be achieved by the following model:

$$\begin{aligned}
 y_i &= \theta_i + \epsilon_i & \epsilon_i &\stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2) & \text{for } i = 1, \dots, n \\
 (D^{(1)}\theta)_j &= \nu_j + \eta_j & \eta_j &\stackrel{\text{i.i.d.}}{\sim} C(0, \tau) & \text{for } j = 1, \dots, n-1 \\
 (D^{(1)}\nu)_l &= \zeta_l & \zeta_l &\stackrel{\text{i.i.d.}}{\sim} C(0, \gamma) & \text{for } l = 1, \dots, n-2.
 \end{aligned} \tag{2.1}$$

We place diffuse Normal priors on θ_1 and ν_1 , and Jeffreys' priors on σ , τ , and γ .

This model is closely related to the well-known *local linear trend* model in time-series analysis (see Section 3.2 of [Durbin and Koopman \(2012\)](#)). The key difference is that the latter assumes normally-distributed errors in place of the Cauchy-distributed errors in (2.1). Based on this analogy, we refer to (2.1) as the *Cauchy local linear trend* model.

Next, we provide an intuition on why (2.1) is appropriate for trends that mix piecewise constant and linear behaviors. First, consider that, if $\gamma = 0$, the model becomes very similar to the 1st-order model from Chapter 1 with an additional constant ν_1 added to $D^{(1)}\theta$, as in $(D^{(1)}\theta)_j = \nu_1 + \eta_j$ with $\eta_j \stackrel{\text{i.i.d.}}{\sim} C(0, \tau)$ for $j = 1, \dots, n-1$. Secondly, if $\tau = 0$, we can combine the last two equations in (2.1) to obtain:

$$D^{(2)}\theta = D^{(1)}\nu = \zeta,$$

where we used the relationship $D^{(1)}D^{(1)}\theta = D^{(2)}\theta$. Therefore, setting $\tau = 0$ in (2.1), we recover the 2nd-order model from Chapter 1. When both γ and τ are very small but strictly positive, most of the ζ_l 's and η_j 's will be very close to zero, with only a few deviating, potentially significantly. These deviations introduce random changes in the slope and level of θ , allowing the model to capture a mix of piecewise constant and linear behaviors.

A related frequentist approach for estimating signals with such mixed behaviors is the *second-order Total Generalized Variation* (TGV) method ([Bredies et al. \(2010\)](#), [Bredies and Holler \(2020\)](#)). However, TGV has a significant drawback: it requires careful tuning of the hyperparameters that penalize 1st- and 2nd-order differences, with no universally accepted method for selecting these parameters. In contrast, model (2.1) offers a Bayesian alternative, where the sparsity in the 1st- and 2nd-order differences is automatically learned through estimating the scale parameters τ and γ .

Posterior Inference

Posterior inference for the Cauchy local linear trend model can be efficiently achieved using Gibbs sampling, leveraging the same speed-ups for banded precision matrices discussed for Algorithm 1. Indeed, both θ and ν have full conditional distributions that are multivariate normals, with precision matrices exhibiting a banded structure with a bandwidth of 3. The Gibbs sampler is detailed in Algorithm 4. The complete conditional distributions needed for the sampler can be obtained by a straightforward simplification of the distributions in Appendix B.2.

Simulations

The *Jumpy Sawtooth* function considered in Section 1.4 is an example of a function with a mixed piecewise linear and constant behavior. Indeed, it can be obtained by summing a *Sawtooth* function with a piecewise constant one. We would therefore expect the Cauchy local linear trend model (2.1) to perform even better than the latent Cauchy differences model (1.18) with $k = 2$.

We perform a convergence study similar to the one in Section 1.4, but only focus on three functions: *Blocks*, *Sawtooth*, and *Jumpy Sawtooth*, and only compare model (2.1) to model (1.18) with $k = 1$ for the *Blocks* case, and with $k = 2$ for the other two cases. Notice that even if we expect (2.1) to improve on (1.18) with $k = 2$ for the *Jumpy Sawtooth*, we expect it to perform slightly worse than (1.18) with $k = 1$ for *Blocks* data, and than (1.18) with $k = 2$ for *Sawtooth* data. Indeed, in these latter two cases, the functions are purely piecewise constant or piecewise linear, so they can be better estimated by setting γ or τ exactly to zero in (2.1).

We consider 10 different values for the number n of observed data points, and choose them in the range from 200 to 2000, and equispaced on the log-scale. For each value of n and each function, we simulate 20 datasets by first evaluating the function at n equispaced design points on $[0, 1]$, and then adding Gaussian noise. The variance of the Gaussian noise is set in such a way that the signal-to-noise ratio is always equal to 7.

Figure 2.1 and Table 2.1 confirm our expectations: the Cauchy local linear trend model improves inference for the *Jumpy Sawtooth*, but is slightly less precise for *Blocks* and *Sawtooth*. Notice that, in the *Jumpy Sawtooth* case and when n is large, there is a significant gap between the two models considered in terms of MSE.

To investigate the MSE gap for large n , we compute the posterior means obtained by applying the Cauchy local linear trend model and the latent Cauchy 2^{nd} -order differences model to noisy *Jumpy Sawtooth* data with $n = 3000$. Two scenarios are considered: one with a high signal-to-noise ratio ($SNR = 7$) and one with a low signal-to-noise ratio ($SNR = 3$). Figure 2.2 compares the posterior means from the two fitted models with the true generating function in both scenarios. We can infer that the MSE is larger for the 2^{nd} -order differences model because some of its point estimates are located in the vertical space between adjacent linear blocks. This behavior is more pronounced in the low SNR scenario.

Algorithm 4: Gibbs Sampler for the Cauchy local linear trend model

Input: data $y = (y_1, \dots, y_n)$.

Initialize $\sigma^{2(0)}, \tau^{2(0)}, \lambda_1^{2(0)}, \dots, \lambda_{n-1}^{2(0)}, \gamma^{2(0)}, \xi_1^{2(0)}, \dots, \xi_{n-2}^{2(0)}, \nu_1^{(0)}, \dots, \nu_{n-1}^{(0)}$ to some positive constants (we let $\sigma^{2(0)} = 10, \tau^{2(0)} = 0.01, \gamma^{2(0)} = 0.001, \lambda_j^{2(0)} = \xi_l^{2(0)} = 1$, and $\nu_j \stackrel{\text{i.i.d.}}{\sim} N(0, 10^{-8})$ for every $j = 1, \dots, n-1, l = 1, \dots, n-2$).

For $m = 1, \dots, M$ (with M a large positive integer):

1. Let

$$\Lambda^{(m-1)} = \text{diag}(\lambda^{2(m-1)})$$

and

$$P^{(m-1)} = \frac{1}{\sigma^{2(m-1)}} I_n + \frac{1}{\tau^{2(m-1)}} D^{(1)\top} \Lambda^{-1(m-1)} D^{(1)},$$

and perform the Cholesky factorization

$$P^{(m-1)} = LL^\top$$

2. Solve

a) $Lv = \frac{1}{\sigma^{2(m-1)}} y + \frac{1}{\tau^{2(m-1)}} D^{(1)\top} \Lambda^{-1(m-1)} \nu^{(m-1)}$

b) $L^\top a = v$

c) $L^\top b = z$, where $z \sim N(\mathbf{0}, I_n)$

3. Let

$$\theta^{(m)} = a + b$$

and

$$\beta^{(m)} = D^{(1)} \theta^{(m)} - \nu^{(m-1)}$$

4. Sample

$$\lambda_j^{2(m)} \stackrel{\text{ind}}{\sim} IG \left(1, \frac{\beta_j^{2(m)} + \tau^{2(m-1)}}{2\tau^{2(m-1)}} \right), \quad j = 1, \dots, n-1$$

$$\tau^{2(m)} \sim IG \left(\frac{n-1}{2}, \sum_{j=1}^{n-1} \frac{\beta_j^{2(m)}}{2\lambda_j^{2(m)}} \right)$$

$$\sigma^{2(m)} \sim IG \left(\frac{n}{2}, \frac{\|y - \theta^{(m)}\|_2^2}{2} \right)$$

5. Sample

$$\gamma^{2(m)} \sim IG \left(\frac{n-2}{2}, \sum_{l=1}^{n-2} \frac{(D^{(1)}\nu^{(m-1)})_l^2}{2\xi_l^{2(m-1)}} \right)$$

$$\xi_l^{2(m)} \stackrel{\text{ind}}{\sim} IG \left(1, \frac{(D^{(1)}\nu^{(m-1)})_l^2 + \gamma^{2(m)}}{2\gamma^{2(m)}} \right), \quad l = 1, \dots, n-2$$

6. Let

$$\Xi^{(m)} = \text{diag}(\xi^{2(m)})$$

and

$$\tilde{P}^{(m)} = \frac{1}{\tau^{2(m)}} \Lambda^{-1(m)} + \frac{1}{\gamma^{2(m)}} D^{(1)\top} \Xi^{-1(m)} D^{(1)},$$

and perform the Cholesky factorization

$$P^{(m)} = LL^\top$$

7. Solve

- a) $Lv = \frac{1}{\tau^{2(m)}} \Lambda^{-1(m)} D^{(1)\top} \theta^{(m)}$
- b) $L^\top a = v$
- c) $L^\top b = z$, where $z \sim N(\mathbf{0}, I_{n-1})$

8. Let

$$\nu^{(m)} = a + b$$

Table 2.1: Slopes of log(mean MSE) vs log(n) for Cauchy local linear trend and latent Cauchy differences

Function	Cauchy Local Linear Trend	Cauchy Differences
Blocks	-1.00	-1.05 (k = 1)
Sawtooth	-0.78	-0.98 (k = 2)
Jumpy Sawtooth	-1.14	-0.64 (k = 2)

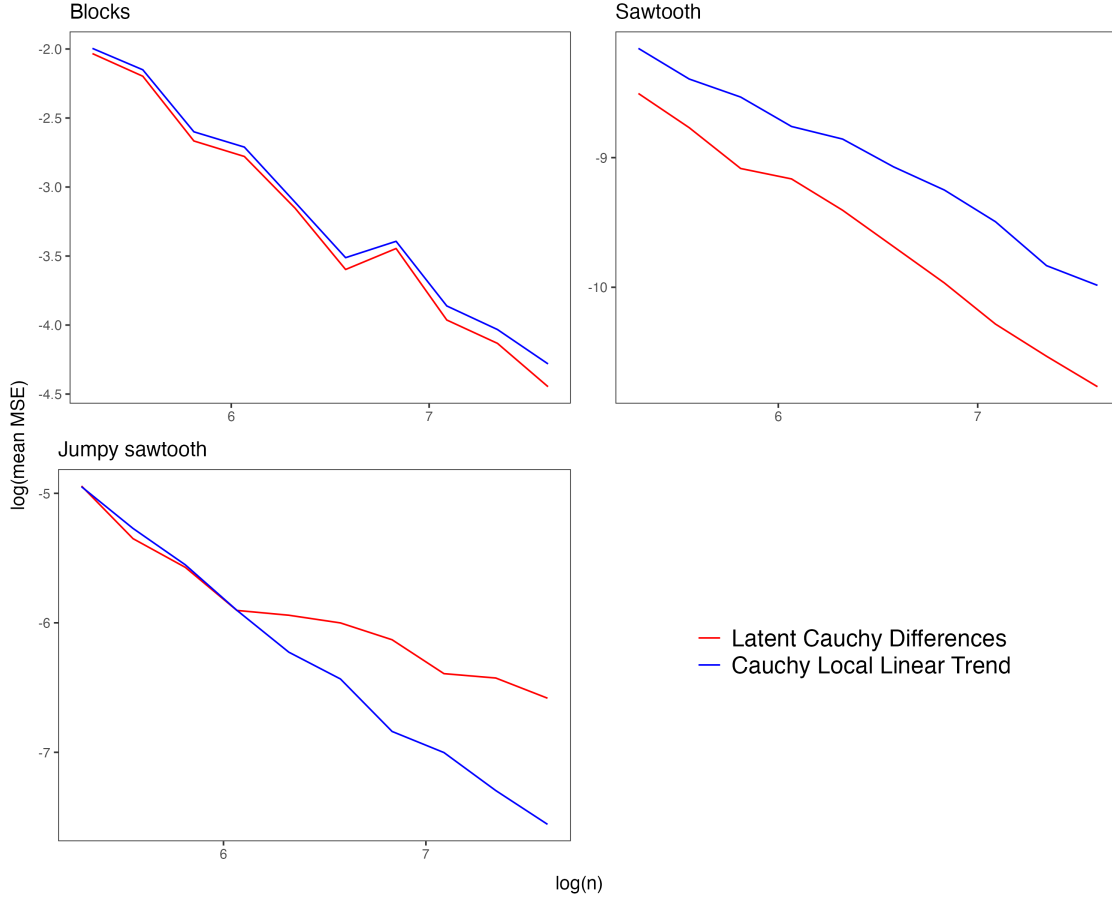


Figure 2.1: Convergence rates. Comparing the Cauchy local linear trend model to the latent Cauchy 1st-order differences (Blocks), and 2nd-order differences (Sawtooth and Jumpy sawtooth).

In the high SNR scenario, the posterior mean of the Cauchy scale obtained from the 2nd-order model is approximately 10^{-5} . For the local linear trend model, the posterior mean for γ is even smaller (10^{-6}), but it is accompanied by a larger estimate for τ (10^{-4}). Similar results are observed for the low SNR case.

The differing fits of the two models can be explained by the fact that the small estimated scale of the 2nd-order model allows it to accurately recover strictly linear segments but limits its ability to adapt to abrupt changes. On the other hand, the Cauchy local linear trend model avoids this issue. A similarly small value for the estimated γ allows to effectively capture linear components, while the presence of the second scale parameter, τ , enables the model to adapt to abrupt jumps.

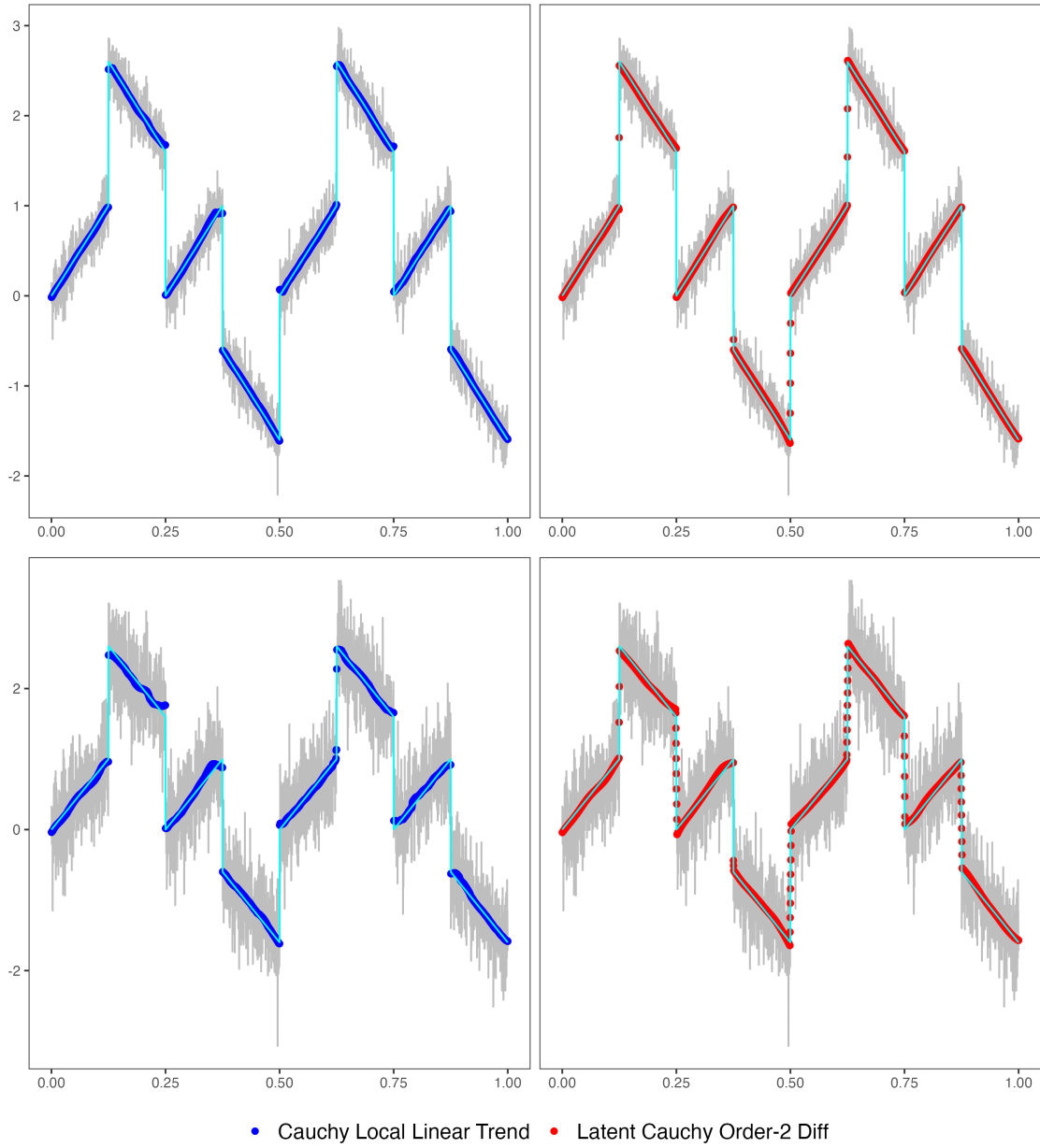


Figure 2.2: Jumpy Sawtooth estimates. Comparison of the posterior means obtained by fitting the Cauchy local linear trend model and the latent Cauchy 2^{nd} -order differences to $n = 3000$ data points, generated by adding Gaussian noise to Jumpy Sawtooth data (signal-to-noise ratio equal to 7 in the top plots, and equal to 3 in the bottom plots). True function in cyan, noisy data in gray.

2.3 Decomposition

Model (2.1) can be re-formulated in a more interpretable way as follows

$$\begin{aligned} y_i &= \theta_i^{(1)} + \theta_i^{(2)} + \epsilon_i & \epsilon_i &\stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2) & \text{for } i = 1, \dots, n \\ (D^{(1)}\theta^{(1)})_j &= \eta_j^{(1)} & \eta_j^{(1)} &\stackrel{\text{i.i.d.}}{\sim} C(0, \tau_1) & \text{for } j = 1, \dots, n-1, \\ (D^{(2)}\theta^{(2)})_l &= \eta_l^{(2)} & \eta_l^{(2)} &\stackrel{\text{i.i.d.}}{\sim} C(0, \tau_2) & \text{for } l = 1, \dots, n-2. \end{aligned} \quad (2.2)$$

The equivalence between (2.2) and (2.1) can be established by decomposing the vector θ in (2.1) as $\theta^{(1)} + \theta^{(2)}$, and by letting $\eta = \eta^{(1)}$, $\nu = D^{(1)}\theta^{(2)}$, and $\zeta = \eta^{(2)}$. The new formulation allows for a more straightforward understanding of why this model is suitable for trends that have a mixed piecewise constant and linear behavior. Indeed, the vector θ is modeled as a sum of two vectors, the first, $\theta^{(1)}$, expected to be approximately piecewise constant, and the second, $\theta^{(2)}$, expected to be approximately piecewise linear.

Using the posterior estimates obtained for the left plots of Figure 2.2, we can compute the corresponding estimates for $\theta^{(1)}$ and $\theta^{(2)}$. Figure 2.3 overlays the posterior means of $\theta^{(1)}$ and $\theta^{(2)}$, and the noisy Jumpy Sawtooth data. As expected, we are able to decompose the estimate for θ into a component that is approximately piecewise constant and one that is approximately piecewise linear.

A more flexible model, able to incorporate, for example, piecewise quadratic components, can be obtained by decomposing θ into the sum of K vectors $\theta^{(1)}, \theta^{(2)}, \dots, \theta^{(K)}$, and by placing Cauchy priors on the k^{th} -order differences of $\theta^{(k)}$, for $k = 1, 2, \dots, K$. This generalization leads to the following model:

$$\begin{aligned} y_i &= \theta_i + \epsilon_i & \epsilon_i &\stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2) \\ \theta_i &= \sum_{k=1}^K \theta_i^{(k)} \\ (D^{(k)}\theta^{(k)})_j &= \eta_j^{(k)} & \eta_j^{(k)} &\stackrel{\text{i.i.d.}}{\sim} C(0, \tau_k) \end{aligned} \quad (2.3)$$

for $i = 1, \dots, n$, $j = 1, \dots, n-k$, and $k = 1, \dots, K$. To complete the model specification, we assume a diffuse Normal prior $N(0, L)$, with L an arbitrarily large integer, on the k^{th} entry of each $\theta^{(k)}$ vector, for $k = 1, \dots, K$. Moreover, we assume that the first $k-1$ entries of $\theta^{(k)}$ are equal to zero for every component with $k > 1$. This zero assumption ensures identifiability of the model's components. Without it, the decomposition of θ into the sum of components $\theta^{(k)}$ would not be unique, as one could add or subtract constants from the individual components without altering the sum. Assuming a flat prior would exacerbate this issue, as it would assign equal prior probability to all such decompositions.

We will next focus on model (2.3) with $K = 3$ components, provide an algorithm to fit it, and apply the method to simulated data.

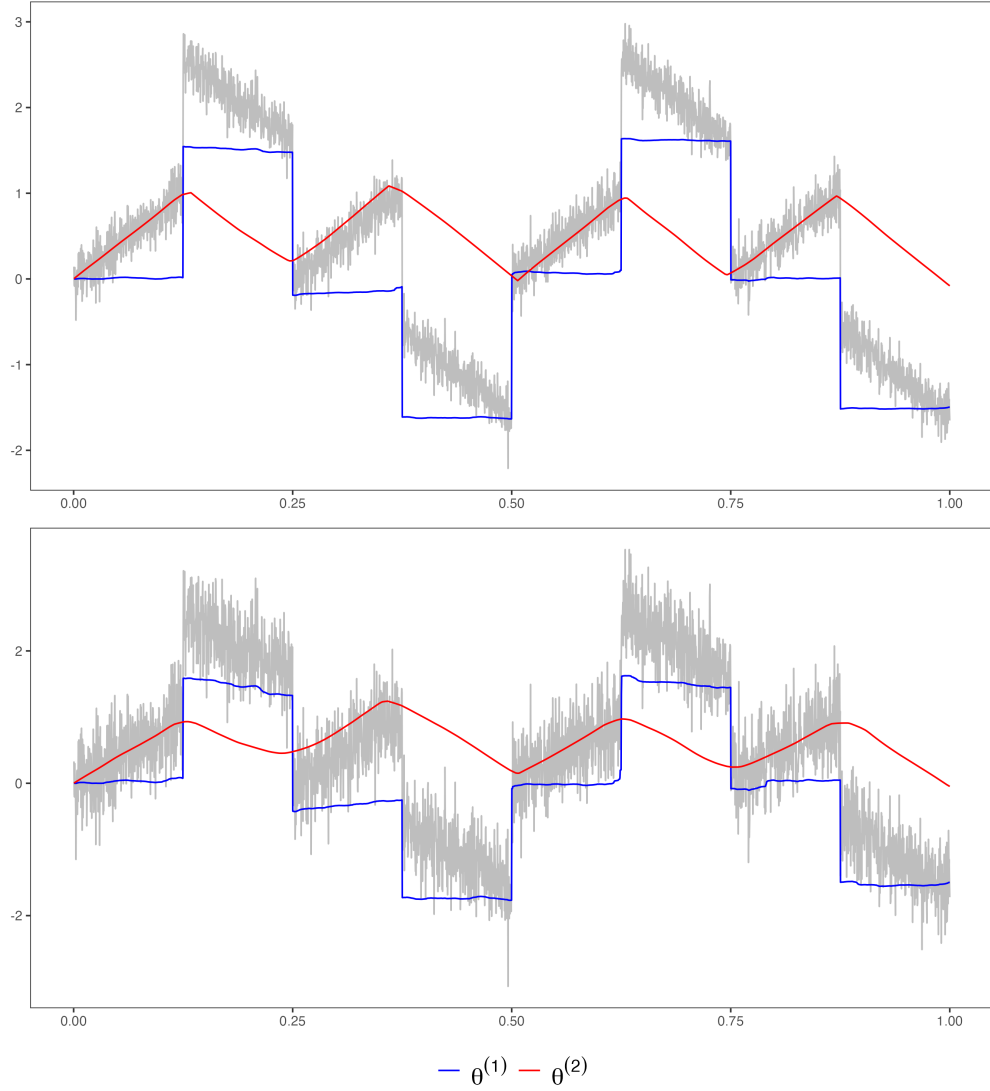


Figure 2.3: Decomposition of noisy Jumpy Sawtooth data via the Cauchy local linear trend model (signal-to-noise ratio equal to 7 in the top plot, and equal to 3 in the bottom plot).

Posterior Inference

Let's consider model (2.3) with $K = 3$, and re-express it by taking first-order differences of the vector θ :

$$\begin{aligned}
 y &= \theta + \epsilon & \epsilon_i &\stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2) & \text{for } i = 1, \dots, n \\
 D^{(1)}\theta &= \eta^{(1)} + D^{(1)}\theta^{(2)} + D^{(2)}\theta^{(3)} & \eta_j^{(1)} &\stackrel{\text{i.i.d.}}{\sim} C(0, \tau_1) & \text{for } j = 1, \dots, n-1 \\
 D^{(1)}(D^{(1)}\theta^{(2)}) &= \eta^{(2)} & \eta_k^{(2)} &\stackrel{\text{i.i.d.}}{\sim} C(0, \tau_2) & \text{for } k = 1, \dots, n-2 \\
 D^{(2)}(D^{(1)}\theta^{(3)}) &= \eta^{(3)} & \eta_l^{(3)} &\stackrel{\text{i.i.d.}}{\sim} C(0, \tau_3) & \text{for } l = 1, \dots, n-3.
 \end{aligned} \tag{2.4}$$

Let's also re-parametrize the model by setting

$$\tilde{\theta}^{(1)} = D^{(1)}\theta^{(2)}$$

$$\tilde{\theta}^{(2)} = D^{(1)}\theta^{(3)}$$

and by redefining η_r and τ_r respectively as η_{r-1} and τ_{r-1} , for $r = 1, 2, 3$. The model then becomes

$$\begin{aligned} y &= \theta + \epsilon & \epsilon_i &\stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2) & \text{for } i = 1, \dots, n \\ D^{(1)}\theta &= \tilde{\theta}^{(1)} + \tilde{\theta}^{(2)} + \eta^{(0)} & \eta_j^{(0)} &\stackrel{\text{i.i.d.}}{\sim} C(0, \tau_0) & \text{for } j = 1, \dots, n-1 \\ D^{(1)}\tilde{\theta}^{(1)} &= \eta^{(1)} & \eta_k^{(1)} &\stackrel{\text{i.i.d.}}{\sim} C(0, \tau_1) & \text{for } k = 1, \dots, n-2 \\ D^{(2)}\tilde{\theta}^{(2)} &= \eta^{(2)} & \eta_l^{(2)} &\stackrel{\text{i.i.d.}}{\sim} C(0, \tau_2) & \text{for } l = 1, \dots, n-3. \end{aligned} \tag{2.5}$$

Notice that our original prior assumption also implies a diffuse $N(0, L)$ prior on the first entry of θ and $\tilde{\theta}^{(1)}$, and on the second entry of $\tilde{\theta}^{(2)}$, while the first entry of $\tilde{\theta}^{(2)}$ is assumed to be equal to zero.

A Gibbs sampler similar to Algorithm 4 can be constructed for model (2.5). This new sampler is detailed in Algorithm 5, with mathematical derivations of the complete conditional distribution provided in Appendix B.2. Notice that in Algorithm 5 we sample $\tilde{\theta}^{(1)}$ and $\tilde{\theta}^{(2)}$ jointly (Steps 1-3). While this joint sampling approach increases computational costs compared to an alternative Gibbs sampler that samples the two vectors separately, it avoids mixing issues observed with the faster alternative. The increased computational cost arises because the precision matrix of the joint complete conditional distribution for $\tilde{\theta}^{(1)}$ and $\tilde{\theta}^{(2)}$ lacks a banded structure. In contrast, the sampler in Chapter 1, Algorithm 4, and the alternative approach benefit from a banded precision matrix, enabling more efficient Cholesky decompositions and shorter computing times.

Although joint sampling of $\tilde{\theta}^{(1)}$ and $\tilde{\theta}^{(2)}$ results in longer computing times, it is preferred over separate sampling due to its superior mixing performance observed in simulations. Addressing the computational efficiency of Algorithm 5 is left for future work.

Simulation

We next illustrate the performance of the Cauchy decomposition model (2.5) in comparison to the Cauchy local linear trend model (2.1) and the latent Cauchy k^{th} -order differences (1.18) with $k = 1, 2, 3$ on a simulated dataset.

The dataset is generated using the Heavisine function, introduced by Donoho and Johnstone (1994) as a benchmark for evaluating wavelet shrinkage methods. The Heavisine function is characterized by a sinusoidal behavior, combined with abrupt jumps. We evaluate the function at 1000 equispaced design points on the interval $[0, 1]$, with Gaussian noise added to achieve a signal-to-noise ratio of 7. The true function, the noisy data, and the posterior means obtained by fitting each model are reported in Figure 2.4.

Algorithm 5: Gibbs Sampler for the Cauchy decomposition

Input: data $y = (y_1, \dots, y_n)$.

Let $\tilde{\theta}_{[-1]}^{(2)}$ and $D_{[-1]}^{(2)}$ denote, respectively, the vector $\tilde{\theta}^{(2)}$ without its first entry and the matrix $D^{(2)}$ without its first column. Set the first entry of $\tilde{\theta}^{(2)}$ to 0, and let

$$X = \begin{pmatrix} I_{n-1} & \begin{pmatrix} \mathbf{0}^\top \\ I_{n-2} \end{pmatrix} \end{pmatrix}.$$

Initialize $\sigma^{2(0)}, \theta^{(0)}, \tau_k^{2(0)}$ and $\lambda_{kj}^{2(0)}$ for $k = 0, 1, 2$ and $j = 1, \dots, n-1-k$ to some positive constants (we let $\sigma^{2(0)} = 10$, $\theta^{(0)} = y$, $\tau_k^{2(0)} = 0.01^k$ and $\lambda_{kj}^{2(0)} = 1$ for every $k = 0, 1, 2$ and $j = 1, \dots, n-1-k$).

For $m = 1, \dots, M$ (with M a large positive integer):

1. Let

$$\Lambda_k^{(m-1)} = \text{diag}(\lambda_k^{2(m-1)}) \quad \text{for } k = 0, 1, 2$$

$$Q_1^{(m-1)} = \tau_1^{-2(m-1)} D^{(1)\top} \Lambda_1^{-1(m-1)} D^{(1)}$$

$$Q_2^{(m-1)} = \tau_2^{-2(m-1)} D_{[-1]}^{(2)\top} \Lambda_2^{-1(m-1)} D_{[-1]}^{(2)}$$

$$\tilde{P}^{(m-1)} = \frac{1}{\tau_0^{2(m-1)}} X^\top \Lambda_0^{-1(m-1)} X + \begin{pmatrix} Q_1^{(m-1)} & \mathbf{0}_{(n-1) \times (n-2)} \\ \mathbf{0}_{(n-2) \times (n-1)} & Q_2^{(m-1)} \end{pmatrix}$$

and perform the Cholesky factorization

$$\tilde{P}^{(m-1)} = LL^\top$$

2. Solve

a) $Lv = \frac{1}{\tau_0^{2(m-1)}} X^\top \Lambda_0^{-1(m-1)} D^{(1)} \theta^{(m-1)}$

b) $L^\top a = v$

c) $L^\top b = z$, where $z \sim N(\mathbf{0}, I_{2n-3})$

3. Let

$$\begin{pmatrix} \tilde{\theta}^{(1)} \\ \tilde{\theta}_{[-1]}^{(2)} \end{pmatrix}^{(m)} = a + b$$

4. Let

$$P^{(m-1)} = \frac{1}{\sigma^{2(m-1)}} I_n + \frac{1}{\tau_0^{2(m-1)}} D^{(1)\top} \Lambda_0^{-1(m-1)} D^{(1)},$$

and perform the Cholesky factorization

$$P^{(m-1)} = LL^\top$$

5. Solve

a) $Lv = \frac{1}{\sigma^{2(m-1)}} y + \frac{1}{\tau_0^{2(m-1)}} D^{(1)\top} \Lambda_0^{-1(m-1)} (\tilde{\theta}^{(1)(m)} + \tilde{\theta}^{(2)(m)})$

b) $L^\top a = v$

c) $L^\top b = z$, where $z \sim N(\mathbf{0}, I_n)$

6. Let

$$\theta^{(m)} = a + b,$$

$$\beta_1^{(m)} = D^{(1)} \theta^{(m)} - (\tilde{\theta}^{(1)(m)} + \tilde{\theta}^{(2)(m)})$$

and

$$\beta_k^{(m)} = D^{(k)} \tilde{\theta}^{(k)(m)} \quad \text{for } k = 1, 2.$$

7. Sample

$$\tau_k^{2(m)} \sim IG \left(\frac{n-1-k}{2}, \sum_{j=1}^{n-1-k} \frac{\beta_{kj}^{2(m)}}{2\lambda_{kj}^{2(m-1)}} \right) \quad k = 0, 1, 2$$

$$\lambda_{kj}^{2(m)} \stackrel{\text{ind}}{\sim} IG \left(1, \frac{\beta_{kj}^{2(m)} + \tau_k^{2(m)}}{2\tau_k^{2(m)}} \right), \quad j = 1, \dots, n-1-k, \text{ and } k = 0, 1, 2$$

$$\sigma^{2(m)} \sim IG \left(\frac{n}{2}, \frac{\|y - \theta^{(m)}\|_2^2}{2} \right)$$

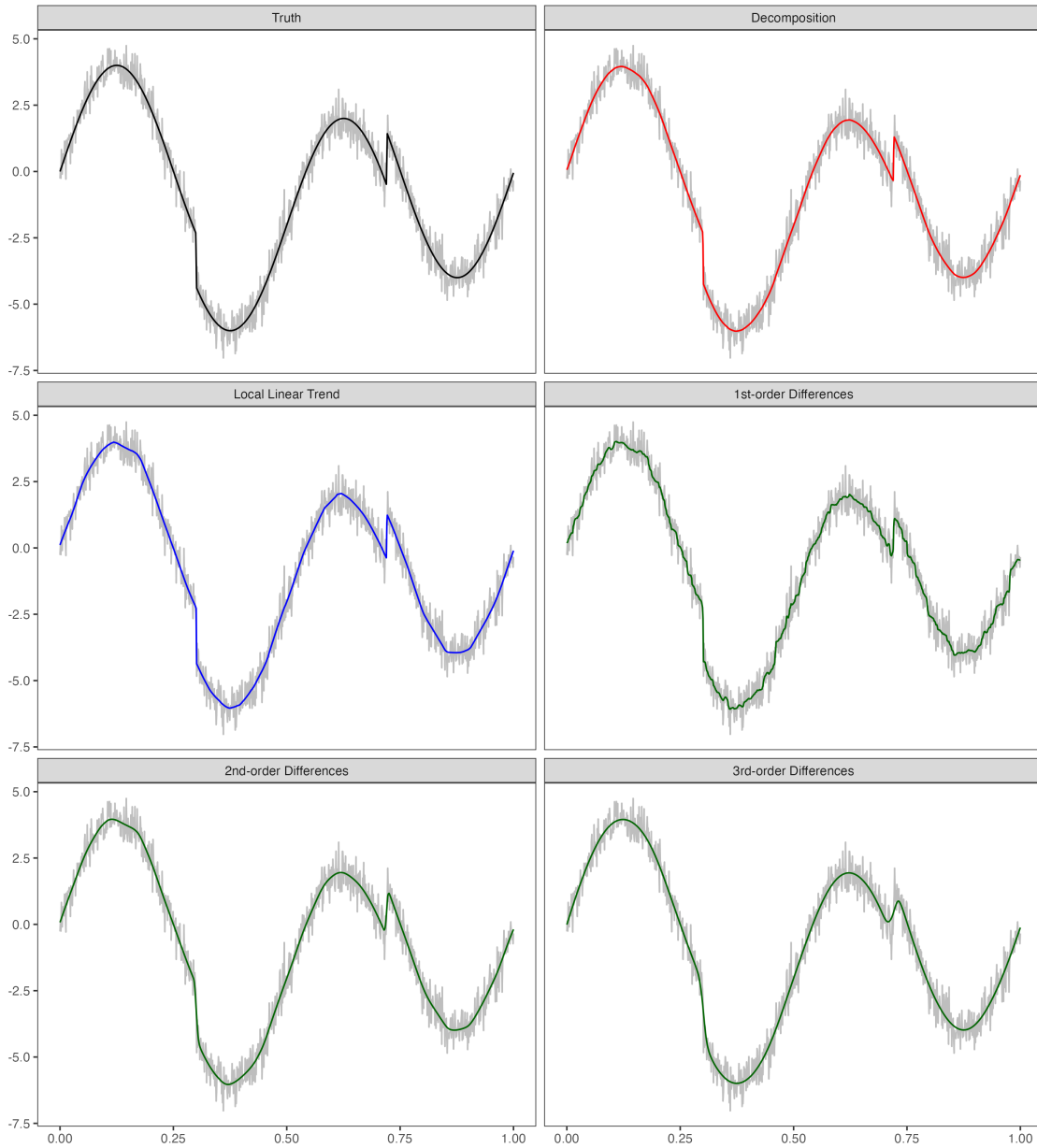


Figure 2.4: Posterior means obtained by fitting different models to noisy Heavisine data. Compared models: Cauchy decomposition, Cauchy local linear trend model, and latent Cauchy k^{th} -order differences with $k = 1, 2, 3$.

The local linear trend estimates are quite similar to the ones obtained from the 2^{nd} -order model. Both effectively recover the true function, but exhibit slight under-smoothing in the sinusoidal regions. In contrast, the 1^{st} and 3^{rd} -order models provide the poorest fits: the former is overly wiggly, while the latter excessively smooths the jumps. The decomposition

method successfully combines the strengths of the other approaches while avoiding their weaknesses. Specifically, the posterior mean adapts well to the smooth sinusoidal sections and accurately identifies the abrupt jumps. In terms of mean squared error, the decomposition method yields the best fit, followed by the local linear model, and then the 2nd-, 3rd-, and 1st-order models.

2.4 Cauchy Process Prior

In Chapter 1, we focused on estimating an unknown sequence $\theta_1, \dots, \theta_n$ from regularly spaced data points $y_1, \dots, y_n$, under the observation model

$$y_i = \theta_i + \epsilon_i \quad \text{with } \epsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2). \quad (2.6)$$

In this Section, we will take a more general perspective, and assume that $\theta_1, \dots, \theta_n$ are the values that a function $\theta(t)$, $t \geq 0$, takes at the design points $t_1 < \dots < t_n$, i.e. $\theta_i = \theta(t_i)$, for $i = 1, \dots, n$. Notice that, in this setup, the design points are not required to be regularly spaced. In the following analysis, we will always assume $t_1 = 0$. This is without loss of generality, since the observed design points can always be shifted to have origin at 0.

In order to specify priors on the infinite-dimensional space $\{\theta(t), t \geq 0\}$, we will take advantage of Cauchy processes. A Cauchy process (see e.g. [Cinlar \(2011\)](#), Chapter 7.2) is a stochastic process $\{C(t), t \geq 0\}$ having the following properties:

- $C(0) = 0$;
- the increments $C(t) - C(s)$ are independent and Cauchy distributed with location at zero and scale equal to $t - s$, for any $t > s \geq 0$.

In Chapter 1, we took advantage of the principle of maximum entropy to motivate the use of *iid* Cauchy priors on the k^{th} -order differences of the vector θ . As we move from a θ defined on a discrete space to one defined on a continuous space, we consider substituting

$$(D^{(k)}\theta)_1, \dots, (D^{(k)}\theta)_{n-k} \stackrel{\text{i.i.d.}}{\sim} C(0, V(n-k)^{-k})$$

with the prior

$$d^k\theta(t) = VdC(t)dt^{k-1}, \quad t > 0. \quad (2.7)$$

For $t = 0$, we consider an improper uniform prior on $d^k\theta(t)$. This approach parallels the relationship between Normal priors on k^{th} -order differences and Wiener processes in continuous time, as discussed in [Rue and Held \(2005\)](#) (Chapter 3) and [Lindgren and Rue \(2008\)](#). To the best of our knowledge, there have been no theoretical advancements extending the use of heavy-tailed priors to irregularly spaced design points. While [Faulkner and Minin \(2018\)](#) extend their Laplace and Horseshoe models on k^{th} -order differences to irregularly spaced design points, they do so by directly applying the theory developed for the Normal case.

In the following, we detail how the prior in (2.7) can be in practice used to perform inference for $k = 1, 2, 3$, given a discrete set of observations $y_1, \dots, y_n$. Notice that these priors can be combined to construct a version of the decomposition model (2.3) that accommodates irregularly spaced design points.

First Order Differences

Equation (2.7) leads to a very simple model for $k = 1$. Under the prior assumption

$$d\theta(t) = VdC(t)$$

the prior on the first order differences of the function $\theta(\cdot)$ is

$$\theta(t) - \theta(s) = V[C(t) - C(s)] \quad (2.8)$$

for $t > s \geq 0$. Assuming we observe some data $y_1, y_2, \dots, y_n$ at the design points $0 = t_1 < t_2 \dots < t_n$, equations (2.6) and (2.8) translate into the following simple model

$$\begin{aligned} y_i &= \theta_i + \epsilon_i, & \epsilon_i &\stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2) \\ \theta_{j+1} &= \theta_j + \eta_j, & \eta_j &\stackrel{\text{ind}}{\sim} C(0, V(t_{j+1} - t_j)) \end{aligned} \quad (2.9)$$

for $i = 1, \dots, n$, and $j = 1, \dots, n - 1$. An improper flat prior is assumed on θ_1 .

Second Order Differences

For $k = 2$, we can write (2.7) as follows

$$\begin{aligned} d\theta(t) &= \nu(t)dt \\ d\nu(t) &= VdC(t) \end{aligned} \quad (2.10)$$

for $t > 0$. For $t = 0$, we consider an improper flat prior on $\theta(t)$ and $\nu(t)$. In Appendix B.3, we prove the following Lemma.

Lemma 2.1. Let $y_1, \dots, y_n$ be observations from model (2.6) at the design points $0 = t_1 < t_2 \dots < t_n$. Assume $\theta(t)$ is endowed with the prior (2.10). Under the approximation

$$\int_s^u \nu(l)dl \approx (u - s)\nu(s),$$

posterior inference for $\theta_i := \theta(t_i)$, $i = 1, \dots, n$, can be obtained by fitting the following model

$$\begin{aligned} y_i &= \theta_i + \epsilon_i & \epsilon_i &\stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2) \\ \theta_{j+2} &= \left(1 + \frac{d_{j+1}}{d_j}\right) \theta_{j+1} - \frac{d_{j+1}}{d_j} \theta_j + V\eta_j & \eta_j &\stackrel{\text{ind}}{\sim} C(0, d_{j+1}d_j) \end{aligned} \quad (2.11)$$

with $j = 1, \dots, n - 2$, and $d_l := t_{l+1} - t_l$, for $l = 1, \dots, n - 1$.

Notice that, if the design points were equispaced, i.e. $d_l = \frac{1}{M}$ for some constant M and for every $l = 1, \dots, n-1$, then the second equation in (2.11) would boil down to $(D^{(2)}\theta)_j = \eta_j$ with $\eta_j \stackrel{\text{i.i.d.}}{\sim} C(0, \frac{V}{M^2})$. This is equivalent to (1.18) with $k = 2$.

Third Order Differences

For $k = 3$, we can write (2.7) as follows

$$\begin{aligned} d\theta(t) &= \phi(t)dt \\ d\phi(t) &= \nu(t)dt \\ d\nu(t) &= VdC(t), \end{aligned} \tag{2.12}$$

for $t > 0$. For $t = 0$, we consider an improper flat prior on $\theta(t)$, $\phi(t)$, and $\nu(t)$. In Appendix B.3, we prove the following Lemma.

Lemma 2.2. Let $y_1, \dots, y_n$ be observations from model (2.6) at the design points $0 = t_1 < t_2 < \dots < t_n$. Assume $\theta(t)$ is endowed with the prior (2.12). Under the approximation

$$\int_s^u \nu(l)dl \approx (u-s)\nu(s),$$

posterior inference for $\theta_i := \theta(t_i)$, $i = 1, \dots, n$, can be obtained by fitting the following model

$$\begin{aligned} y_i &= \theta_i + \epsilon_i & \epsilon_i &\stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2) \\ \theta_{j+3} &= (1 + b_j)\theta_{j+2} - (b_j + c_j)\theta_{j+1} + c_j\theta_j + V\eta_j \end{aligned} \tag{2.13}$$

where, for $j = 1, \dots, n-3$,

$$\begin{aligned} b_j &:= \frac{d_{j+2}}{d_{j+1}} \left(1 + \frac{d_{j+2} + d_{j+1}}{d_{j+1} + d_j} \right), \\ c_j &:= \frac{d_{j+2}}{d_j} \frac{d_{j+2} + d_{j+1}}{d_{j+1} + d_j}, \\ \eta_j &\stackrel{\text{ind}}{\sim} C \left(0, \frac{d_{j+2}^2 d_{j+1}}{2} + \frac{d_{j+2} d_j^2}{2} \frac{d_{j+2} + d_{j+1}}{d_{j+1} + d_j} \right), \end{aligned}$$

and $d_l := t_{l+1} - t_l$, for $l = 1, \dots, n-1$.

Notice that, if the design points were equispaced, i.e. $d_l = \frac{1}{M}$ for some constant M and for every $l = 1, \dots, n-1$, then the state equation in (2.13) would boil down to $(D^{(2)}\theta)_j = \eta_j$ with $\eta_j \stackrel{\text{i.i.d.}}{\sim} C(0, \frac{V}{M^3})$. This is equivalent to (1.18) with $k = 3$.

Posterior Inference

Posterior inference for the first order model can be performed via a modified version of Algorithm 1. Indeed, model (2.9) differs from model (1.18) with $k = 1$ only in terms of state error: the random variable $\eta_j \stackrel{\text{i.i.d.}}{\sim} C(0, \tau)$ of (1.18) is replaced by $\eta_j \stackrel{\text{ind}}{\sim} C(0, Vd_j)$ in (2.9), for every $j = 1, \dots, n-1$. The values of $d_j, j = 1, \dots, n-1$, are known, while V is unknown and assumed to be $V \propto 1/V$. Algorithm 1 can be swiftly adapted to account for this change.

A similar, but less straightforward, reasoning can be applied for the second order model. Notice that the second equation in (2.11) can be re-expressed as

$$\frac{1}{d_j} \left(\frac{\theta_{j+2} - \theta_{j+1}}{d_{j+1}} - \frac{\theta_{j+1} - \theta_j}{d_j} \right) \stackrel{\text{ind}}{\sim} C(0, V).$$

Thus, one could define a matrix $\tilde{D}^{(2)}$ such that

$$\left(\tilde{D}^{(2)} \theta \right)_j = \frac{1}{d_j} \left(\frac{\theta_{j+2} - \theta_{j+1}}{d_{j+1}} - \frac{\theta_{j+1} - \theta_j}{d_j} \right)$$

for every $j = 1, \dots, n-1$. Algorithm 1 with $k = 2$ could therefore be adapted to fit (2.11) if we simply replaced $D^{(2)}$ with $\tilde{D}^{(2)}$. In practice, this algorithm suffers from computational issues since $\tilde{D}^{(2)}$ is characterized by extremely large entries when some of the d_j 's are very small. This can lead to non-positive definite matrix errors during the Cholesky decomposition step in Algorithm 1. A partial fix involves scaling the matrix $\tilde{D}^{(2)}$ by an appropriately chosen constant value M . This leads to

$$\frac{1}{M} \left(\tilde{D}^{(2)} \theta \right)_j \stackrel{\text{ind}}{\sim} C\left(0, \frac{V}{M}\right).$$

Choices of M that help stabilize computations include the mean of $\frac{1}{d_j d_{j+1}}$, the mean of $\frac{1}{d_j^2}$, and the mean of $\frac{2}{d_j} \left(\frac{1}{d_{j+1}} + \frac{1}{d_j} \right)$, evaluated over $j = 1, \dots, n-1$. However, this adjustment is only a partial solution. For large datasets, the non-positive definite matrix error can still occur. Developing a stable algorithm to fit this model remains an open problem for future research. Similar computational issues also affect Gibbs sampling for the third order model (2.13).

2.5 Latent Cauchy Autoregressive Model

Model (1.18) places a Cauchy prior on the k^{th} -order differences of the vector θ . As noted in (1.20), this model can be equivalently expressed in the following form

$$\begin{aligned} y_i &= \mu + \theta_i + \epsilon_i, & \epsilon_i &\stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2) \\ \theta_{j+k} &= \sum_{l=1}^k \phi_l \theta_{j+k-l} + \eta_j, & \eta_j &\stackrel{\text{i.i.d.}}{\sim} C(0, \tau). \end{aligned} \tag{2.14}$$

In Chapter 1, $\mu = 0$, and $\phi_1, \dots, \phi_k$ were fixed values, deterministically obtained as a function of the differencing matrix $D^{(k)}$. As a natural extension, we propose to unfix $\mu, \phi_1, \dots, \phi_k$, and let them be random variables whose values will be learned from the data. This change implies moving from modeling the k^{th} -order differences of θ as a Cauchy random variable to assuming that θ has an autoregressive (AR) structure of order k with additive Cauchy noise. This modification leads to a model with even weaker assumptions on θ , and has therefore the potential for a wider applicability. We will refer to (2.14) with unknown μ and $\phi := (\phi_1, \dots, \phi_k)$ as *latent Cauchy AR(k) model*. Notice that, when $\sigma = 0$, (2.14) simplifies to the standard AR model with Cauchy-distributed errors.

In this Section, we present a modified version of the Gibbs sampler from Chapter 1 to accommodate unknown μ and ϕ . We then discuss how the new model is relevant for time series data originating from a latent process whose behavior can be, most of the time, characterized by a linear differential equation. Finally, we evaluate the performance of the proposed method on simulated datasets.

Posterior Inference

Next, we adapt the Gibbs sampler from Chapter 1 to provide posterior inference for the latent Cauchy AR(k) model. Assuming no prior information on μ and ϕ , we place an improper uniform prior on these parameters. In order to adapt the Gibbs sampler to the new scenario we need to:

- Substitute the (fixed) differencing matrix $D^{(k)}$ with the parametrized matrix $D_\phi^{(k)}$ constructed as follows

$$(D_\phi^{(k)}\theta)_j = \theta_{j+k} - \sum_{l=1}^k \phi_l \theta_{j+k-l} \quad (2.15)$$

for every $j = 1, \dots, n-k$. The parametrized matrix $D_\phi^{(k)}$ has same dimensions as $D^{(k)}$, i.e. $(n-k) \times n$.

- Derive the complete conditional distribution of ϕ and μ . As shown in Appendix B.1, these distributions are

$$p(\phi|\cdot) = N \left(\Omega \sum_{j=1}^{n-k} \frac{\theta_{j+k}}{\lambda_j^2} \theta_{(j+k-1):j}, \tau^2 \Omega \right) \quad (2.16)$$

where

$$\Omega = \left(\sum_{j=1}^{n-k} \frac{\theta_{(j+k-1):j} \theta_{(j+k-1):j}^\top}{\lambda_j^2} \right)^{-1}, \quad (2.17)$$

and

$$p(\mu|\cdot) = N \left(\frac{1}{n} \sum_{i=1}^n (y_i - \theta_i), \frac{\sigma^2}{n} \right). \quad (2.18)$$

- Adapt the complete conditional distribution of σ^2 and θ , by substituting y with $y - \mu \mathbf{1}_n$, with $\mathbf{1}_n$ being the $n \times 1$ vector of ones.

We can now modify Algorithm 1 into a Gibbs sampler for the latent Cauchy AR(k) model. The resulting algorithm is reported as Algorithm 6.

Algorithm 6: Gibbs Sampler for the latent Cauchy AR(k) model

Input: data $y = (y_1, \dots, y_n)$, and order k .

Define $D_\phi^{(k)}$ as in (2.15).

Initialize $\sigma^{2(0)}, \tau^{2(0)}, \lambda_1^{2(0)}, \dots, \lambda_{n-k}^{2(0)}, \mu^{(0)}, \phi_1^{(0)}, \dots, \phi_k^{(0)}$ to some positive constants (we let $\sigma^{2(0)} = 10$, $\tau^{2(0)} = 0.01$, $\lambda_j^{2(0)} = 1$ for $j = 1, \dots, n - k$, $\mu^{(0)} = \frac{1}{n} \sum_{i=1}^n y_i$, and $\phi^{(0)}$ equal to ϕ from Chapter 1).

For $m = 1, \dots, M$ (with M a large positive integer):

1. Let

$$\Lambda^{(m-1)} = \text{diag}(\lambda^{2(m-1)}),$$

and

$$P^{(m-1)} = \frac{1}{\sigma^{2(m-1)}} I_n + \frac{1}{\tau^{2(m-1)}} D_{\phi^{(m-1)}}^{(k)\top} \Lambda^{-1(m-1)} D_{\phi^{(m-1)}}^{(k)},$$

and perform the Cholesky factorization

$$P^{(m-1)} = LL^\top$$

2. Solve

a) $Lv = \frac{1}{\sigma^{2(m-1)}}(y - \mu^{(m-1)} \mathbf{1}_n)$

b) $L^\top a = v$

c) $L^\top b = z$, where $z \sim N(\mathbf{0}, I_n)$

3. Let

$$\theta^{(m)} = a + b,$$

$$\beta^{(m)} = D_{\phi^{(m-1)}}^{(k)} \theta^{(m)}$$

$$\Omega^{(m)} = \left(\sum_{j=1}^{n-k} \frac{\theta_{(j+k-1):j}^{(m)} \theta_{(j+k-1):j}^{(m)\top}}{\lambda_j^{2(m-1)}} \right)^{-1}$$

4. Sample

$$\begin{aligned}
\phi^{(m)} &\sim N \left(\Omega^{(m)} \sum_{j=1}^{n-k} \frac{\theta_{j+k}^{(m)}}{\lambda_j^{2(m-1)}} \theta_{(j+k-1):j}^{(m)}, \tau^{2(m-1)} \Omega^{(m)} \right) \\
\lambda_j^{2(m)} &\stackrel{\text{ind}}{\sim} IG \left(1, \frac{\beta_j^{2(m)} + \tau^{2(m-1)}}{2\tau^{2(m-1)}} \right), \quad j = 1, \dots, n-k \\
\tau^{2(m)} &\sim IG \left(\frac{n-k}{2}, \sum_{i=1}^{n-k} \frac{\beta_i^{2(m)}}{2\lambda_i^{2(m)}} \right) \\
\sigma^{2(m)} &\sim IG \left(\frac{n}{2}, \frac{\|y - \mu^{(m-1)} \mathbf{1}_n - \theta^{(m)}\|_2^2}{2} \right) \\
\mu^{(m)} &\sim N \left(\frac{1}{n} \sum_{i=1}^n (y_i - \theta_i^{(m)}), \frac{\sigma^{2(m)}}{n} \right)
\end{aligned}$$

Differential Equations with Shocks

Assume we have a time series $y_1, \dots, y_n$ consisting of noisy observations from a latent process. The latent Cauchy AR(k) model is particularly suitable for problems where the process is expected to evolve according to a specific linear differential equation *most of the time*, but where infrequent shocks, altering the expected dynamics, are allowed to happen. To understand why this model is suitable, consider the following:

- Suppose $\tilde{\theta}(t)$ is a continuous-time process that evolves according to a linear differential equation, observed at regular time points $t = 1, \dots, n$. The corresponding discrete-time representation of the differential equation can be written as

$$\tilde{\theta}_{t+k} = \phi_0 + \sum_{l=1}^k \phi_l \tilde{\theta}_{t+k-l} \quad (2.19)$$

for some $k \in \mathbb{N}$ and $\phi_0, \phi_1, \dots, \phi_k \in \mathbb{R}$. Equation (2.19) is equivalent to

$$\tilde{\theta}_{t+k} - \mu = \sum_{l=1}^k \phi_l (\tilde{\theta}_{t+k-l} - \mu)$$

as long as $\mu = \frac{\phi_0}{1 - \sum_{l=1}^k \phi_l}$ is well-defined. Letting $\theta_t = \tilde{\theta}_t - \mu$, we can describe the process

$\tilde{\theta}(t)$ in discrete time as $\theta_t + \mu$, with

$$\theta_{t+k} = \sum_{l=1}^k \phi_l \theta_{t+k-l} \quad (2.20)$$

for every $t = 1, \dots, n$.

- For small τ , the Cauchy distribution $C(0, \tau)$ is characterized by a large mass at zero and heavy tails. Therefore, adding noise $\eta_t \stackrel{\text{i.i.d.}}{\sim} C(0, \tau)$ to (2.19) means adding a random quantity that is often close to zero, but that is sometimes different, also significantly, from zero. This implies that Equation (2.19) holds approximately true most of the time, but shocks can occur, in the sense that some θ_t 's might deviate from the expected behavior, even considerably.

As a concrete example, consider the sinusoidal function $\tilde{\theta}(t)$, defined as

$$\begin{aligned} \tilde{\theta}(t) &= \theta(t) + \mu \\ \theta(t) &= \alpha_1 \cos(\omega t) + \alpha_2 \sin(\omega t) \end{aligned} \quad (2.21)$$

for some real values μ, α_1, α_2 , and ω . Notice that $\theta(t)$ can be equivalently expressed as a solution to the differential equation

$$\theta(t)'' = -\omega^2 \theta(t). \quad (2.22)$$

In discrete time, (2.22) satisfies

$$\theta_{t+2} = 2 \cos \omega \theta_{t+1} - \theta_t, \quad (2.23)$$

which is equivalent to (2.20) with $k = 2$, $\phi_1 = 2 \cos \omega$, and $\phi_2 = -1$. By adding Cauchy noise (with small scale) to (2.23), i.e. letting

$$\theta_{t+2} = 2 \cos \omega \theta_{t+1} - \theta_t + \eta_t \quad \eta_t \stackrel{\text{i.i.d.}}{\sim} C(0, \tau), \quad (2.24)$$

we can generate, and model, sequences that are approximately sinusoidal with occasional amplitude changes. Figure 2.5 shows data generated according to (2.24) with $\tau = 0.01$ and $\cos \omega = 0.83$.

The estimation of signals that satisfy differential equations has been addressed in a frequentist setting in Goldenshluger and Nemirovski (1997) and in Chapter 4 of Nemirovski (2000). Both of these references emphasize the limitations of frequentist estimators designed for ℓ_p -smooth functions (e.g. trend filtering) in recovering such signals. In simulations, we observe that our estimators for Chapter 1 also have limited applicability in this context, whereas model (2.14) shows very good performance.

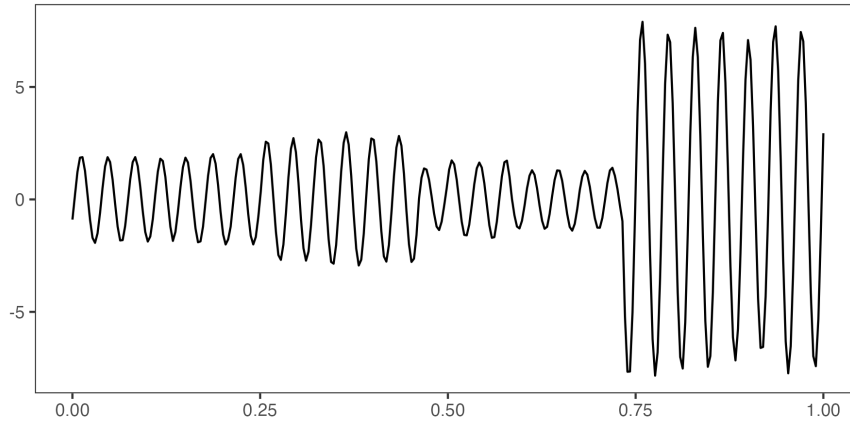


Figure 2.5: Sinusoidal with amplitude changes

Simulations

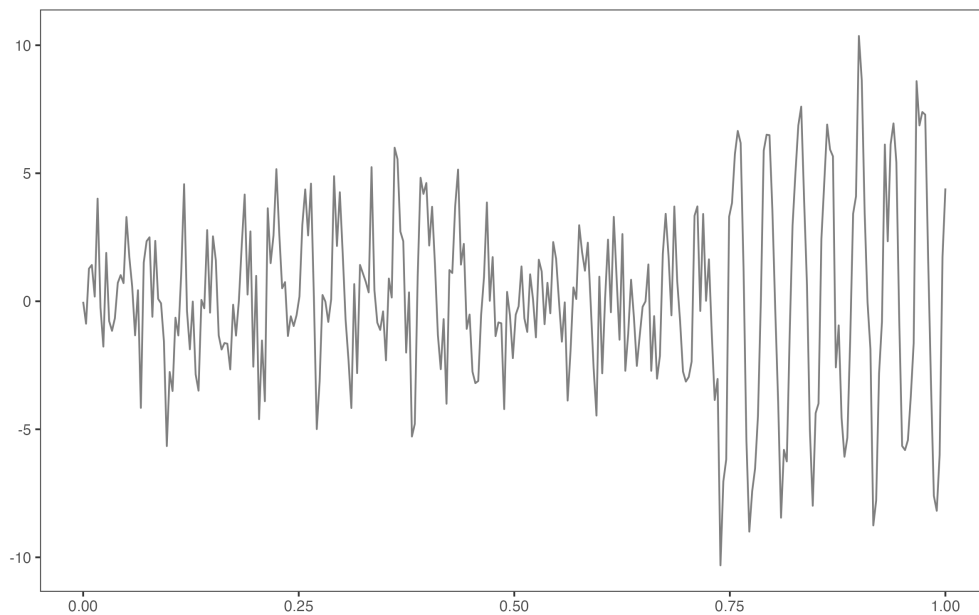
Let's go back to the data in Figure 2.5, and add i.i.d. Gaussian noise so that the signal-to-noise ratio equals 2. The resulting data are very noisy and are reported in Figure 2.6a. Fitting the model from Chapter 1 with $k = 2$, produces posterior estimates that are nearly flat, failing to capture the underlying signal. Increasing k to 3 improves the recovery of the signal, though it remains less accurate compared to the latent Cauchy AR(2) model. The posterior estimates from these methods are illustrated in Figure 2.6b. Using the latent Cauchy AR(2) model, the posterior means for ϕ_1, ϕ_2 , and σ^2 approximate the true values of 1.66, -1, and 2.31 with remarkable precision, deviating by less than 10^{-3} . The corresponding posterior standard deviations are 0.005, 0.005, and 0.204, respectively. The posterior mean for μ (0.15, with a standard deviation of 0.088) is also close to the true value of 0. The latent Cauchy AR(2) model achieves a 99% improvement in MSE compared to the 2^{nd} -order differencing model, and an 83% improvement compared to the 3^{rd} -order differencing model.

We also consider data generated by adding Gaussian noise to the following AR(2) process with shocks

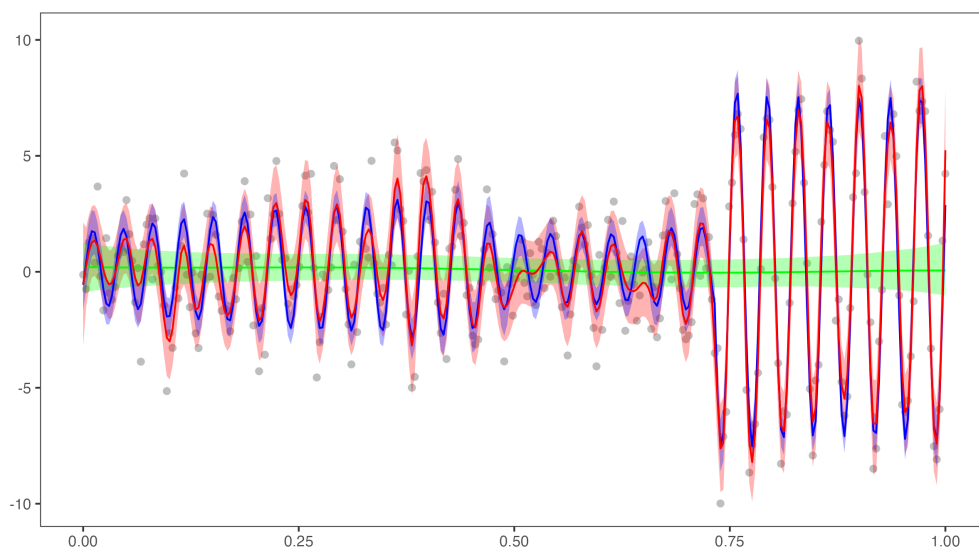
$$\theta_{t+2} = 1.38\theta_{t+1} - 0.69\theta_t + \begin{cases} 1 & \text{if } t = 0.5, \\ -3 & \text{if } t = 0.75, \\ 0 & \text{otherwise.} \end{cases} \quad (2.25)$$

Process (2.25) is displayed at the top of Figure 2.7. Notice that, in this case, the shocks are not generated by adding Cauchy random variables to the AR(2) process, but by including two deterministic breaks. We are interested in understanding whether model (2.14) can adapt to such sparse, non-Cauchy shocks. The noisy data (signal-to-noise ratio = 2) and posterior means obtained by the latent Cauchy AR(2) model and by the latent Cauchy 2^{nd} - and 3^{rd} -order differences are shown in the bottom two plots of Figure 2.7.

Both models from Chapter 1 exhibit over-smoothing and fail to capture the small oscilla-



(a) Noisy data from sinusoidal with amplitude changes



Latent Cauchy Model AR(2) Order-2 Diff Order-3 Diff

(b) Posterior means and 95% credible intervals of three different methods: latent Cauchy AR(2), latent Cauchy 2^{nd} - and 3^{rd} -order differences.

Figure 2.6: Cauchy estimation for noisy data from sinusoidal with amplitude changes

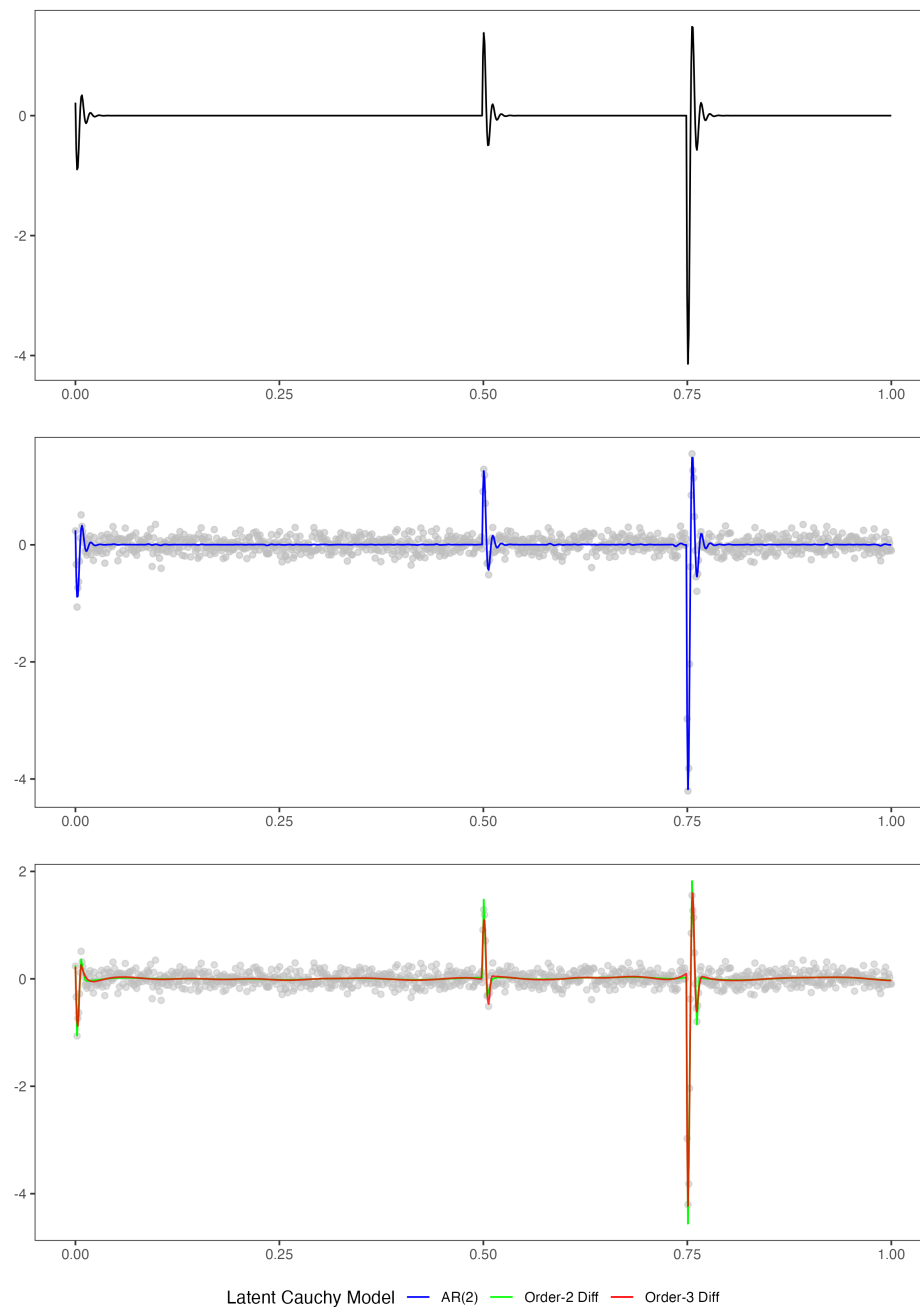


Figure 2.7: From top to bottom: truth, posterior means from the latent Cauchy AR(2) model, and posterior means from the latent Cauchy 2nd- and 3rd-order differences model

tions around the prominent peaks. On the other hand, the latent Cauchy AR(2) model effectively adapts to these variations. Additionally, its posterior estimates for the AR parameters are highly accurate, with posterior means for ϕ_1 and ϕ_2 of 1.39 and -0.68, and corresponding standard deviations of 0.004 and 0.007. This model achieves a 92% improvement in MSE compared to the 2nd-order differencing model, and an 88% improvement compared to the 3rd-order differencing model.

To conclude, we consider a scenario where the signal f to be recovered is an *amplitude modulation* of an ℓ_p smooth function g , defined as

$$f(t) = g(t) \sin(\omega t + c),$$

and the frequency ω is large. This function class is described in Chapter 4 of [Nemirovski \(2000\)](#). We construct f by setting g to be the *Hills* function from Chapter 1, and by letting $\omega = 600$ and $c = 0$. Noisy data are generated by adding Gaussian noise (signal-to-noise ratio = 2) to f evaluated at 1000 points in the interval $[0, 1]$. We fit the the latent Cauchy AR(2) and AR(3) models, as well as the 2nd- and 3rd-order differences models.

Figure 2.8 shows f , the noisy data, and the posterior means from the four models. Notably, both the differencing methods heavily over-smooth the data. In contrast, the AR models provide fits that closely match the true function. Interestingly, increasing the AR model order from 2 to 3 does not lead to any noticeable improvement in the fit, with the AR(2) model achieving an 8% reduction in MSE compared to the AR(3) model. The improvement in MSE relative to the differencing methods is far more pronounced: 98% in comparison to the 2nd-order model, and 95% in comparison to the 3rd-order model.

2.6 Higher Dimensions

In Chapter 1, we assumed that each observation y_i , $i = 1, \dots, n$, in (1.18), has dimension equal to one. A possible extension of (1.18) would be to adapt the model to handle observations with higher dimension. In particular, the extension to bi-dimensional observations would be relevant for applications such as image denoising, where one is interested in removing noise from an image, which might be characterized by sharp edges and abrupt color changes, or by smooth color transitions. An extension of (1.18) to two dimensions would seem suitable to the purpose. We will next introduce a bi-dimensional extension of (1.18) with $k = 1$, discuss how to perform posterior inference, and demonstrate the method's performance on two simulated datasets.

A 2-Dimensional Model

Consider an $m \times n$ regular grid of points, indexed by $i = 1, \dots, m$ and $j = 1, \dots, n$. Assume that for each ij -point we have a corresponding observation y_{ij} . We will consider the following

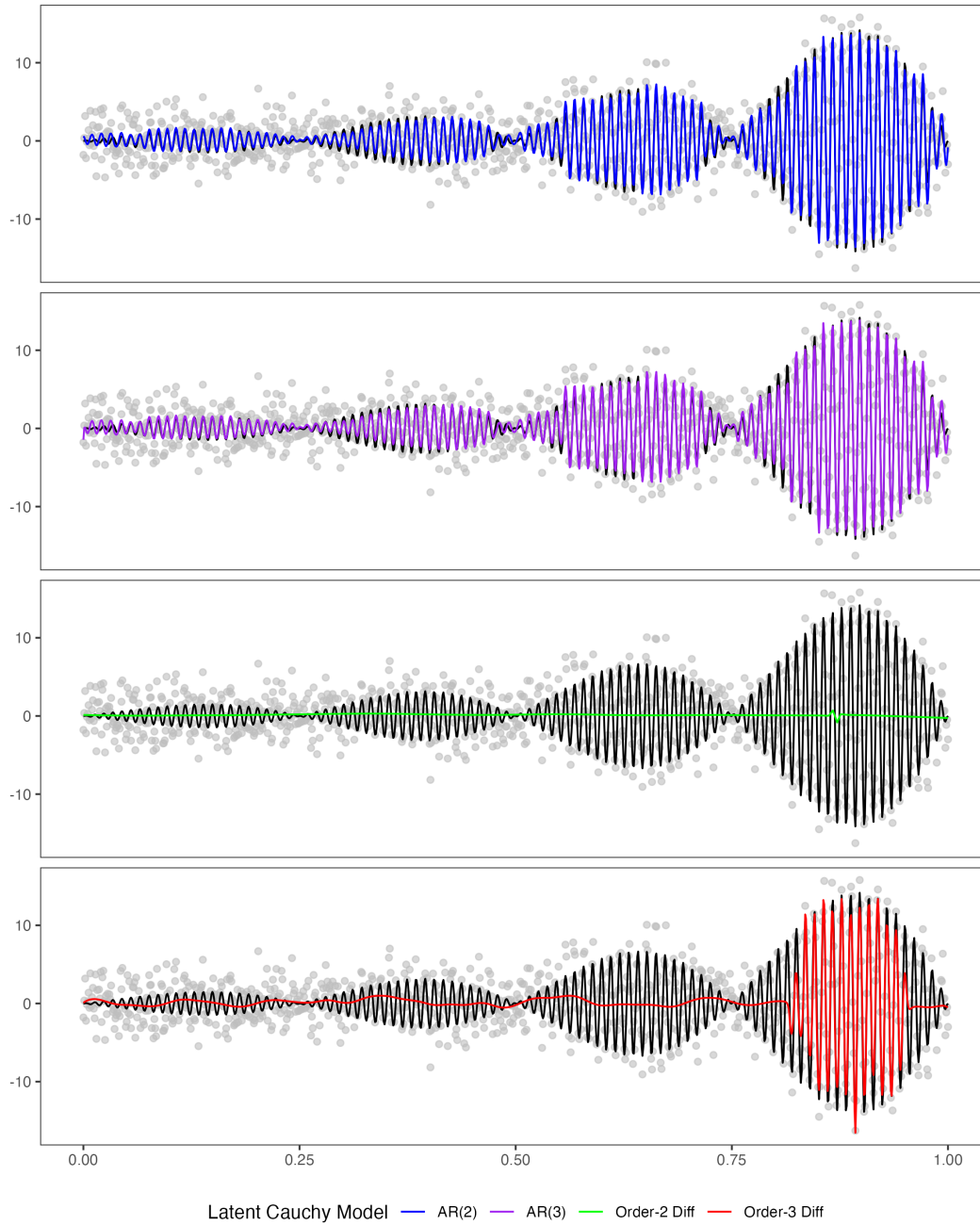


Figure 2.8: Amplitude modulated signal (black), noisy data (gray), and posterior means from the latent Cauchy AR(2), AR(3), 2^{nd} -, and 3^{rd} -order differences model

model:

$$\begin{aligned} y_{ij} &= \theta_{ij} + \epsilon_{ij}, & \epsilon_{ij} &\stackrel{\text{i.i.d}}{\sim} N(0, \sigma^2) \\ \Delta\theta_{ij} &= \eta_{ij}, & \eta_{ij} &\stackrel{\text{i.i.d}}{\sim} C(0, \tau) \end{aligned} \quad (2.26)$$

where $\Delta\theta_{ij}$ is defined as

$$\begin{aligned} \theta_{i,j} - \theta_{i-1,j-1} + \theta_{i,j-1} + \theta_{i-1,j} & \text{ if } i, j > 1 \\ \theta_{i,j} - \theta_{i,j-1} & \text{ if } i = 1, j > 1 \\ \theta_{i,j} - \theta_{i-1,j} & \text{ if } i > 1, j = 1. \end{aligned} \quad (2.27)$$

If both i and j equal to 1, then we will let $\Delta\theta_{ij} = \theta_{ij}$ and consider a flat prior on it instead of the Cauchy prior in (2.26). This model extends (1.18) with order $k = 1$ to the case of two-dimensional observations. We expect this model to be suitable for denoising images with sharp edges and abrupt color changes.

A frequentist analogue of (2.26) is the Hardy-Krause denoising method introduced by Fang et al. (2021). Various definitions of the differencing operator $\Delta\theta$, which lead to different estimators, have also been explored in the literature. In the context of image denoising, common differencing operators include $\Delta_1\theta_{ij} = \theta_{i+1,j} - \theta_{ij}$ and $\Delta_2\theta_{ij} = \theta_{i,j+1} - \theta_{ij}$. For textbook treatments of this topic, see Li (2009) and Winkler (2012). The prior in (2.26), up to potential boundary differences, has been explored by Suuronen et al. (2022) and Chada et al. (2019), where it is referred to as the *two-dimensional Cauchy sheet*. While Chada et al. (2019) establishes theoretical guarantees for this prior, Suuronen et al. (2022) provides a review of various Cauchy-based methods, including (2.26), and discusses algorithms for posterior inference in inverse problems. Our approach to inference for (2.26) diverges from that of Suuronen et al. (2022) in how the scale parameter τ is handled. While Suuronen et al. (2022) treat τ as a user-defined parameter, we treat it as an unknown parameter and perform inference to estimate it.

Posterior Inference

Posterior inference for (2.26) can be performed using a Gibbs sampler that boils down to a modified version of Algorithm 1. Let $y = (y_{11}, y_{12}, \dots, y_{1n}, y_{21}, \dots, y_{mn})$ and $\theta = (\theta_{11}, \theta_{12}, \dots, \theta_{1n}, \theta_{21}, \dots, \theta_{mn})$, and define the matrix D such that, for every $i = 1, \dots, m$ and $j = 1, \dots, n$,

$$(D\theta)_{ij} = \Delta\theta_{ij}.$$

Then, Algorithm 1 with $k = 1$ can be applied to fit (2.26) by simply replacing $D^{(1)}$ with D . The new matrix D has a structure similar to $D^{(1)}$: it is sparse and banded, with bandwidth equal to 2, but also includes an additional subdiagonal band of the same width. Therefore, the precision matrix of the full conditional distribution of θ remains sparse but exhibits three bands: one on the main diagonal, one above it, and one below it, each with a width of 3.

Simulations

We will apply (2.26) to two simulated datasets. In both cases, we consider a 20×20 grid of points. Data are generated by adding i.i.d. Gaussian noise with standard deviation equal to 0.7 to two different 20×20 matrices. The first matrix is always equal to 0, except for its top-left 10×10 sub-matrix, that is equal to -1. The second matrix has a structure of alternating rectangles having constant values equal to -2, -1, 0 or 1. The original matrices, their noisy versions, and the posterior means obtained by fitting (2.26) are reported in Figure 2.9. The posterior estimates are able to accurately reconstruct the original matrices.

2.7 Considerations for Model Choice

In Chapters 1 and 2, we introduced a set of Cauchy-based methods and algorithms designed for applications where the goal is to estimate a signal from noisy data without making restrictive assumptions about the signal's structure.

The latent Cauchy k^{th} -order differences model (1.18) from Chapter 1 is particularly well-suited for scenarios where the signal exhibits a piecewise constant ($k = 1$), piecewise linear ($k = 2$), or piecewise quadratic ($k = 3$) structure. This model strikes a balance between flexibility and simplicity by focusing on specific structural patterns in the signal.

In contrast, the Cauchy local linear trend model (2.1), the Cauchy decomposition (2.3), and the latent Cauchy AR (2.14) model discussed in Chapter 2 offer greater flexibility by further relaxing the smoothness assumptions on the signal. These models are particularly useful when the signal is expected to exhibit more complex or irregular behavior, albeit at the expense of introducing additional parameters.

Practitioners choosing between these models should consider how much flexibility needs to be built into the model. In situations where a piecewise polynomial structure is anticipated, model (1.18) should be preferred due to its computational simplicity and efficiency. Opting for more complex models from Chapter 2 in such cases may lead to unnecessary complexity and a slight loss of precision. Conversely, when the signal exhibits more intricate or irregular behavior that cannot be captured by simpler assumptions, the advanced models provide greater adaptability and can effectively uncover nuanced patterns in the data.

An important avenue for future research is the development of systematic model selection approaches. These methods could streamline the decision-making process by automatically identifying the model with the strongest empirical support, thereby helping practitioners balance complexity and performance without manual intervention.

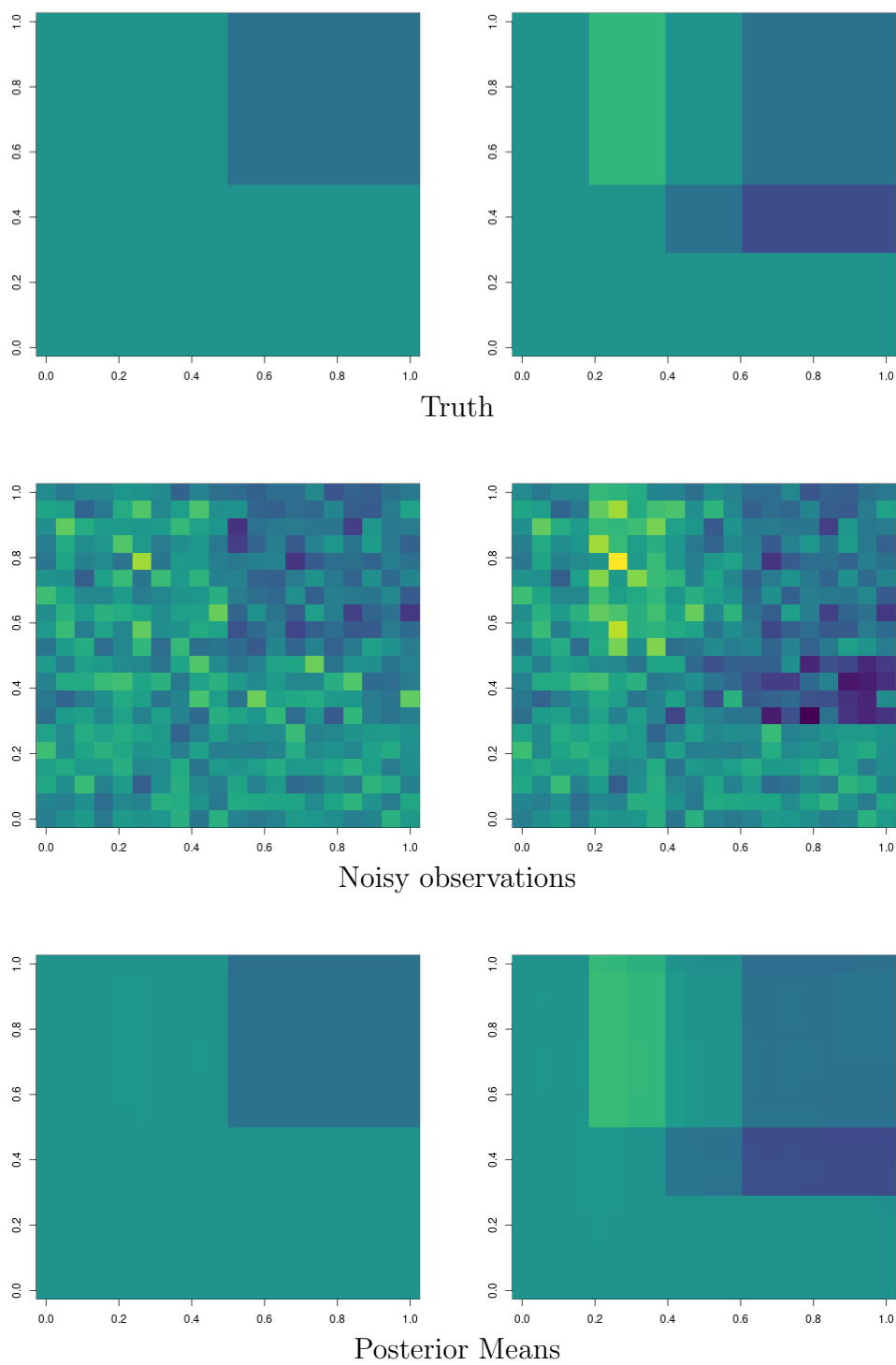


Figure 2.9: Image denoising through (2.26)

Chapter 3

Theoretical Analysis for the Normal Means Model

3.1 Introduction

The goal of this Chapter is to assess the frequentist properties of the Bayesian methods from Chapter 1. Let $\hat{\theta}_1(k), \dots, \hat{\theta}_n(k)$ denote the posterior mean estimator for $\theta_1, \dots, \theta_n$ based on our Cauchy model on k^{th} -order differences (1.18). We would like to prove that the risk

$$\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \left(\hat{\theta}_i(k) - \theta_i \right)^2 \right]$$

is small and comparable to the risk achievable by other related estimators. Here, the expectation is taken with respect to the data $y_1, \dots, y_n$. To make this concrete, consider the two classes:

$$\Theta_1 = \left\{ \theta \in \mathbb{R}^n : \sum_j |(D^{(k)}\theta)_j| \leq Vn^{1-k} \right\} \text{ and } \Theta_0 = \left\{ \theta \in \mathbb{R}^n : \sum_j \mathbb{1}\{(D^{(k)}\theta)_j \neq 0\} \leq q \right\}.$$

Under the squared error loss:

$$L(\tilde{\theta}, \theta) = \frac{1}{n} \sum_{i=1}^n \left(\tilde{\theta}_i - \theta_i \right)^2,$$

the minimax rates for Θ_1 and Θ_0 are given by $r_n(\Theta_1) := n^{-2k/(2k+1)}$ and $r_n(\Theta_0) := (\log n)/n$, respectively (with V and q treated as fixed constants). These minimax rates are established in Tibshirani (2014), Guntuboyina et al. (2020), and Raskutti et al. (2011).

Our conjecture for the Cauchy prior-based estimator is that for $p \in \{0, 1\}$:

$$\sup_{\theta \in \Theta_p} \mathbb{E} L \left(\hat{\theta}(k), \theta \right) \lesssim r_n(\Theta_p) \tag{3.1}$$

where $\lesssim$ hides multiplicative constants that can depend on V , q and σ .

A simplified version of (3.1) is the following: let $\hat{\theta}(k, \tau, \sigma)$ denote the posterior mean under the Cauchy prior when τ and σ are treated as fixed, known values. We conjecture that there exist a value of τ (say, τ^*) for which:

$$\sup_{\theta \in \Theta_p} \mathbb{E}L\left(\hat{\theta}(k, \tau^*, \sigma), \theta\right) \lesssim r_n(\Theta_p). \quad (3.2)$$

The conjecture (3.2) should be easier to prove compared to (3.1) because the value of τ^* can be chosen based on knowledge of θ .

In this Chapter, we take a first step towards these conjectures by considering $k = 0$ and analyzing (3.2) for $p = 0$, i.e., for the class Θ_0 when $k = 0$. This specific case has already been addressed in the literature (Ghosh and Chakrabarti (2017)), where results are provided for a more general class of priors, including the Cauchy as a special case. For the Cauchy prior, the marginal distribution of y_i is given by the Voigt profile, for which many analytical properties are known. We shall provide an alternative proof of (3.2) for $p = 0$ and $k = 0$ using properties of the Voigt profile. We leave the proof of (3.1) for $p = 0$ and higher values of k as a subject for future work.

3.2 The Sparse Normal Means Model

Setting $k = 0$ in model (1.18) and assuming $\theta \in \Theta_0$ boils down to assuming a Cauchy prior within the context of the *sparse Normal means* model. This model assumes that, for $i = 1, \dots, n$,

$$y_i = \theta_i + \epsilon_i, \quad \epsilon_i \stackrel{\text{i.i.d}}{\sim} N(0, \sigma^2), \quad (3.3)$$

and that the fraction of non-zero θ_i 's is small and tends to zero as $n \rightarrow \infty$. More precisely, it assumes that the vector $\theta = (\theta_1, \dots, \theta_n)$ belongs to

$$\ell_0[q_n] := \left\{ \theta \in \mathbb{R}^n : \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{\theta_i \neq 0\} \leq q_n \right\}. \quad (3.4)$$

Vectors belonging to $\ell_0[q_n]$ are referred to as *nearly black* signals, while the combination of (3.3) and (3.4) defines the *sparse Normal means* model.

This model finds application in numerous fields where the signal of interest is high-dimensional but predominantly sparse. In such cases, we observe the signal once, corrupted by noise. Applications include image denoising, where the goal is to recover a clean image from noisy pixel data, and gene expression studies, where the aim is to identify a small subset of genes with significant activity among a vast number of possibilities.

From a Bayesian perspective, we want to choose a prior on θ that is appropriate given the characteristics of the problem considered. Two types of priors are commonly used for the sparse Normal means model:

- Two-groups priors, i.e. two-component mixtures of a distribution $\delta_{\{0\}}$ degenerate at 0 and an absolutely continuous distribution F over $\mathbb{R}$, leading to

$$\theta_i \stackrel{\text{i.i.d.}}{\sim} (1 - w)\delta_{\{0\}} + wF. \quad (3.5)$$

- One-group shrinkage priors, absolutely continuous distributions over $\mathbb{R}$, characterized by a large mass around 0 and heavy tails. Most priors in this class can be expressed as scale-mixtures on Normals

$$\begin{aligned} \theta_i \mid \lambda_i^2, \tau^2 &\stackrel{\text{ind}}{\sim} N(0, \lambda_i^2 \tau^2) \\ \lambda_i^2 &\stackrel{\text{i.i.d.}}{\sim} \pi_1 \\ \tau^2 &\sim \pi_2. \end{aligned} \quad (3.6)$$

Johnstone and Silverman (2004) provide theoretical guarantees for two-groups priors when F is a heavy-tailed distribution and $\sigma^2 = 1$. Despite the theoretical appeal of such priors, the discrete nature of the $\delta_{\{0\}}$ component poses computational challenges in more complicated models, such as (1.18) for $k \geq 1$. Many one-group shrinkage priors of the form (3.6) have been considered in the literature. Some examples include: the Horseshoe prior (Carvalho et al. (2009)), the Laplace prior (Park and Casella (2008)), and the generalized double Pareto prior (Armagan et al. (2013)). The Cauchy prior also belongs to this family. The parameter τ in these priors is often referred to as the *global* shrinkage parameter, as it captures the overall level of sparsity in the data, while $\lambda_1^2, \dots, \lambda_n^2$ are *local* shrinkage parameters, and they allow large signals to escape the global pull to zero induced by a small τ^2 . Theoretical guarantees for this family of priors, under the assumption $\sigma^2 = 1$, have been analyzed by Ghosh and Chakrabarti (2017), and by Paul and Chakrabarti (2022). Ghosh and Chakrabarti (2017) show that Bayes estimates based on a large class of one-group shrinkage priors are asymptotically minimax up to some multiplicative constant, as long as the global parameter τ^2 is chosen appropriately given the sparsity level, which is unknown in practice. Moreover, they prove that the multiplicative constant is equal to 1 for a subclass of *horseshoe-type* priors. Priors in this subclass can be expressed as (3.6) with $\lambda_i^2 \propto (\lambda_i^2)^{-\frac{3}{2}} L(\lambda_i^2)$, and $L : (0, \infty) \rightarrow (0, \infty)$ a measurable, non-constant, slowly varying function satisfying $\lim_{x \rightarrow \infty} L(\alpha x)/L(x) = 1$, for any fixed $\alpha > 0$. The authors also show that, when the sparsity level is unknown, an empirical Bayes version of the estimator is minimax up to a multiplicative constant. Their work extends the study of van der Pas et al. (2014) on minimaxity of the Horseshoe prior. Further theoretical analysis on posterior contraction rates and uncertainty quantification for the Horseshoe prior is provided by van der Pas et al. (2017a) and van der Pas et al. (2017b).

In this Chapter, we will provide an alternative proof of the minimax result from Ghosh and Chakrabarti (2017), specializing to one distribution belonging to the horseshoe-type family: the Cauchy prior.

3.3 Theoretical Guarantees for Cauchy Prior

Let's consider the normal means model (3.3) with Cauchy prior

$$\begin{aligned} y_i &= \theta_i + \epsilon_i, & \epsilon_i &\stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2) \\ \theta_i &\stackrel{\text{i.i.d.}}{\sim} C(0, \tau). \end{aligned} \quad (3.7)$$

Notice that, for every $i = 1, \dots, n$, the marginal likelihood of y_i is given by the convolution of a Normal and a Cauchy distribution, and is therefore equal to the Voigt profile

$$p(y_i) = \int N(y_i; \theta_i, \sigma^2) C(\theta_i; 0, \tau) d\theta_i = V(y_i; \sigma, \tau).$$

Using Tweedie's formula, the posterior means for model (3.7) can be obtained as

$$\mathbb{E}(\theta_i \mid y_i) = y_i + \sigma^2 \frac{p'(y_i)}{p(y_i)}. \quad (3.8)$$

As described in Appendix A.2, the Voigt profile can be expressed in terms of the real part of the Faddeeva function (see Equation A.3). Moreover, Lemma A.1 provides an expression of the first order derivative of the Voigt profile in terms of the real and imaginary part of the Faddeeva function (see Equation A.6). Plugging (A.3) and (A.6) into (3.8), we obtain the following expression for the posterior means

$$\mathbb{E}(\theta_i \mid y_i) = \tau \frac{L\left(\frac{y_i}{\sqrt{2}\sigma}, \frac{\tau}{\sqrt{2}\sigma}\right)}{K\left(\frac{y_i}{\sqrt{2}\sigma}, \frac{\tau}{\sqrt{2}\sigma}\right)}, \quad (3.9)$$

where K and L are, respectively, the real and imaginary parts of the Faddeeva function (see Appendix A.2 for the definitions).

Supporting Lemmas

We will next introduce a Lemma providing bounds on the absolute value of (3.9). The proof takes advantage of characteristics of the real and imaginary part of the Faddeeva function.

Lemma 3.1. Consider model (3.7). Let $\Gamma(a, b)$ denote the upper incomplete gamma function, defined as $\Gamma(a, b) := \int_b^\infty e^{-t} t^{a-1} dt$. For each fixed $y_i \in \mathbb{R}$, and $\tau > 0$, we have

$$|\mathbb{E}(\theta_i \mid y_i)| \leq y_i \quad (3.10)$$

and

$$\mathbb{E}(\theta_i \mid y_i) \leq \sqrt{2} y_i e^{\frac{y_i^2}{2\sigma^2}} \frac{\tau}{\sigma} \left[\frac{e^{-\frac{\tau^2}{2\sigma^2}}}{\Gamma\left(\frac{1}{2}, \frac{\tau^2}{2\sigma^2}\right)} - \frac{\tau}{\sqrt{2}\sigma} \right] \quad (3.11)$$

for every $i = 1, \dots, n$.

Proof. To prove Lemma 3.1 we are going to use Equations (47) and (48) from Haubold and John (1979) which give an alternative expression for the real and imaginary part of the Faddeeva function as

$$K(t, u) = \frac{u}{\sqrt{\pi}} e^{-t^2} \int_0^\infty \frac{e^{-\alpha u^2}}{\sqrt{1+\alpha}} \exp\left(\frac{t^2}{1+\alpha}\right) d\alpha$$

and

$$L(t, u) = \frac{t}{\sqrt{\pi}} e^{-t^2} \int_0^\infty \frac{e^{-\alpha u^2}}{(1+\alpha)^{3/2}} \exp\left(\frac{t^2}{1+\alpha}\right) d\alpha.$$

Letting $z = 1/(1+\alpha)$, we obtain:

$$K(t, u) = \frac{u}{\sqrt{\pi}} e^{-t^2+u^2} \int_0^1 e^{zt^2-\frac{u^2}{z}} z^{-\frac{3}{2}} dz$$

and

$$L(t, u) = \frac{t}{\sqrt{\pi}} e^{-t^2+u^2} \int_0^1 e^{zt^2-\frac{u^2}{z}} z^{-\frac{1}{2}} dz.$$

It follows that the posterior mean $\mathbb{E}(\theta_i | y_i)$ can be re-expressed as

$$\tau \frac{L(\frac{y_i}{\sqrt{2}\sigma}, \frac{\tau}{\sqrt{2}\sigma})}{K(\frac{y_i}{\sqrt{2}\sigma}, \frac{\tau}{\sqrt{2}\sigma})} = y_i \frac{\int_0^1 e^{z\frac{y_i^2}{2\sigma^2}-\frac{\tau^2}{2\sigma^2 z}} z^{-\frac{1}{2}} dz}{\int_0^1 e^{z\frac{y_i^2}{2\sigma^2}-\frac{\tau^2}{2\sigma^2 z}} z^{-\frac{3}{2}} dz}. \quad (3.12)$$

Notice that the ratio on the right hand side of (3.12) is at most equal to 1 since $z \in (0, 1)$, and therefore

$$|\mathbb{E}(\theta_i | y_i)| \leq |y_i|.$$

We will now obtain the second bound in Lemma 3.1 by first considering the case $y_i > 0$, and noticing that (3.12) implies

$$\mathbb{E}(\theta_i | y_i) \leq y_i e^{\frac{y_i^2}{2\sigma^2}} \frac{\int_0^1 e^{-\frac{\tau^2}{2\sigma^2 z}} z^{-\frac{1}{2}} dz}{\int_0^1 e^{-\frac{\tau^2}{2\sigma^2 z}} z^{-\frac{3}{2}} dz}. \quad (3.13)$$

The argument in the denominator on the right-hand side of (3.13) is proportional to the distribution of an inverse gamma variable with shape $\frac{1}{2}$ and scale $\frac{\tau^2}{2\sigma^2}$. Let $\Gamma(a, b)$ be the upper incomplete gamma function, and $\Gamma(a)$ be the incomplete gamma function, respectively defined as

$$\Gamma(a, b) := \int_b^\infty e^{-t} t^{a-1} dt \quad \text{and} \quad \Gamma(a) := \Gamma(a, 0).$$

Using the fact that the CDF $F(x; \alpha, \beta)$ of an inverse gamma variable with parameters α and β is equal to $\frac{\Gamma(\alpha, \frac{\beta}{x})}{\Gamma(\alpha)}$, we re-write the denominator on the right-hand side of (3.13) as

$$\begin{aligned} \int_0^1 e^{-\frac{\tau^2}{2\sigma^2 z}} z^{-\frac{3}{2}} dz &= \frac{\sqrt{2}\sigma\Gamma(\frac{1}{2})}{\tau} \int_0^1 \frac{\tau}{\sqrt{2}\sigma\Gamma(\frac{1}{2})} z^{-\frac{1}{2}-1} e^{-\frac{\tau^2}{2\sigma^2 z}} dz \\ &= \frac{\sqrt{2}\sigma\Gamma(\frac{1}{2})}{\tau} \frac{\Gamma(\frac{1}{2}, \frac{\tau^2}{2\sigma^2})}{\Gamma(\frac{1}{2})} \\ &= \frac{\sqrt{2}\sigma}{\tau} \Gamma\left(\frac{1}{2}, \frac{\tau^2}{2\sigma^2}\right). \end{aligned} \quad (3.14)$$

For the numerator on the right-hand side of (3.13), we use the change of variable $t = \frac{\tau^2}{2\sigma^2 z}$ and the definition of incomplete gamma function to obtain

$$\begin{aligned} \int_0^1 e^{-\frac{\tau^2}{2\sigma^2 z}} z^{-\frac{1}{2}} dz &= \frac{\tau}{\sqrt{2}\sigma} \int_{\tau^2/2\sigma^2}^{\infty} e^{-t} t^{-\frac{3}{2}} dt \\ &= \frac{\tau}{\sqrt{2}\sigma} \Gamma\left(-\frac{1}{2}, \frac{\tau^2}{2\sigma^2}\right). \end{aligned} \quad (3.15)$$

The following recurrence relation can be obtained for the upper incomplete gamma function using integration by part

$$\Gamma(a, b) = (a-1)\Gamma(a-1, b) + b^{a-1}e^{-b}. \quad (3.16)$$

Setting $a = \frac{1}{2}$ and $b = \frac{\tau^2}{2\sigma^2}$ in (3.16), and reordering, we obtain

$$\Gamma\left(-\frac{1}{2}, \frac{\tau^2}{2\sigma^2}\right) = 2\frac{\sqrt{2}\sigma}{\tau} e^{-\frac{\tau^2}{2\sigma^2}} - 2\Gamma\left(\frac{1}{2}, \frac{\tau^2}{2\sigma^2}\right),$$

which can be used to re-express (3.15) as

$$\int_0^1 e^{-\frac{\tau^2}{2\sigma^2 z}} z^{-\frac{1}{2}} dz = 2 \left[e^{-\frac{\tau^2}{2\sigma^2}} - \frac{\tau}{\sqrt{2}\sigma} \Gamma\left(\frac{1}{2}, \frac{\tau^2}{2\sigma^2}\right) \right]. \quad (3.17)$$

Plugging (3.14) and (3.17) into (3.13), we get

$$\begin{aligned} \mathbb{E}(\theta_i | y_i) &\leq y_i e^{\frac{y_i^2}{2\sigma^2}} \frac{2\tau}{\sqrt{2}\sigma} \frac{e^{-\frac{\tau^2}{2\sigma^2}} - \frac{\tau}{\sqrt{2}\sigma} \Gamma\left(\frac{1}{2}, \frac{\tau^2}{2\sigma^2}\right)}{\Gamma\left(\frac{1}{2}, \frac{\tau^2}{2\sigma^2}\right)} \\ &= \sqrt{2} y_i e^{\frac{y_i^2}{2\sigma^2}} \frac{\tau}{\sigma} \left[\frac{e^{-\frac{\tau^2}{2\sigma^2}}}{\Gamma\left(\frac{1}{2}, \frac{\tau^2}{2\sigma^2}\right)} - \frac{\tau}{\sqrt{2}\sigma} \right]. \end{aligned} \quad (3.18)$$

The bound for the case $y_i \leq 0$ can be derived analogously. $\square$

Bound (3.11) implies that, for small τ ,

$$|\mathbb{E}(\theta_i | y_i)| \lesssim \sqrt{\frac{2}{\pi}} y_i e^{\frac{y_i^2}{2\sigma^2}} \frac{\tau}{\sigma} \quad (3.19)$$

since, as $\tau \downarrow 0$,

$$\left[\frac{e^{-\frac{\tau^2}{2\sigma^2}}}{\Gamma\left(\frac{1}{2}, \frac{\tau^2}{2\sigma^2}\right)} - \frac{\tau}{\sqrt{2}\sigma} \right] \rightarrow \frac{1}{\Gamma(\frac{1}{2})} = \sqrt{\frac{1}{\pi}}.$$

We will next adapt Lemma 3 of [Ghosh and Chakrabarti \(2017\)](#), specializing it to the Cauchy prior and allowing the observation variance σ^2 to have a value that is possibly different from 1.

Lemma 3.2. Consider model (3.7). For each fixed $\tau \in (0, 1)$, $c > 1$, and every $i = 1, \dots, n$,

$$|\mathbb{E}(\theta_i | y_i) - y_i| \leq h(y_i, \tau) \quad (3.20)$$

where $h(\cdot, \tau)$ is a non-negative real-valued function such that, for any $\rho > c$,

$$\lim_{\tau \downarrow 0} \sup_{|y_i| > \sqrt{-2\rho\sigma^2 \log \tau}} h(y_i, \tau) = 0. \quad (3.21)$$

Proof. Using (3.12), we obtain

$$|\mathbb{E}(\theta_i | y_i) - y_i| = \left| y_i \left(\frac{\int_0^1 e^{z \frac{y_i^2}{2\sigma^2} - \frac{\tau^2}{2\sigma^2 z}} z^{-\frac{1}{2}} dz}{\int_0^1 e^{z \frac{y_i^2}{2\sigma^2} - \frac{\tau^2}{2\sigma^2 z}} z^{-\frac{3}{2}} dz} - 1 \right) \right|$$

Letting $t = 1 - z$, the equation above becomes

$$|\mathbb{E}(\theta_i | y_i) - y_i| = \left| y_i \frac{\int_0^1 e^{-\frac{ty_i^2}{2\sigma^2} - \frac{\gamma^2 t}{2\sigma^2(1-t)}} (1-t)^{-\frac{3}{2}} t dt}{\int_0^1 e^{-\frac{ty_i^2}{2\sigma^2} - \frac{\gamma^2 t}{2\sigma^2(1-t)}} (1-t)^{-\frac{3}{2}} dt} \right| \quad (3.22)$$

We will split the integral in the numerator into the sum of two integrals, and separately consider the case $t \in (0, \eta)$ and the case $t \in [\eta, 1)$, where $\eta \in (0, 1)$ is fixed.

- Case $t \in (0, \eta)$. Letting $k = \frac{1}{\tau^2}(\frac{1}{t} - 1)$, we get

$$\begin{aligned}
 \left| y_i \frac{\int_0^\eta e^{-\frac{ty_i^2}{2\sigma^2} - \frac{\tau^2 t}{2\sigma^2(1-t)}} (1-t)^{-\frac{3}{2}} t dt}{\int_0^1 e^{-\frac{ty_i^2}{2\sigma^2} - \frac{\tau^2 t}{2\sigma^2(1-t)}} (1-t)^{-\frac{3}{2}} dt} \right| &= \left| y_i \frac{\int_{\frac{1}{\tau^2}(\frac{1}{\eta}-1)}^\infty e^{-\frac{y_i^2}{2\sigma^2(\tau^2 k+1)} - \frac{1}{2\sigma^2 k}} [k(\tau^2 k+1)]^{-\frac{3}{2}} dk}{\int_0^\infty e^{-\frac{y_i^2}{2\sigma^2(\tau^2 k+1)} - \frac{1}{2\sigma^2 k}} k^{-\frac{3}{2}} (\tau^2 k+1)^{-\frac{1}{2}} dk} \right| \\
 &\leq \left| y_i \frac{\int_{\frac{1}{\tau^2}(\frac{1}{\eta}-1)}^\infty e^{-\frac{y_i^2}{2\sigma^2(\tau^2 k+1)} - \frac{1}{2\sigma^2 k}} [k(\tau^2 k+1)]^{-\frac{3}{2}} dk}{\int_{\frac{k_0}{\tau^2}}^\infty e^{-\frac{y_i^2}{2\sigma^2(\tau^2 k+1)} - \frac{1}{2\sigma^2 k}} k^{-\frac{3}{2}} (\tau^2 k+1)^{-\frac{1}{2}} dk} \right| \\
 &\leq \left| \frac{y_i \int_{\frac{1}{\tau^2}(\frac{1}{\eta}-1)}^\infty e^{-\frac{y_i^2}{2\sigma^2(\tau^2 k+1)} - \frac{1}{2\sigma^2 k}} [k(\tau^2 k+1)]^{-\frac{3}{2}} dk}{c_0 \int_{\frac{k_0}{\tau^2}}^\infty e^{-\frac{y_i^2}{2\sigma^2(\tau^2 k+1)} - \frac{1}{2\sigma^2 k}} k^{-\frac{3}{2}} (\tau^2 k+1)^{-\frac{1}{2}} dk} \right|,
 \end{aligned}$$

where, in the last two steps, we used the fact that there exists $c_0 > 0$ such that $e^{-\frac{1}{2\sigma^2 k}} > c_0$ for $k = \frac{k_0}{\tau} > 0$.

Then we can further substitute $u = \frac{y_i^2}{\tau^2 k+1}$ and $s = \frac{1}{1+k_0}$, and obtain the bound

$$\begin{aligned}
 \left| y_i \frac{\int_0^\eta e^{-\frac{ty_i^2}{2\sigma^2} - \frac{\tau^2 t}{2\sigma^2(1-t)}} (1-t)^{-\frac{3}{2}} t dt}{\int_0^1 e^{-\frac{ty_i^2}{2\sigma^2} - \frac{\tau^2 t}{2\sigma^2(1-t)}} (1-t)^{-\frac{3}{2}} dt} \right| &\leq \left| \frac{y_i \int_0^{\eta y_i^2} e^{-\frac{u}{2\sigma^2}} \left(\frac{u}{y_i^2}\right)^{\frac{3}{2}} u^{-2} \left(\frac{y_i^2}{u} - 1\right)^{-\frac{3}{2}} du}{c_0 \int_0^{s y_i^2} e^{-\frac{u}{2\sigma^2}} \left(\frac{u}{y_i^2}\right)^{\frac{1}{2}} u^{-2} \left(\frac{y_i^2}{u} - 1\right)^{-\frac{3}{2}} du} \right| \\
 &= \left| \frac{1}{c_0 y_i} \frac{\int_0^{\eta y_i^2} e^{-\frac{u}{2\sigma^2}} u \left(1 - \frac{u}{y_i^2}\right)^{-\frac{3}{2}} du}{\int_0^{s y_i^2} e^{-\frac{u}{2\sigma^2}} \left(1 - \frac{u}{y_i^2}\right)^{-\frac{3}{2}} du} \right|.
 \end{aligned}$$

Noticing that when $0 < u < \eta y_i^2$, we have $1 - \eta < 1 - \frac{u}{y_i^2} < 1$, and when $0 < u < s y_i^2$, we have $1 - s < 1 - \frac{u}{y_i^2} < 1$, we can further upper bound the quantity of interest as

$$\begin{aligned}
 \left| y_i \frac{\int_0^\eta e^{-\frac{ty_i^2}{2\sigma^2} - \frac{\tau^2 t}{2\sigma^2(1-t)}} (1-t)^{-\frac{3}{2}} t dt}{\int_0^1 e^{-\frac{ty_i^2}{2\sigma^2} - \frac{\tau^2 t}{2\sigma^2(1-t)}} (1-t)^{-\frac{3}{2}} dt} \right| &\leq \left| \frac{1}{c_0 y_i (1-\eta)^{\frac{3}{2}}} \frac{\int_0^{\eta y_i^2} e^{-\frac{u}{2\sigma^2}} u du}{\int_0^{s y_i^2} e^{-\frac{u}{2\sigma^2}} du} \right| \\
 &\leq \left| \frac{1}{c_0 y_i (1-\eta)^{\frac{3}{2}}} \frac{\int_0^\infty e^{-\frac{u}{2\sigma^2}} u du}{\int_0^{s y_i^2} e^{-\frac{u}{2\sigma^2}} du} \right| \tag{3.23} \\
 &= C^* \left[\left| y_i \int_0^{s y_i^2} e^{-\frac{u}{2\sigma^2}} du \right| \right]^{-1} \\
 &=: h_1(y_i, \tau),
 \end{aligned}$$

where C^* is a constant independent of y_i and τ .

- Case $t \in [\eta, 1)$. Fixing $\delta \in (0, 1)$, we obtain

$$\begin{aligned} \left| y_i \frac{\int_{\eta}^1 e^{-\frac{ty_i^2}{2\sigma^2} - \frac{\tau^2 t}{2\sigma^2(1-t)}} (1-t)^{-\frac{3}{2}} t dt}{\int_0^1 e^{-\frac{ty_i^2}{2\sigma^2} - \frac{\tau^2 t}{2\sigma^2(1-t)}} (1-t)^{-\frac{3}{2}} dt} \right| &\leq \left| y_i \frac{e^{-\frac{\eta(1-\delta)y_i^2}{2\sigma^2}} \int_{\eta}^1 e^{-\frac{\tau^2 t}{2\sigma^2(1-t)}} (1-t)^{-\frac{3}{2}} dt}{\int_0^{\eta\delta} e^{-\frac{\tau^2 t}{2\sigma^2(1-t)}} (1-t)^{-\frac{3}{2}} dt} \right| \\ &= \left| y_i \frac{e^{-\frac{\eta(1-\delta)y_i^2}{2\sigma^2}} \int_0^{\frac{1}{\tau^2}(\frac{1}{\eta}-1)} e^{-\frac{1}{2\sigma^2 k}} k^{-\frac{3}{2}} (1+\tau^2 k)^{-\frac{1}{2}} dk}{\int_{\frac{1}{\tau^2}(\frac{1}{\eta\delta}-1)}^{\infty} e^{-\frac{1}{2\sigma^2 k}} k^{-\frac{3}{2}} (1+\tau^2 k)^{-\frac{1}{2}} dk} \right| \end{aligned}$$

where we use again the substitution $k = \frac{1}{\tau^2}(\frac{1}{t} - 1)$ in the last step.

Taking advantage of the fact that, for $k \geq \frac{1}{\tau^2}(\frac{1}{\eta\delta} - 1)$, we have $\frac{k\tau^2}{1+k\tau^2} \geq 1 - \eta\delta$, we upper bound the expression above as follows

$$\begin{aligned} \left| y_i \frac{\int_{\eta}^1 e^{-\frac{ty_i^2}{2\sigma^2} - \frac{\tau^2 t}{2\sigma^2(1-t)}} (1-t)^{-\frac{3}{2}} t dt}{\int_0^1 e^{-\frac{ty_i^2}{2\sigma^2} - \frac{\tau^2 t}{2\sigma^2(1-t)}} (1-t)^{-\frac{3}{2}} dt} \right| &\leq \left| y_i \frac{e^{-\frac{\eta(1-\delta)y_i^2}{2\sigma^2}} \int_0^{\infty} e^{-\frac{1}{2\sigma^2 k}} k^{-\frac{3}{2}} dk}{\sqrt{\frac{1-\eta\delta}{\tau^2}} \int_{\frac{1}{\tau^2}(\frac{1}{\eta\delta}-1)}^{\infty} e^{-\frac{1}{2\sigma^2 k}} k^{-2} dk} \right| \\ &= \left| \frac{\tau y_i e^{-\frac{\eta(1-\delta)y_i^2}{2\sigma^2}} \sqrt{2\pi\sigma}}{\sqrt{1-\eta\delta} 2\sigma^2 \left(1 - e^{-\frac{\tau^2 \eta\delta}{2\sigma^2(1-\eta\delta)}} \right)} \right| \quad (3.24) \\ &= \left| \sqrt{\frac{\pi}{2(1-\eta\delta)}} \frac{\tau}{\sigma} y_i \frac{e^{-\frac{\eta(1-\delta)y_i^2}{2\sigma^2}}}{1 - e^{-\frac{\tau^2 \eta\delta}{2\sigma^2(1-\eta\delta)}}} \right| \\ &=: h_2(y_i, \tau). \end{aligned}$$

Let $h(y_i, \tau) := h_1(y_i, \tau) + h_2(y_i, \tau)$. Then, putting (3.22), (3.23), and (3.24) together, we obtain (3.20). Furthermore, notice that $h_1(y_i, \tau)$ is decreasing in $|y_i|$ for any $|y_i| > 0$, and that $h_2(y_i, \tau)$ is decreasing in $|y_i|$ for any $|y_i| \geq \frac{\sigma}{\sqrt{\eta(1-\delta)}}$. Using these facts, we obtain

$$\lim_{\tau \downarrow 0} \sup_{|y_i| > \sqrt{-2\rho\sigma^2 \log \tau}} h(y_i, \tau) = \begin{cases} 0 & \text{if } \rho > \frac{1}{\eta(1-\delta)} \\ \infty & \text{otherwise.} \end{cases} \quad (3.25)$$

Given any $c > 1$, we can choose $\eta, \delta \in (0, 1)$ such that $c = \frac{1}{\eta(1-\delta)}$, therefore obtaining (3.21). $\square$

Bound on Mean Squared Error of the Posterior Mean

We will next show that the Cauchy prior is asymptotically minimax, as long as τ^2 is chosen appropriately given the (known) sparsity level. We will use the notation $A_n = o(B_n)$ if $\lim_{n \rightarrow \infty} \frac{A_n}{B_n} = 0$, and $A_n \sim B_n$ if $\lim_{n \rightarrow \infty} \frac{A_n}{B_n} = 1$.

Theorem 3.1. Consider model (3.7), assume $\theta \in \ell_0[q_n]$, and let $\hat{\theta} := \mathbb{E}(\theta \mid y)$. Then, as $\tau \downarrow 0$ and $n \rightarrow \infty$,

$$\sup_{\theta \in \ell_0[q_n]} \mathbb{E} \|\hat{\theta} - \theta\|^2 \leq 2q_n \left(\sigma^2 \log \frac{1}{\tau} \right) \rho^2 (1 + o(1)) + 2(n - q_n) \left(\tau \log \frac{1}{\tau} \right) o(1), \quad (3.26)$$

where $\rho > 1$ is an arbitrary constant. Moreover, if $\tau = \frac{q_n}{n}$ or $\tau = \frac{q_n}{n} \sqrt{\log \left(\frac{n}{q_n} \right)}$, and $\sigma^2 = 1$,

$$\sup_{\theta \in \ell_0[q_n]} \mathbb{E} \|\hat{\theta} - \theta\|^2 \sim 2q_n \log \left(\frac{n}{q_n} \right). \quad (3.27)$$

In order to bound the mean squared error (MSE) of $\hat{\theta}$, we will proceed by considering the MSE as a sum of two components: the error arising from the estimation of means that are different from zero, and the error coming from the estimation of zero means. We will let $\tilde{q}_n = \#\{i : \theta_i \neq 0\}$, and index the means so that the first $\tilde{q}_n$ entries of θ are non-zero means. We can then write the overall MSE as

$$\mathbb{E} \|\hat{\theta} - \theta\|^2 = \sum_{i=1}^{\tilde{q}_n} \mathbb{E}_{\theta_i} (\hat{\theta}_i - \theta_i)^2 + \sum_{i=\tilde{q}_n+1}^n \mathbb{E}_0 (\hat{\theta}_i^2). \quad (3.28)$$

We will first study the sum involving the zero means, and treat the other one next.

Zero Means

We will start by splitting the expectation in the zero-means sum of (3.28) as

$$\mathbb{E}_0(\hat{\theta}_i^2) = \mathbb{E}_0(\hat{\theta}_i^2) 1_{|y_i| \leq \zeta_\tau} + \mathbb{E}_0(\hat{\theta}_i^2) 1_{|y_i| > \zeta_\tau} \quad (3.29)$$

where $\zeta_\tau = \sqrt{-2\sigma^2 \log \tau}$. Using bound (3.11) for the first term in (3.29), we obtain

$$\begin{aligned} \mathbb{E}_0(\hat{\theta}_i^2) 1_{|y_i| \leq \zeta_\tau} &= 2 \int_0^{\zeta_\tau} \hat{\theta}_i^2 \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{y_i^2}{2\sigma^2}} dy_i \\ &\leq 2 \int_0^{\zeta_\tau} \left\{ \sqrt{2} y_i e^{\frac{y_i^2}{2\sigma^2}} \frac{\tau}{\sigma} \left[\frac{e^{-\frac{\tau^2}{2\sigma^2}}}{\Gamma\left(\frac{1}{2}, \frac{\tau^2}{2\sigma^2}\right)} - \frac{\tau}{\sqrt{2}\sigma} \right] \right\}^2 \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{y_i^2}{2\sigma^2}} dy_i \\ &= 2 \sqrt{\frac{2}{\pi\sigma^2}} \tau^2 \left[\frac{e^{-\frac{\tau^2}{2\sigma^2}}}{\Gamma\left(\frac{1}{2}, \frac{\tau^2}{2\sigma^2}\right)} - \frac{\tau}{\sqrt{2}\sigma} \right]^2 \int_0^{\zeta_\tau} \frac{y_i^2}{\sigma^2} e^{\frac{y_i^2}{2\sigma^2}} dy_i. \end{aligned}$$

We take advantage of the identity $\frac{d}{dy_i} y_i e^{\frac{y_i^2}{2\sigma^2}} = \frac{y_i^2}{\sigma^2} e^{\frac{y_i^2}{2\sigma^2}} + e^{\frac{y_i^2}{2\sigma^2}}$ to re-write the last line as

$$\mathbb{E}_0(\hat{\theta}_i^2) 1_{|y_i| \leq \zeta_\tau} \leq 2\sqrt{\frac{2}{\pi\sigma^2}} \tau^2 \left[\frac{e^{-\frac{\tau^2}{2\sigma^2}}}{\Gamma\left(\frac{1}{2}, \frac{\tau^2}{2\sigma^2}\right)} - \frac{\tau}{\sqrt{2}\sigma} \right]^2 \left(\zeta_\tau e^{\frac{\zeta_\tau^2}{2\sigma^2}} - \int_0^{\zeta_\tau} e^{\frac{y_i^2}{2\sigma^2}} dy_i \right),$$

which can be further upper bounded by dropping the integral term (since it is always positive), leading to

$$\mathbb{E}_0(\hat{\theta}_i^2) 1_{|y_i| \leq \zeta_\tau} \leq 2\sqrt{\frac{2}{\pi\sigma^2}} \tau^2 \left[\frac{e^{-\frac{\tau^2}{2\sigma^2}}}{\Gamma\left(\frac{1}{2}, \frac{\tau^2}{2\sigma^2}\right)} - \frac{\tau}{\sqrt{2}\sigma} \right]^2 \zeta_\tau \frac{1}{\tau}.$$

As we noticed in the Proof of Lemma 3.1, the term in brackets goes to $\sqrt{\frac{1}{\pi}}$ as $\tau \downarrow 0$. This implies that as $\tau \downarrow 0$

$$\mathbb{E}_0(\hat{\theta}_i^2) 1_{|y_i| \leq \zeta_\tau} \leq \frac{4}{\pi} \sqrt{\frac{\log(1/\tau)}{\pi}} \tau. \quad (3.30)$$

We will now use bound (3.10) and the definition of upper incomplete gamma function to bound the last term in (3.29). Let $t = \frac{y_i^2}{2\sigma^2}$, then

$$\begin{aligned} \mathbb{E}_0 \hat{\theta}_i^2 1_{|y_i| > \zeta_\tau} &\leq \mathbb{E}_0 y_i^2 1_{|y_i| > \zeta_\tau} \\ &= 2 \int_{\zeta_\tau}^{\infty} y_i^2 \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y_i^2}{2\sigma^2}} dy_i \\ &= \frac{2\sigma^2}{\sqrt{\pi}} \int_{-\log \tau}^{\infty} \sqrt{t} e^{-t} dt \\ &= \frac{2\sigma^2}{\sqrt{\pi}} \Gamma\left(\frac{3}{2}, -\log \tau\right) \end{aligned} \quad (3.31)$$

Bounds (3.30) and (3.31) can be used to upper bound (3.29) as

$$\begin{aligned} \mathbb{E}_0(\hat{\theta}_i^2) &\leq \frac{4}{\pi} \sqrt{\frac{\log \frac{1}{\tau}}{\pi}} \tau + \frac{2\sigma^2}{\sqrt{\pi}} \Gamma\left(\frac{3}{2}, \log \frac{1}{\tau}\right) \\ &= 2\tau \log \frac{1}{\tau} \left[\frac{2}{\pi} \frac{1}{\sqrt{\pi \log \frac{1}{\tau}}} + \frac{\sigma^2}{\sqrt{\pi \log \frac{1}{\tau}}} \frac{\Gamma\left(\frac{3}{2}, \log \frac{1}{\tau}\right)}{\tau \sqrt{\log \frac{1}{\tau}}} \right]. \end{aligned} \quad (3.32)$$

Using the definition of upper incomplete gamma function, we obtain that

$$\frac{\Gamma(a, b)}{b^{a-1} e^{-b}} \rightarrow 1 \text{ as } b \rightarrow \infty,$$

which implies that

$$\frac{\Gamma\left(\frac{3}{2}, \log \frac{1}{\tau}\right)}{\tau \sqrt{\log \frac{1}{\tau}}} \rightarrow 1 \text{ as } \tau \downarrow 0. \quad (3.33)$$

Therefore, the term inside the brackets in (3.32) converges to 0 as $\tau \downarrow 0$, we can thus write

$$\mathbb{E}_0(\hat{\theta}_i^2) \leq \left(2\tau \log \frac{1}{\tau}\right) o(1) \text{ as } \tau \downarrow 0. \quad (3.34)$$

Non-zero Means

We will now bound the first sum on the right-hand side of (3.28). Using the Cauchy-Schwartz inequality, we get

$$\mathbb{E}_{\theta_i}(\hat{\theta}_i - \theta_i)^2 \leq \left[\sqrt{\mathbb{E}_{\theta_i}(\hat{\theta}_i - y_i)^2} + \sigma \right]^2. \quad (3.35)$$

so we will focus on finding an upper bound for $\mathbb{E}_{\theta_i}(\hat{\theta}_i - y_i)^2$. We fix $c > 1$, and choose any $\rho > c$. By applying Lemma 3.2 and bound (3.10), we obtain that, as $\tau \downarrow 0$,

$$\begin{aligned} \mathbb{E}_{\theta_i}(\hat{\theta}_i - y_i)^2 &= \mathbb{E}_{\theta_i}[(\hat{\theta}_i - y_i)^2 1_{|y_i| \leq \rho \zeta_\tau}] + \mathbb{E}_{\theta_i}[(\hat{\theta}_i - y_i)^2 1_{|y_i| > \rho \zeta_\tau}] \\ &\leq y_i^2 1_{|y_i| \leq \rho \zeta_\tau} + h^2(y_i, \tau) 1_{|y_i| > \rho \zeta_\tau} \\ &\leq \rho^2 \zeta_\tau^2 (1 + o(1)). \end{aligned} \quad (3.36)$$

Sum of zero and non-zero means

Putting everything together, for $\tau \downarrow 0$, we can bound the MSE as follows

$$\begin{aligned} \mathbb{E}_\theta \|\hat{\theta} - \theta\|^2 &= \sum_{i=1}^{\tilde{q}_n} \mathbb{E}_{\theta_i}(\hat{\theta}_i - \theta_i)^2 + \sum_{i=\tilde{q}_n+1}^n \mathbb{E}_0(\hat{\theta}_i)^2 \\ &\leq \tilde{q}_n \rho^2 \left(2\sigma^2 \log \frac{1}{\tau}\right) (1 + o(1)) + (n - \tilde{q}_n) \left(2\tau \log \frac{1}{\tau}\right) o(1). \end{aligned} \quad (3.37)$$

Remembering that $\tilde{q}_n \leq q_n$ and $q_n = o(n)$, and taking the supremum over $\theta \in \ell_0[q_n]$, we obtain (3.26).

Setting $\sigma^2 = 1$, and $\tau = \frac{q_n}{n}$ or $\tau = \frac{q_n}{n} \sqrt{\log \frac{n}{q_n}}$ in (3.26), we obtain

$$\sup_{\theta \in \ell_0[q_n]} \mathbb{E} \|\hat{\theta} - \theta\|^2 \leq 2\rho^2 q_n \left(\log \frac{n}{q_n} \right) (1 + o(1)). \quad (3.38)$$

Donoho et al. (1992) showed that, for nearly black vectors, the minimax error rate under the ℓ_2 norm is given by

$$\inf_{\tilde{\theta}} \sup_{\theta \in \ell_0[q_n]} \mathbb{E} \|\tilde{\theta} - \theta\|^2 = 2q_n \left(\log \frac{n}{q_n} \right) (1 + o(1)) \quad (3.39)$$

as $n \rightarrow \infty$. Therefore, as $n \rightarrow \infty$, we have that

$$1 \leq \frac{\sup_{\theta \in \ell_0[q_n]} \mathbb{E} \|\hat{\theta} - \theta\|^2}{\inf_{\tilde{\theta}} \sup_{\theta \in \ell_0[q_n]} \mathbb{E} \|\tilde{\theta} - \theta\|^2} \leq \rho^2(1 + o(1)). \quad (3.40)$$

Taking the limit inferior and superior as $n \rightarrow \infty$ in (3.40), we obtain

$$1 \leq \liminf_{n \rightarrow \infty} \frac{\sup_{\theta \in \ell_0[q_n]} \mathbb{E} \|\hat{\theta} - \theta\|^2}{\inf_{\tilde{\theta}} \sup_{\theta \in \ell_0[q_n]} \mathbb{E} \|\tilde{\theta} - \theta\|^2} \leq \limsup_{n \rightarrow \infty} \frac{\sup_{\theta \in \ell_0[q_n]} \mathbb{E} \|\hat{\theta} - \theta\|^2}{\inf_{\tilde{\theta}} \sup_{\theta \in \ell_0[q_n]} \mathbb{E} \|\tilde{\theta} - \theta\|^2} \leq \rho^2. \quad (3.41)$$

Since $\rho > c > 1$ is an arbitrary constant, and $\hat{\theta}$ does not depend on it, we can take the infimum over all possible $\rho > c > 1$ in (3.41), and obtain

$$\sup_{\theta \in \ell_0[q_n]} \mathbb{E} \|\hat{\theta} - \theta\|^2 \sim \inf_{\tilde{\theta}} \sup_{\theta \in \ell_0[q_n]} \mathbb{E} \|\tilde{\theta} - \theta\|^2. \quad (3.42)$$

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Appendix A

Supplementary materials for Chapter 1

A.1 Derivation of conditional distributions for Algorithm 1

Consider the model specified in 1.18 with the reparametrization in 1.22. The complete conditional distribution of λ^2 is

$$\begin{aligned}
 p(\lambda^2|\cdot) &\propto p\left(D^{(k)}\theta|\lambda^2, \tau^2\right) p\left(\lambda^2\right) \\
 &= \prod_{j=1}^{n-k} \left[N\left((D^{(k)}\theta)_j; 0, \tau^2 \lambda_j^2\right) IG\left(\lambda_j^2; 0.5, 0.5\right) \right] \\
 &\propto \prod_{j=1}^{n-k} \left[\frac{1}{\lambda_j} \exp\left\{-\frac{(D^{(k)}\theta)_j^2}{2\tau^2 \lambda_j^2}\right\} \frac{1}{\lambda_j^3} \exp\left\{-\frac{1}{2\lambda_j^2}\right\} \right] \\
 &= \prod_{j=1}^{n-k} \left[\frac{1}{\lambda_j^4} \exp\left\{-\frac{(D^{(k)}\theta)_j^2 + \tau^2}{2\tau^2 \lambda_j^2}\right\} \right],
 \end{aligned}$$

which implies that the λ_j^2 's are conditionally independent with

$$p(\lambda_j^2|\cdot) = IG\left(1, \frac{(D^{(k)}\theta)_j^2 + \tau^2}{2\tau^2}\right)$$

for $j = 1, \dots, n - k$.

The complete conditional distribution of τ^2 and of σ^2 are respectively

$$\begin{aligned}
p(\tau^2|\cdot) &\propto p\left(D^{(k)}\theta|\lambda^2, \tau^2\right) p\left(\tau^2\right) \\
&\propto \frac{1}{\tau^2} \prod_{j=1}^{n-k} \left[\frac{1}{\tau} \exp \left\{ -\frac{(D^{(k)}\theta)_j^2}{2\tau^2\lambda_j^2} \right\} \right] \\
&= \left(\frac{1}{\tau^2} \right)^{\frac{n-k}{2}+1} \exp \left\{ -\frac{1}{\tau^2} \sum_{j=1}^{n-k} \frac{(D^{(k)}\theta)_j^2}{2\lambda_j^2} \right\} \\
&\propto IG \left(\frac{n-k}{2}, \sum_{j=1}^{n-k} \frac{(D^{(k)}\theta)_j^2}{2\lambda_j^2} \right).
\end{aligned}$$

and

$$\begin{aligned}
p(\sigma^2|\cdot) &\propto p(y|\theta, \sigma^2) p(\sigma^2) \\
&\propto \frac{1}{\sigma^n} \exp \left\{ -\frac{\|y - \theta\|_2^2}{2\sigma^2} \right\} \frac{1}{\sigma^2} \\
&\propto IG \left(\frac{n}{2}, \frac{\|y - \theta\|_2^2}{2} \right).
\end{aligned}$$

The complete conditional distribution of θ is obtained as

$$\begin{aligned}
p(\theta|\cdot) &\propto p(y|\theta, \sigma^2) p(\theta|\lambda^2, \tau^2) \\
&= N(y; \theta, \sigma^2 I_n) N \left(\begin{pmatrix} \theta_{1:k} \\ D^{(k)}\theta \end{pmatrix}; \mathbf{0}, \begin{pmatrix} L\mathbf{I}_k & 0 \\ 0 & \tau^2\Lambda \end{pmatrix} \right) \\
&\propto \exp \left\{ -\frac{\|y - \theta\|_2^2}{2\sigma^2} - \frac{\theta^T D^{(k)T} \Lambda^{-1} D^{(k)} \theta}{2\tau^2} \right\} \exp \left\{ -\frac{\theta_{1:k}^T \theta_{1:k}}{2L^2} \right\}
\end{aligned}$$

where $\mathbf{0}$ is the n -dimensional vector of zeros, while $\mathbf{I}_k$ is the identity matrix of dimension $k \times k$. Since L is arbitrarily large, the factor depending on L is approximately constant and equal to 1, and can therefore be dropped together with every other factor not depending on

θ . The complete conditional distribution simplifies to

$$\begin{aligned}
 p(\theta|\cdot) &\propto \exp \left\{ -\frac{\|y - \theta\|_2^2}{2\sigma^2} - \frac{\theta^T D^{(k)T} \Lambda^{-1} D^{(k)} \theta}{2\tau^2} \right\} \\
 &\propto \exp \left\{ -\frac{1}{2} \left[\frac{\theta^T \theta}{\sigma^2} + \frac{\theta^T D^{(k)T} \Lambda^{-1} D^{(k)} \theta}{\tau^2} \right] + \frac{y^T \theta}{\sigma^2} \right\} \\
 &\propto \exp \left\{ -\frac{1}{2} \left(\theta - \frac{1}{\sigma^2} \Sigma y \right)^T \Sigma^{-1} \left(\theta - \frac{1}{\sigma^2} \Sigma y \right) \right\} \\
 &\propto N \left(\frac{1}{\sigma^2} \Sigma y, \Sigma \right),
 \end{aligned}$$

where $\Sigma = \left(\frac{1}{\sigma^2} I + \frac{1}{\tau^2} D^{(k)T} \Lambda^{-1} D^{(k)} \right)^{-1}$.

A.2 The Voigt Profile

The Voigt profile is the convolution of the normal density centered at zero:

$$x \mapsto \frac{1}{\sqrt{2\pi}\sigma} \exp \left(-\frac{x^2}{2\sigma^2} \right)$$

and the Cauchy density centered at zero:

$$x \mapsto \frac{\tau}{\pi(\tau^2 + x^2)}.$$

In other words,

$$\begin{aligned}
 V(x; \sigma, \tau) &:= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp \left(-\frac{(x-t)^2}{2\sigma^2} \right) \frac{\tau}{\pi(\tau^2 + t^2)} dt \\
 &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp \left(-\frac{t^2}{2\sigma^2} \right) \frac{\tau}{\pi(\tau^2 + (x-t)^2)} dt.
 \end{aligned} \tag{A.1}$$

The Voigt profile can be written in terms of the function:

$$K(x, y) := \frac{y}{\pi} \int_{-\infty}^{\infty} \frac{e^{-t^2}}{y^2 + (x-t)^2} dt. \tag{A.2}$$

Indeed, it is easy to show (by the simple change of variable $t = u/(\sqrt{2}\sigma)$ in the integral in (A.1)) that

$$V(x; \sigma, \tau) = \frac{1}{\sqrt{2\pi}\sigma} K \left(\frac{x}{\sqrt{2}\sigma}, \frac{\tau}{\sqrt{2}\sigma} \right) \tag{A.3}$$

The function $K(x, y)$ is itself closely related to the function $L(x, y)$ given by

$$L(x, y) := \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{(x-t)e^{-t^2}}{y^2 + (x-t)^2} dt. \quad (\text{A.4})$$

$K(x, y)$ and $L(x, y)$ are simply the real and imaginary parts of the Faddeeva Function:

$$w(z) = \frac{i}{\pi} \int_{-\infty}^{\infty} \frac{e^{-t^2}}{z-t} dt$$

because, when $z = x + iy$,

$$\frac{1}{z-t} = \frac{1}{(x-t) + iy} = \frac{x-t-iy}{(x-t)^2 + y^2}.$$

If the normal density is centered at μ_N and the Cauchy density is centered at μ_C , the convolution is centered at $\mu_V = \mu_N + \mu_C$, that is

$$V(x; \mu_V, \sigma, \tau) = \frac{1}{\sqrt{2\pi}\sigma} K\left(\frac{x-\mu_V}{\sqrt{2}\sigma}, \frac{\tau}{\sqrt{2}\sigma}\right). \quad (\text{A.5})$$

For notational convenience, in this section we will only consider the centered Voigt profile $V(x; \sigma, \tau)$, but all the proofs can be easily extended to $V(x; \mu_V, \sigma, \tau)$. The derivatives of $V(x; \sigma, \tau)$ with respect to x can be expressed in terms of the $K(\cdot, \cdot)$ and $L(\cdot, \cdot)$ functions as shown in the following lemma.

Lemma A.1. The following formulae hold:

$$\frac{d}{dx} V(x; \sigma, \tau) = \frac{1}{\sigma^3 \sqrt{2\pi}} \left(\tau L\left(\frac{x}{\sigma\sqrt{2}}, \frac{\tau}{\sigma\sqrt{2}}\right) - x K\left(\frac{x}{\sigma\sqrt{2}}, \frac{\tau}{\sigma\sqrt{2}}\right) \right), \quad (\text{A.6})$$

and

$$\frac{d^2}{dx^2} V(x; \sigma, \tau) = \frac{-1}{\sigma^5 \sqrt{2\pi}} \left\{ (\tau^2 - x^2 + \sigma^2) K\left(\frac{x}{\sigma\sqrt{2}}, \frac{\tau}{\sigma\sqrt{2}}\right) + 2x\tau L\left(\frac{x}{\sigma\sqrt{2}}, \frac{\tau}{\sigma\sqrt{2}}\right) - \tau\sigma \sqrt{\frac{2}{\pi}} \right\}. \quad (\text{A.7})$$

Proof of Lemma A.1. To see why (A.6) is true, first note that

$$\frac{d}{dx} V(x; \sigma, \tau) = \frac{1}{2\sigma^2 \sqrt{\pi}} K^{(1,0)}\left(\frac{x}{\sqrt{2}\sigma}, \frac{\tau}{\sqrt{2}\sigma}\right)$$

where $K^{(1,0)} = \frac{\partial}{\partial x} K(x, y)$. Now use the following formula (taken from [Heinzel \(1978\)](#))

$$K^{(1,0)}(x, y) = 2(yL(x, y) - xK(x, y)).$$

For (A.7), first note that

$$\frac{d^2}{dx^2}V(x, \tau, \sigma) = \frac{1}{2\sigma^3\sqrt{2\pi}}K^{(2,0)}\left(\frac{x}{\sigma\sqrt{2}}, \frac{\tau}{\sigma\sqrt{2}}\right),$$

where $K^{(2,0)} = \frac{\partial^2}{\partial x^2}K(x, y)$. Now use the following formula (taken again from [Heinzel \(1978\)](#))

$$K^{(2,0)}(x, y) = -4 \left[\left(y^2 - x^2 + \frac{1}{2} \right) K(x, y) + 2xyL(x, y) - \frac{y}{\sqrt{\pi}} \right].$$

□

Lemma A.2. For $v \in (-\infty, \infty)$ and $s, \tau > 0$, consider the density

$$g(u) := \frac{N(v; u, s^2)C(u; 0, \tau)}{V(v; s, \tau)} \quad \text{for } u \in (-\infty, \infty).$$

Then

$$\int_{-\infty}^{\infty} ug(u)du = \frac{\tau L\left(\frac{v}{s\sqrt{2}}, \frac{\tau}{s\sqrt{2}}\right)}{K\left(\frac{v}{s\sqrt{2}}, \frac{\tau}{s\sqrt{2}}\right)}, \quad (\text{A.8})$$

and

$$\int_{-\infty}^{\infty} u^2 g(u)du = \frac{\tau s \sqrt{2/\pi}}{K\left(\frac{v}{s\sqrt{2}}, \frac{\tau}{s\sqrt{2}}\right)} - \tau^2 \quad (\text{A.9})$$

where $K(\cdot, \cdot)$ and $L(\cdot, \cdot)$ are the Voigt functions (A.2) and (A.4) respectively.

Proof of Lemma A.2. We first write

$$\int_{-\infty}^{\infty} ug(u)du = \frac{1}{V(v; s, \tau)} \int_{-\infty}^{\infty} uN(v; u, s^2) \frac{\tau}{\pi(u^2 + \tau^2)} du$$

and then use the formula:

$$\frac{uN(v; u, s^2)}{s^2} = \frac{vN(v; u, s^2)}{s^2} + \frac{d}{dv}N(v; u, s^2).$$

This gives

$$\begin{aligned} \int_{-\infty}^{\infty} ug(u)du &= \frac{1}{V(v; s, \tau)} \int_{-\infty}^{\infty} \left(vN(v; u, s^2) + s^2 \frac{d}{dv}N(v; u, s^2) \right) \frac{\tau}{\pi(u^2 + \tau^2)} du \\ &= \frac{v}{V(v; s, \tau)} \int_{-\infty}^{\infty} N(v; u, s^2) \frac{\tau}{\pi(u^2 + \tau^2)} du + \frac{s^2}{V(v; s, \tau)} \int_{-\infty}^{\infty} \frac{d}{dv}N(v; u, s^2) \frac{\tau}{\pi(u^2 + \tau^2)} du \\ &= v + \frac{s^2}{V(v; s\tau)} \frac{d}{dv}V(v; s, \tau) \end{aligned}$$

where, for the second term above, we interchanged $\frac{d}{dv}$ and $\int_{-\infty}^{\infty}$. The formula (A.6) (and (A.3)) now proves (A.8).

For (A.9), we write

$$\int_{-\infty}^{\infty} u^2 g(u) du = \frac{1}{V(v; s, \tau)} \int_{-\infty}^{\infty} u^2 N(v; u, s^2) \frac{\tau}{\pi(u^2 + \tau^2)} du$$

and then use the following formula:

$$u^2 N(v; u, s^2) = s^4 \left(\frac{d^2}{dv^2} N(v; u, s^2) \right) + 2uv N(v; u, s^2) + (s^2 - v^2) N(v; u, s^2).$$

This gives

$$\begin{aligned} \int_{-\infty}^{\infty} u^2 g(u) du &= \frac{1}{V(v; s, \tau)} \left\{ s^4 \int \left(\frac{d^2}{dv^2} N(v; u, s^2) \right) \frac{\tau}{\pi(u^2 + \tau^2)} du \right. \\ &\quad \left. + 2v \int u N(v; u, s^2) \frac{\tau}{\pi(u^2 + \tau^2)} du + (s^2 - v^2) \int N(v; u, s^2) \frac{\tau}{\pi(u^2 + \tau^2)} du \right\}. \end{aligned}$$

Bringing the $\frac{d^2}{dv^2}$ outside the first integral, we obtain

$$\int_{-\infty}^{\infty} u^2 g(u) du = \frac{s^4}{V(v; s, \tau)} \left[\frac{d^2}{dv^2} V(v; s, \tau) \right] + 2v \int u g(u) du + (s^2 - v^2).$$

The formula (A.9) is now deduced by using (A.7) (for the second derivative term above) and (A.8) (for the second term in the right hand side above). $\square$

A.3 Mixture of Normal Approximation for Filtering

Let's consider approximating g in (1.28) using a mixture of Normal distributions:

$$g(\eta; y, s, \tau) \approx \sum_{j=1}^l \alpha_j(y, s, \tau) N(\eta; \mu_j(y, s, \tau), \sigma_j^2(y, s, \tau)).$$

Then the recursion (1.28) becomes

$$\begin{aligned} p(x_{t+1} \mid y_{1:t+1}) &= \\ &\sum_{i=1}^{n_t} \sum_{j=1}^l \frac{\alpha_j(y_{t+1} - \phi^T m_{t|t}^{(i)}, s^{(i)}, \tau) w_t^{(i)} V(y_{t+1} - \phi^T m_{t|t}^{(i)}; s^{(i)}, \tau)}{\sum_{i=1}^{n_t} w_t^{(i)} V(y_{t+1} - \phi^T m_{t|t}^{(i)}; s^{(i)}, \tau)} N(x_{t+1}; m_{t+1|t+1}^{(i,j)}, V_{t+1|t+1}^{(i,j)}) \end{aligned} \quad (\text{A.10})$$

where

$$m_{t+1|t+1}^{(i,j)} := \check{m}_{t+1}^{(i)} \left(\mu_j(y_{t+1} - \phi^T m_{t|t}^{(i)}, s^{(i)}, \tau) \right)$$

and

$$V_{t+1|t+1}^{(i,j)} = \check{V}_{t+1}^{(i)} + \frac{\sigma_j^2(y_{t+1} - \phi^T m_{t|t}^{(i)}, s, \tau)}{(\sigma^2 + \phi^T V_{t|t}^{(i)} \phi)^2} \begin{pmatrix} \sigma^4 & -\sigma^2 \phi^T V_{t|t}^{(i)}[-k] \\ -\sigma^2 V_{t|t}^{(i)}[-k, \cdot] \phi & V_{t|t}^{(i)}[-k, \cdot] \phi \phi^T V_{t|t}^{(i)}[-k, \cdot] \end{pmatrix}$$

The number of Gaussian components in the mixture (A.10) equals $n_{t+1} = l n_t$. Because we initialize (as in (1.25)) with $k_p = 1$, we have $n_t = l^{t-k}$ for $t \geq k$. This clearly becomes very large for $l \geq 2$. One way to control the number of components in the mixture is to use a standard reduction procedure such as Runnall's algorithm (see e.g., Runnalls (2007) or Crouse et al. (2011, Algorithm 3)). A description of Runnall's algorithm is given as Algorithm 7.

Given the filtering approximation (1.24), the approximate likelihood is calculated (similarly as in (1.31)):

$$\prod_{t=k}^{n-1} \left[\sum_{i=1}^{n_t} w_t^{(i)} V \left(y_{t+1} - \phi^T m_{t|t}^{(i)}; \sqrt{\phi^T V_{t|t}^{(i)} \phi + \sigma^2}, \tau \right) \right].$$

A.4 Generalization and Proof of Lemma 1.1

Lemma A.3. Fix $t \geq k$. Assume that the filtering distribution $p(x_t \mid y_{1:t})$ is given by $\sum_{i=1}^{n_t} w_t^{(i)} N(m_{t|t}^{(i)}, V_{t|t}^{(i)})$. First assume that $k \geq 2$. For $i = 1, \dots, n_t$, let

$$m_{t|t}^{(i)} = \begin{pmatrix} m_{t|t}^{(i)}[-k] \\ m_{t|t}^{(i)}[k] \end{pmatrix} \quad \text{and} \quad V_{t|t}^{(i)} = \begin{pmatrix} V_{t|t}^{(i)}[-k, -k] & V_{t|t}^{(i)}[-k, k] \\ V_{t|t}^{(i)}[k, -k] & V_{t|t}^{(i)}[k, k] \end{pmatrix}.$$

Also, let

$$f_{t|t}^{(i)}(u) := m_{t|t}^{(i)}[k] + V_{t|t}^{(i)}[k, -k] \left(V_{t|t}^{(i)}[-k, -k] \right)^{-1} (u - m_{t|t}^{(i)}[-k]) \quad \text{for } (k-1) \times 1 \text{ vectors } u$$

and

$$r_{t|t}^{(i)} := \sqrt{V_{t|t}^{(i)}[k, k] - V_{t|t}^{(i)}[k, -k] \left(V_{t|t}^{(i)}[-k, -k] \right)^{-1} V_{t|t}^{(i)}[-k, k]}.$$

Further, let

$$b(x_{t+1}) = b(\theta_{t+1}, \dots, \theta_{t-k+2}) := \frac{\theta_{t+1}}{\phi_k} - \frac{\phi_{k-1}\theta_{t-k+2} + \dots + \phi_1\theta_t}{\phi_k},$$

and

$$\gamma_t^{(i)}(x_{t+1}) \propto w_t^{(i)} N(x_{t+1}[-1]; m_{t|t}^{(i)}[-k], V_{t|t}^{(i)}[-k, -k]) V(f_{t|t}^{(i)}(x_{t+1}[-1]) - b(x_{t+1}); r_{t|t}^{(i)}, \tau / |\phi_k|)$$

Algorithm 7: Runnall's Algorithm

Input: A normal mixture density $\sum_{i=1}^n \alpha_i N(\cdot, m_i, V_i)$ with n components.

Target: To obtain, for each $1 \leq l < n$, a normal mixture approximation to this density with l mixture components.

Initialization: Set the current mixture to the given original mixture $\sum_{i=1}^n \alpha_i N(\cdot, m_i, V_i)$.

Recursion: Suppose the current mixture density is $\sum_{i=1}^l w_i N(\cdot; \mu_i, \Sigma_i)$. Repeat the following until $l = 1$ or l equals the desired number of components. For every pair (i, j) with $i \neq j$, compute the following quantities:

$$\begin{aligned}\mu_{i+j} &= \frac{w_i \mu_i + w_j \mu_j}{w_i + w_j} \\ \Sigma_{i+j} &= \frac{w_i}{w_i + w_j} (\Sigma_i + (\mu_i - \mu_{i+j})(\mu_i - \mu_{i+j})^T) \\ &\quad + \frac{w_j}{w_i + w_j} (\Sigma_j + (\mu_j - \mu_{i+j})(\mu_j - \mu_{i+j})^T) \\ c_{i,j} &= (w_i + w_j) \log \det \Sigma_{i+j} - w_i \log \det \Sigma_i - w_j \log \det \Sigma_j.\end{aligned}$$

Find the pair $(i, j), i \neq j$ for which $c_{i,j}$ is the smallest. Then replace the current mixture density:

$$\sum_{i=1}^l w_i N(\cdot; \mu_i, \Sigma_i) = w_i N(\cdot; \mu_i, \Sigma_i) + w_j N(\cdot; \mu_j, \Sigma_j) + \sum_{u \notin \{i,j\}} w_u N(\cdot; \mu_u, \Sigma_u)$$

by the modified mixture:

$$(w_i + w_j) N(\cdot; \mu_{i+j}, \Sigma_{i+j}) + \sum_{u \notin \{i,j\}} w_u N(\cdot; \mu_u, \Sigma_u)$$

which has one fewer component.

where the constant underlying the proportionality above is such that $\sum_{i=1}^{n_t} \gamma_t^{(i)}(x_{t+1}) = 1$. Then

$$\begin{aligned} & p(\theta_{t-k+1} \mid x_{t+1}, y_{1:t}) \\ &= \sum_{i=1}^{n_t} \gamma_t^{(i)}(x_{t+1}) g \left(\theta_{t-k+1} - b(x_{t+1}); f_{t|t}^{(i)}(x_{t+1}[-1]) - b(x_{t+1}), r_{t|t}^{(i)}, \frac{\tau}{|\phi_k|} \right) \end{aligned} \quad (\text{A.11})$$

where the density g is defined as in (1.27).

For $k = 1$, the formula is given by

$$p(\theta_t \mid \theta_{t+1}, y_{1:t}) = \sum_{i=1}^{n_t} \gamma_t^{(i)}(\theta_{t+1}) g \left(\theta_t - \frac{\theta_{t+1}}{\phi_1}; m_{t|t}^{(i)} - \frac{\theta_{t+1}}{\phi_1}, \sqrt{V_{t|t}^{(i)}}, \frac{\tau}{|\phi_1|} \right) \quad (\text{A.12})$$

with $\gamma_t^{(i)}(\theta_{t+1}) \propto w_t^{(i)} V \left(m_{t|t}^{(i)} - \frac{\theta_{t+1}}{\phi_1}; \sqrt{V_{t|t}^{(i)}}, \frac{\tau}{|\phi_1|} \right)$.

Proof of Lemma A.3. We shall provide the proof of $k \geq 2$ (the argument for $k = 1$ is similar and simpler and is hence omitted). We write

$$p(\theta_{t-k+1} \mid x_{t+1}, y_{1:t}) = \frac{p(\theta_{t+1} \mid \theta_{(t-k+1):t}) p(\theta_{t-k+1} \mid \theta_{(t-k+2):t}, y_{1:t})}{\int p(\theta_{t+1} \mid \theta_{(t-k+1):t}) p(\theta_{t-k+1} \mid \theta_{(t-k+2):t}, y_{1:t}) d\theta_{t-k+1}} \quad (\text{A.13})$$

Under the filtering assumption $p(x_t \mid y_{1:t}) \approx \sum_{i=1}^{n_t} w_t^{(i)} N(m_{t|t}^{(i)}, V_{t|t}^{(i)})$, it can be checked that

$$\begin{aligned} & p(\theta_{t-k+1} \mid \theta_{(t-k+2):t}, y_{1:t}) \\ &= \sum_{i=1}^{n_t} \left(\frac{w_t^{(i)} N(\theta_{t:(t-k+2)}; m_{t|t}^{(i)}[-k], V_{t|t}^{(i)}[-k, -k])}{\sum_{j=1}^{n_t} w_t^{(j)} N(\theta_{t:(t-k+2)}; m_{t|t}^{(j)}[-k], V_{t|t}^{(j)}[-k, -k])} \right) N(\theta_{t-k+1}; f_{t|t}^{(i)}(\theta_{t:(t-k+2)}), (r_{t|t}^{(i)})^2) \end{aligned}$$

where $f_{t|t}^{(i)}(\theta_t, \dots, \theta_{t-k+2})$ and $r_{t|t}^{(i)}$ are as defined in the statement of the lemma.

Using the notation

$$\beta_t^{(i)}(\theta_{t:(t-k+2)}) := \frac{w_t^{(i)} N(\theta_{t:(t-k+2)}; m_{t|t}^{(i)}[-k], V_{t|t}^{(i)}[-k, -k])}{\sum_{j=1}^{n_t} w_t^{(j)} N(\theta_{t:(t-k+2)}; m_{t|t}^{(j)}[-k], V_{t|t}^{(j)}[-k, -k])}$$

for $1 \leq i \leq n_t$, we get

$$p(\theta_{t-k+1} \mid \theta_{(t-k+2):t}, y_{1:t}) = \sum_{i=1}^{n_t} \beta_t^{(i)}(\theta_{t:(t-k+2)}) N(f_{t|t}^{(i)}(\theta_{t:(t-k+2)}); \theta_{t-k+1}, (r_{t|t}^{(i)})^2). \quad (\text{A.14})$$

Further

$$p(\theta_{t+1} \mid \theta_{(t-k+1):t}) = C(\theta_{t+1}; \phi_1 \theta_t + \dots + \phi_k \theta_{t-k+1}, \tau) = \frac{1}{|\phi_k|} C \left(\theta_{t-k+1}; b(x_{t+1}), \frac{\tau}{|\phi_k|} \right) \quad (\text{A.15})$$

where $b(x_{t+1})$ is defined in the statement of the lemma. Plugging the expressions (A.14) and (A.15) in (A.13), we obtain

$$\begin{aligned} & p(\theta_{t-k+1} \mid x_{t+1}, y_{1:t}) \\ &= \frac{\sum_{i=1}^{n_t} \beta_t^{(i)}(\theta_{t:(t-k+2)}) N(f_{t|t}^{(i)}(\theta_{t:(t-k+2)}); \theta_{t-k+1}, (r_{t|t}^{(i)})^2) C\left(\theta_{t-k+1}; b(x_{t+1}), \frac{\tau}{|\phi_k|}\right)}{\sum_{i=1}^{n_t} \beta_t^{(i)}(\theta_{t:(t-k+2)}) V\left(f_{t|t}^{(i)}(\theta_{t:(t-k+2)}) - b(x_{t+1}); r_{t|t}^{(i)}, \tau/|\phi_k|\right)}, \end{aligned}$$

where, for the denominator, we used $\int N(y; u, \sigma^2) C(u; \mu, \tau) du = V(y - \mu; \sigma, \tau)$. Let

$$\gamma_t^{(i)}(x_{t+1}) := \frac{\beta_t^{(i)}(\theta_{t:(t-k+2)}) V\left(f_{t|t}^{(i)}(\theta_{t:(t-k+2)}) - b(x_{t+1}); r_{t|t}^{(i)}, \tau/|\phi_k|\right)}{\sum_{j=1}^{n_t} \beta_t^{(j)}(\theta_{t:(t-k+2)}) V\left(f_{t|t}^{(j)}(\theta_{t:(t-k+2)}) - b(x_{t+1}); r_{t|t}^{(j)}, \tau/|\phi_k|\right)}$$

and note that it coincides with the formula for $\gamma_t^{(i)}(x_{t+1})$ given in the statement of the lemma. We then write

$$\begin{aligned} & p(\theta_{t-k+1} \mid x_{t+1}, y_{1:t}) \\ &= \sum_{i=1}^{n_t} \gamma_t^{(i)}(x_{t+1}) \frac{N(f_{t|t}^{(i)}(\theta_{t:(t-k+2)}); \theta_{t-k+1}, (r_{t|t}^{(i)})^2) C\left(\theta_{t-k+1}; b(x_{t+1}), \frac{\tau}{|\phi_k|}\right)}{V\left(f_{t|t}^{(i)}(\theta_{t:(t-k+2)}) - b(x_{t+1}); r_{t|t}^{(i)}, \tau/|\phi_k|\right)} \end{aligned}$$

The proof of Lemma A.3 is completed by using the formula (1.27) for the density g . $\square$

Appendix B

Supplementary materials for Chapter 2

B.1 Derivation of conditional distributions for the latent Cauchy AR(k) model

The complete conditional for ϕ is obtained as

$$\begin{aligned}
 p(\phi \mid \cdot) &\propto p(\theta \mid \phi, \tau^2, \lambda^2) p(\phi) \\
 &\propto \prod_{j=1}^{n-k} \exp \left\{ -\frac{\left(\theta_{j+k} - \sum_{l=1}^k \phi_l \theta_{j+k-l} \right)^2}{2\tau^2 \lambda_j^2} \right\} \\
 &\propto \exp \left\{ -\frac{1}{2\tau^2} \sum_{j=1}^{n-k} \frac{\phi^\top \theta_{(j+k-1):j} \theta_{(j+k-1):j}^\top \phi}{\lambda_j^2} + \sum_{j=1}^{n-k} \frac{\theta_{j+k}}{\tau^2 \lambda_j^2} \theta_{(j+k-1):j}^\top \phi \right\} \\
 &\propto N \left(\Omega \sum_{j=1}^{n-k} \frac{\theta_{j+k}}{\lambda_j^2} \theta_{(j+k-1):j}, \tau^2 \Omega \right)
 \end{aligned}$$

where

$$\Omega = \left(\sum_{j=1}^{n-k} \frac{\theta_{(j+k-1):j} \theta_{(j+k-1):j}^\top}{\lambda_j^2} \right)^{-1}.$$

The complete conditional for μ is obtained as

$$\begin{aligned}
p(\mu \mid \cdot) &\propto p(y \mid \mu, \theta, \sigma^2) p(\mu) \\
&\propto \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n [\mu - (y_i - \theta_i)]^2 \right\} \\
&\propto \exp \left\{ -\frac{1}{2\sigma^2} \left[n\mu^2 - 2\mu \sum_{i=1}^n (y_i - \theta_i) \right] \right\} \\
&\propto \exp \left\{ -\frac{n}{2\sigma^2} \left[\mu^2 - 2\mu \frac{1}{n} \sum_{i=1}^n (y_i - \theta_i) \right] \right\} \\
&\propto N \left(\frac{1}{n} \sum_{i=1}^n (y_i - \theta_i), \frac{\sigma^2}{n} \right).
\end{aligned}$$

B.2 Derivation of conditional distributions for the Cauchy decomposition

Consider model (2.5). Let $\beta_0 := (D^{(1)}\theta) - (\tilde{\theta}^{(1)} + \tilde{\theta}^{(2)})$, $\beta_1 := D^{(1)}\tilde{\theta}^{(1)}$, and $\beta_2 := D^{(2)}\tilde{\theta}^{(2)}$. For $k = 0, 1, 2$, the complete conditional distribution of vector λ_k^2 is

$$\begin{aligned}
p(\lambda_k^2 \mid \cdot) &\propto p(\beta_k \mid \lambda_k^2, \tau_k^2) p(\lambda_k^2) \\
&= \prod_{j=1}^{n-1-k} [N(\beta_{kj}; 0, \tau_k^2 \lambda_{kj}^2) IG(\lambda_{kj}^2; 0.5, 0.5)] \\
&\propto \prod_{j=1}^{n-1-k} \frac{1}{\lambda_{kj}} \exp \left\{ -\frac{\beta_{kj}^2}{2\tau_k^2 \lambda_{kj}^2} \right\} \frac{1}{\lambda_{kj}^3} \exp \left\{ -\frac{1}{2\lambda_{kj}^2} \right\} \\
&= \prod_{j=1}^{n-1-k} \frac{1}{\lambda_{kj}^4} \exp \left\{ -\frac{\beta_{kj}^2 + \tau_k^2}{2\tau_k^2 \lambda_{kj}^2} \right\}.
\end{aligned}$$

which implies that the λ_{kj}^2 's are conditionally independent with

$$p(\lambda_{kj}^2 \mid \cdot) = IG \left(1, \frac{\beta_{kj}^2 + \tau_k^2}{2\tau_k^2} \right)$$

for $j = 1, \dots, n-1-k$ and $k = 0, 1, 2$. For $k = 0, 1, 2$, the complete conditional distribution of τ_k^2 is

$$\begin{aligned}
 p(\tau_k^2 \mid \cdot) &\propto p(\beta_k \mid \lambda_k^2, \tau_k^2) p(\tau_k^2) \\
 &\propto \frac{1}{\tau_k^2} \prod_{j=1}^{n-1-k} \left[\frac{1}{\tau_k} \exp \left\{ -\frac{\beta_{kj}^2}{2\tau_k^2 \lambda_{kj}^2} \right\} \right] \\
 &= \left(\frac{1}{\tau_k^2} \right)^{\frac{n-1-k}{2}+1} \exp \left\{ -\frac{1}{\tau_k^2} \sum_{j=1}^{n-1-k} \frac{\beta_{kj}^2}{2\lambda_{kj}^2} \right\} \\
 &\propto IG \left(\frac{n-1-k}{2}, \sum_{j=1}^{n-1-k} \frac{\beta_{kj}^2}{2\lambda_{kj}^2} \right).
 \end{aligned}$$

The complete conditional distribution of σ^2 is

$$\begin{aligned}
 p(\sigma^2 \mid \cdot) &\propto p(y \mid \theta, \sigma^2) p(\sigma^2) \\
 &\propto \frac{1}{\sigma^n} \exp \left\{ -\frac{\|y - \theta\|_2^2}{2\sigma^2} \right\} \frac{1}{\sigma^2} \\
 &\propto IG \left(\frac{n}{2}, \frac{\|y - \theta\|_2^2}{2} \right).
 \end{aligned}$$

The complete conditional distribution of the vector θ is obtained as

$$\begin{aligned}
 p(\theta \mid \cdot) &\propto p(y \mid \theta, \sigma^2) p(\theta \mid \tilde{\theta}^{(1)}, \tilde{\theta}^{(2)}, \lambda_0^2, \tau_0^2) \\
 &\propto N(y; \theta, \sigma^2) N \left(\begin{pmatrix} \theta_1 \\ D^{(1)}\theta \end{pmatrix}; \begin{pmatrix} 0 \\ \tilde{\theta}^{(1)} + \tilde{\theta}^{(2)} \end{pmatrix}, \begin{pmatrix} L & 0 \\ 0 & \tau_0^2 \Lambda_0 \end{pmatrix} \right) \\
 &\propto \exp \left\{ -\frac{\|y - \theta\|_2^2}{2\sigma^2} \right\} \exp \left\{ -\frac{\theta_1^2}{2L^2} \right\} \\
 &\quad \exp \left\{ -\frac{\left(D^{(1)}\theta - \tilde{\theta}^{(1)} - \tilde{\theta}^{(2)} \right)^\top \Lambda_0^{-1} \left(D^{(1)}\theta - \tilde{\theta}^{(1)} - \tilde{\theta}^{(2)} \right)}{2\tau_0^2} \right\}.
 \end{aligned}$$

Since the integer L is arbitrarily large, the second factor in the last line is approximately constant and equal to 1, and can therefore be dropped together with every other factor not

depending on θ . The complete conditional distribution simplifies to

$$\begin{aligned}
p(\theta \mid \cdot) &\propto \exp \left\{ -\frac{\theta^\top \theta}{2\sigma^2} + \frac{y^\top \theta}{\sigma^2} - \frac{\theta^\top D^{(1)\top} \Lambda_0^{-1} D^{(1)} \theta}{2\tau_0^2} + \frac{(\tilde{\theta}^{(1)} + \tilde{\theta}^{(2)})^\top \Lambda_0^{-1} D^{(1)} \theta}{\tau_0^2} \right\} \\
&\propto \exp \left\{ -\frac{1}{2} \left[\theta - \Sigma \left(\frac{1}{\sigma^2} y + \frac{1}{\tau_0^2} D^{(1)\top} \Lambda_0^{-1} (\tilde{\theta}^{(1)} + \tilde{\theta}^{(2)}) \right) \right]^\top \Sigma^{-1} \right. \\
&\quad \left. \left[\theta - \Sigma \left(\frac{1}{\sigma^2} y + \frac{1}{\tau_0^2} D^{(1)\top} \Lambda_0^{-1} (\tilde{\theta}^{(1)} + \tilde{\theta}^{(2)}) \right) \right] \right\} \\
&\propto N(\mu, \Sigma),
\end{aligned}$$

where

$$\mu = \Sigma \left(\frac{1}{\sigma^2} y + \frac{1}{\tau_0^2} D^{(1)\top} \Lambda_0^{-1} (\tilde{\theta}^{(1)} + \tilde{\theta}^{(2)}) \right)$$

and

$$\Sigma = \left(\frac{1}{\sigma^2} I + \frac{1}{\tau_0^2} D^{(1)\top} \Lambda_0^{-1} D^{(1)} \right)^{-1}.$$

The joint complete conditional distribution of the two vectors $\tilde{\theta}^{(1)}$ and $\tilde{\theta}^{(2)}$ is

$$\begin{aligned}
p \left(\begin{pmatrix} \tilde{\theta}^{(1)} \\ \tilde{\theta}^{(2)} \end{pmatrix} \mid \cdot \right) &\propto p(D^{(1)}\theta \mid \tilde{\theta}^{(1)}, \tilde{\theta}^{(2)}, \lambda_0^2, \tau_0^2) p(\tilde{\theta}^{(1)} \mid \lambda_1^2, \tau_1^2) p(\tilde{\theta}^{(2)} \mid \lambda_2^2, \tau_2^2) \\
&\propto N(D^{(1)}\theta; \tilde{\theta}^{(1)} + \tilde{\theta}^{(2)}, \tau_0^2 \Lambda_0) N \left(\begin{pmatrix} \tilde{\theta}_1^{(1)} \\ D^{(1)} \tilde{\theta}^{(1)} \end{pmatrix}; 0, \begin{pmatrix} L & 0 \\ 0 & \tau_1^2 \Lambda_1 \end{pmatrix} \right) \\
&\quad N \left(\begin{pmatrix} \tilde{\theta}_2^{(2)} \\ D^{(2)} \tilde{\theta}^{(2)} \end{pmatrix}; 0, \begin{pmatrix} L & 0 \\ 0 & \tau_2^2 \Lambda_2 \end{pmatrix} \right) \delta(\tilde{\theta}_1^{(2)}) \\
&\propto \exp \left\{ -\frac{1}{2\tau_0^2} (D^{(1)}\theta - \tilde{\theta}^{(1)} - \tilde{\theta}^{(2)})^\top \Lambda_0^{-1} (D^{(1)}\theta - \tilde{\theta}^{(1)} - \tilde{\theta}^{(2)}) \right\} \\
&\quad \exp \left\{ -\frac{1}{2\tau_1^2} \tilde{\theta}^{(1)\top} D^{(1)\top} \Lambda_1^{-1} D^{(1)} \tilde{\theta}^{(1)} - \frac{1}{2\tau_2^2} \tilde{\theta}^{(2)\top} D^{(2)\top} \Lambda_2^{-1} D^{(2)} \tilde{\theta}^{(2)} \right\} \\
&\quad \exp \left\{ -\frac{\tilde{\theta}_1^{(1)2} + \tilde{\theta}_2^{(2)2}}{2L^2} \right\} \delta(\tilde{\theta}_1^{(2)}),
\end{aligned}$$

Since the integer L is arbitrarily large, the factor depending on L is approximately constant and equal to 1, and can therefore be dropped together with every other factor not depending on $\tilde{\theta}^{(1)}$ and $\tilde{\theta}^{(2)}$. Define $X := \begin{pmatrix} \mathbf{I}_{n-1} & \begin{pmatrix} 0 \\ \mathbf{I}_{n-2} \end{pmatrix} \end{pmatrix}$. Moreover, let $\tilde{\theta}_{[-1]}^{(2)}$ refer to vector $\tilde{\theta}^{(2)}$

without its first entry, and $D_{[-1]}^{(2)}$ refer to matrix $D^{(2)}$ without its first column. The complete conditional distribution simplifies to

$$\begin{aligned}
p\left(\begin{pmatrix} \tilde{\theta}^{(1)} \\ \tilde{\theta}^{(2)} \end{pmatrix} \middle| \cdot\right) &\propto \delta(\tilde{\theta}_1^{(2)}) \exp\left\{-\frac{1}{2}\begin{pmatrix} \tilde{\theta}^{(1)} & \tilde{\theta}_{[-1]}^{(2)} \end{pmatrix} [\tau_0^{-2} X^\top \Lambda_0^{-1} X + \right. \\
&\quad \left. \begin{pmatrix} \tau_1^{-2} D^{(1)\top} \Lambda_1^{-1} D^{(1)} & 0 \\ 0 & \tau_2^{-2} D_{[-1]}^{(2)\top} \Lambda_2^{-1} D_{[-1]}^{(2)} \end{pmatrix}] \begin{pmatrix} \tilde{\theta}^{(1)} \\ \tilde{\theta}_{[-1]}^{(2)} \end{pmatrix}\right\} \\
&\quad \exp\left\{\frac{1}{\tau_0^2} \theta^\top D^{(1)\top} \Lambda_0^{-1} X \begin{pmatrix} \tilde{\theta}^{(1)} \\ \tilde{\theta}_{[-1]}^{(2)} \end{pmatrix}\right\} \\
&\propto \delta(\tilde{\theta}_1^{(2)}) N\left(\begin{pmatrix} \tilde{\theta}^{(1)} \\ \tilde{\theta}_{[-1]}^{(2)} \end{pmatrix}; \frac{1}{\tau_0^2} \tilde{\Sigma} X^\top \Lambda_0^{-1} D^{(1)} \theta, \tilde{\Sigma}\right),
\end{aligned}$$

where

$$\tilde{\Sigma} = \left(\frac{1}{\tau_0^2} X^\top \Lambda_0^{-1} X + \begin{pmatrix} \tau_1^{-2} D^{(1)\top} \Lambda_1^{-1} D^{(1)} & 0 \\ 0 & \tau_2^{-2} D_{[-1]}^{(2)\top} \Lambda_2^{-1} D_{[-1]}^{(2)} \end{pmatrix} \right)^{-1}.$$

B.3 Proofs for Section 2.4

Proof of Lemma 2.1. Under (2.10), the increments of $\theta(\cdot)$ and $\nu(\cdot)$ are a priori equal to

$$\begin{aligned}
\theta(t) - \theta(s) &= \int_s^t \nu(u) du \\
&= \int_s^t (\nu(u) - \nu(s) + \nu(s)) du \\
&= (t-s)\nu(s) + V \int_s^t (C(u) - C(s)) du \\
\nu(t) - \nu(s) &= V(C(t) - C(s))
\end{aligned} \tag{B.1}$$

for $t > s \geq 0$. If the noisy data $y_1, \dots, y_n$ are observed at the design points $0 = t_1 < \dots < t_n$, and we let $\theta_i := \theta(t_i)$ and $\nu_i := \nu(t_i)$, for $i = 1, \dots, n$, we can use (B.1) to obtain the following state equation

$$\begin{pmatrix} \theta_{l+1} \\ \nu_{l+1} \end{pmatrix} = \begin{pmatrix} 1 & d_l \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \theta_l \\ \nu_l \end{pmatrix} + V \begin{pmatrix} \int_{t_l}^{t_{l+1}} (C(u) - C(t_l)) du \\ C(t_{l+1}) - C(t_l) \end{pmatrix} \tag{B.2}$$

where $d_l := t_{l+1} - t_l$ and $l = 1, \dots, n-1$. Under the approximation

$$\int_s^u \nu(l) dl \approx (u-s)\nu(s),$$

the increments in (B.1) simplify to

$$\begin{aligned}\theta(t) - \theta(s) &= (t - s)\nu(s) \\ \nu(t) - \nu(s) &= V(C(t) - C(s)),\end{aligned}$$

and the state equation (B.2) becomes

$$\begin{pmatrix} \theta_{l+1} \\ \nu_{l+1} \end{pmatrix} = \begin{pmatrix} 1 & d_l \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \theta_l \\ \nu_l \end{pmatrix} + V \begin{pmatrix} 0 \\ C(t_{l+1}) - C(t_l) \end{pmatrix}. \quad (\text{B.3})$$

The last equation can be equivalently written in terms of $(\theta_{j+2}, \theta_{j+1})$ for $j = 1, \dots, n - 2$. Indeed, from (B.3), we obtain that, for $l = 1, \dots, n - 1$ and $j = 1, \dots, n - 2$,

$$\nu_l = \frac{\theta_{l+1} - \theta_l}{d_l}$$

and

$$\frac{\theta_{j+2} - \theta_{j+1}}{d_{j+1}} - \frac{\theta_{j+1} - \theta_j}{d_j} = V(C(t_{j+1}) - C(t_j)).$$

Solving for θ_{j+2} ,

$$\theta_{j+2} = \theta_{j+1} + \frac{d_{j+1}}{d_j}(\theta_{j+1} - \theta_j) + V\eta_j,$$

where $\eta_j \stackrel{\text{ind}}{\sim} C(0, d_{j+1}d_j)$. □

Proof of Lemma 2.2. Under (2.12), the increments of $\theta(\cdot)$, $\phi(\cdot)$ and $\nu(\cdot)$ are a priori equal to

$$\begin{aligned}\theta(t) - \theta(s) &= \int_s^t \phi(u) du \\ &= (t - s)\phi(s) + \int_s^t (\phi(u) - \phi(s)) du \\ &= (t - s)\phi(s) + \int_s^t \int_s^u \nu(v) dv du \\ &= (t - s)\phi(s) + \int_s^t \int_s^u (\nu(v) - \nu(s) + \nu(s)) dv du \\ &= (t - s)\phi(s) + \nu(s) \int_s^t (u - s) du + V \int_s^t \int_s^u (C(v) - C(s)) dv du \\ &= (t - s)\phi(s) + \frac{(t - s)^2}{2} \nu(s) + V \int_s^t \int_s^u (C(v) - C(s)) dv du \\ \phi(t) - \phi(s) &= \int_s^t \nu(u) du \\ &= (t - s)\nu(s) + V \int_s^t (C(u) - C(s)) du \\ \nu(t) - \nu(s) &= V(C(t) - C(s))\end{aligned} \quad (\text{B.4})$$

for $t > s \geq 0$. If the noisy data $y_1, \dots, y_n$ are observed at the design points $0 = t_1 < \dots < t_n$, and we let $\theta_i := \theta(t_i)$ and $\nu_i := \nu(t_i)$, for $i = 1, \dots, n$, we can use (B.4) to obtain the following state equation

$$\begin{pmatrix} \theta_{l+1} \\ \phi_{l+1} \\ \nu_{l+1} \end{pmatrix} = \begin{pmatrix} 1 & d_l & d_l^2/2 \\ 0 & 1 & d_l \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \theta_l \\ \phi_l \\ \nu_l \end{pmatrix} + V \begin{pmatrix} \int_{t_l}^{t_{l+1}} \int_{t_l}^u (C(v) - C(t_l)) dv du \\ \int_{t_l}^{t_{l+1}} (C(u) - C(t_l)) du \\ C(t_{l+1}) - C(t_l) \end{pmatrix} \quad (\text{B.5})$$

where $d_l := t_{l+1} - t_l$ and $l = 1, \dots, n-1$. Under the approximation

$$\int_s^u \nu(l) dl \approx (u - s)\nu(s),$$

the increments in (B.4) simplify to

$$\begin{aligned} \theta(t) - \theta(s) &= (t - s)\phi(s) + \frac{(t - s)^2}{2}\nu(s) \\ \phi(t) - \phi(s) &= (t - s)\nu(s) \\ \nu(t) - \nu(s) &= V(C(t) - C(s)), \end{aligned}$$

and the state equation (B.5) becomes

$$\begin{pmatrix} \theta_{l+1} \\ \phi_{l+1} \\ \nu_{l+1} \end{pmatrix} = \begin{pmatrix} 1 & d_l & d_l^2/2 \\ 0 & 1 & d_l \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \theta_l \\ \phi_l \\ \nu_l \end{pmatrix} + V \begin{pmatrix} 0 \\ 0 \\ C(t_{l+1}) - C(t_l) \end{pmatrix}. \quad (\text{B.6})$$

The last equation can be equivalently written in terms of $(\theta_{j+3}, \theta_{j+2}, \theta_{j+1})$ for $j = 1, \dots, n-3$. From (B.6), we obtain that, for $l = 1, \dots, n-2$,

$$\begin{cases} \phi_{l+1} - \phi_l = d_l \nu_l \\ \phi_{l+1} - \phi_l = \frac{\theta_{l+2} - \theta_{l+1}}{d_{l+1}} - \frac{\theta_{l+1} - \theta_l}{d_l} - \frac{1}{2}(d_{l+1}\nu_{l+1} - d_l\nu_l). \end{cases}$$

From the two equalities above, we obtain

$$\frac{\frac{\theta_{l+2} - \theta_{l+1}}{d_{l+1}} - \frac{\theta_{l+1} - \theta_l}{d_l}}{\frac{d_{l+1} + d_l}{2}} = \frac{d_{l+1}\nu_{l+1} + d_l\nu_l}{d_{l+1} + d_l}.$$

Therefore, for $j = 1, \dots, n-3$, the following holds:

$$\begin{aligned} & \frac{\frac{\theta_{j+3} - \theta_{j+2}}{d_{j+2}} - \frac{\theta_{j+2} - \theta_{j+1}}{d_{j+1}}}{\frac{d_{j+2} + d_{j+1}}{2}} - \frac{\frac{\theta_{j+2} - \theta_{j+1}}{d_{j+1}} - \frac{\theta_{j+1} - \theta_j}{d_j}}{\frac{d_{j+1} + d_j}{2}} = \frac{d_{j+2}\nu_{j+2} + d_{j+1}\nu_{j+1}}{d_{j+2} + d_{j+1}} - \frac{d_{j+1}\nu_{j+1} + d_j\nu_j}{d_{j+1} + d_j} \\ & = \frac{d_{j+2}}{d_{j+2} + d_{j+1}}(\nu_{j+2} - \nu_{j+1}) + \frac{d_j}{d_{j+1} + d_j}(\nu_{j+1} - \nu_j) \\ & \stackrel{\text{ind}}{\sim} V \cdot C \left(0, \frac{d_{j+2}d_{j+1}}{d_{j+2} + d_{j+1}} + \frac{d_j^2}{d_{j+1} + d_j} \right). \end{aligned}$$

Solving for θ_{j+3} , we obtain

$$\theta_{j+3} = \theta_{j+2} + \frac{d_{j+2}}{d_{j+1}}(\theta_{j+2} - \theta_{j+1}) + d_{j+2} \frac{d_{j+2} + d_{j+1}}{d_{j+1} + d_j} \left(\frac{\theta_{j+2} - \theta_{j+1}}{d_{j+1}} - \frac{\theta_{j+1} - \theta_j}{d_j} \right) + V\eta_j,$$

where

$$\eta_j \stackrel{\text{ind}}{\sim} C \left(0, \frac{d_{j+2}^2 d_{j+1}}{2} + \frac{d_{j+2} d_j^2}{2} \frac{d_{j+2} + d_{j+1}}{d_{j+1} + d_j} \right).$$

□